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ALGEBRA OF CONSCIENCE

*A Comparative Analysis of
Western and Soviet Ethical Systems*

with a
FOREWORD
by
ANATOL RAPOPORT



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FOREWORD

In this book two ethical systems are described in the language of mathematics. Ordinarily mathematics is thought to be a science of quantity. Indeed, manipulation of quantities constitutes much of mathematics. Elementary applied mathematics deals with reckoning and measurement, where concrete quantities are objects of attention, such as counting sheep or weighing corn. But the operations on these quantities are performed with the help of symbols, from which concrete referents have been 'abstracted out': $3 + 5 = 8$ regardless of whether the symbols stand for numbers of sheep or tons of corn. Thus, the first principle that exhibits the power of mathematics is *abstraction*. It is one of the three pillars on which the edifice of mathematics rests.

Another pillar is *precision*. Ordinarily, man communicates by words. Words serve communication to the extent that they refer to things, events, states of affairs, feelings of the speaker, and so on. These are the *meanings* attributed to words. Communication is successful to the extent that the meanings coded upon words by the speaker correspond to the meanings decoded by the hearer. As is well known, the degree of this correspondence varies enormously in different contexts of discourse and with the backgrounds or attitudes of the speakers and hearers. Mathematics is a language in which the meanings of the symbols (the 'words' of this language) are absolutely precise. This precision is achieved by abstraction. Abstract terms are defined by their relations to other terms and by nothing else. Since circular definitions are avoided, every mathematical system must contain terms that cannot be further defined.

Words of ordinary language are also ultimately related to words that are not, in fact, need not be defined, presumably because they refer to something directly intuited by the speakers and hearers. The undefined terms of pure mathematics have no referents in the 'real world'. In *applied* mathematics, connections between symbols and the real world are established by relating the 'rock bottom' terms to elements of 'reality', e.g., points in space, moments of time, units of mass, etc. In this way, applied mathematics can be used in investigations of and communication about the real world, but always with some loss of precision. How much precision is

lost depends on how much is 'abstracted out' when elements of the real world are coded upon mathematical symbols.

The third pillar of the mathematical edifice is *deduction*. Mathematics is a vast elaboration of strict logic. By means of logical operations on mathematical symbols, relations can be deduced from initial assumptions that could not have been deduced without this apparatus. The meanings coded upon the symbols of a mathematical system and the assumptions about their relations to each other constitute a mathematical model. If the meanings and the relations correspond closely to their referents in the real world, the conclusions arrived at by mathematical deduction can serve as credible explanations of observed phenomena and even as reliable predictions of phenomena not yet observed. These are the remarkable achievements of the exact, i.e., mathematicized sciences.

It may seem strange at first thought that concepts relevant to ethics can be mathematicized, since these concepts usually refer to attitudes, imperatives, moral judgments, and the like rather than to precise quantitative relations, usually thought of as the main (even exclusive) concerns of mathematics. Nevertheless, there have been attempts to quantify such concepts. Bentham's utilitarian principle, 'the greatest good of the greatest number' comes to mind. The approach in this book is different. The principal tool here is not the mathematics of quantity but the mathematics of structure. Quantification does enter in the calculation of the 'ethical status' of individuals or situations, but only as a by-product of structural analysis. It serves to compare the deductions of the model with observations.

The meanings coded upon the 'rock bottom' terms in Lefebvre's models of ethical systems are 'good' and 'evil', depicted by '1' and '0' respectively. These concepts are, of course, fundamental in all ethical systems, and in specific cultural settings, their meanings are presumably intuited. In a mathematical model, however, they must remain undefined. They are only distinguished.

Of course, the tremendous range of meanings assigned to 'good' and 'evil' by people of different cultural backgrounds and temperaments precludes formulations of mathematical models that capture these meanings in all their richness. But if clarity of discussion is a value (as it is in all scientific contexts) and if the deductive power of mathematical reasoning is to be put to a test in this, at first thought inappropriate context, one must begin *somewhere*. One thing that can be said with certainty is that in the context of ethics, 'good' and 'evil' must be distinguished. Lefebvre's models go no further in this respect. The significance of this approach as a contribution to

a theory of ethics must be judged by how much can be done on this sparse foundation.

The next fundamental concepts are 'awareness', 'confrontation' and 'compromise'. A person experiences his feelings, beliefs, attitudes, etc. directly. Possibly, a dog also experiences his feelings (e.g., hunger, rage) directly. A human being, however, has also higher orders of awareness. He can be aware of his awareness. 'This is good' is an expression of a feeling. 'I know that this is good' expresses an awareness of a feeling. 'I believe that he thinks that I am his enemy' expresses awareness of still higher order. As far as we know, only human beings are capable of 'multiple-tier' awareness of this sort. Thus, a person has an 'inner world' of which he is directly aware. He may also be aware of his awareness of his inner world, have an image of someone else's inner world, an image of someone else's image of his own inner world, etc.

Lefebvre uses a particular algebraic formalism to represent this multiple-level awareness as well as friendly or hostile relations between individuals, the individuals' images of these relations, and evaluations of them as 'good' or 'evil'.

The two ethical systems represented by means of this formalism are grounded respectively in two pairs of fundamental assumptions. 'Confrontation between good and evil is good; compromise between good and evil is evil' is at the base of the first ethical system. 'Confrontation between good and evil is evil; compromise between good and evil is good' lies at the base of the second.

Here the reader should be warned against a possible misunderstanding that may arise from the usual connotations of the words 'confrontation' and 'compromise'. Ordinarily, one thinks of confrontation and compromise in relation to persons with different outlooks or interests, whereby confrontation is associated with conflict and compromise with conflict resolution. Lefebvre, however, refers to 'confrontation' or 'compromise' between good and evil *within the same person*. Thus the assumptions 'confrontation between good and evil is good' and 'compromise between good and evil is evil' means that this person refuses to rationalize an evil act by its presumed good consequences. The opposite assumption means that the person is able or willing to do so.

It turns out, as a logical consequence of the model, that welcoming 'confrontation' and rejecting 'compromise' between good and evil leads to a readiness to compromise with an adversary, while welcoming 'compromise' and rejecting 'confrontation' between good and evil leads to an uncom-

promising attitude towards an adversary. Lefebvre remarks that this result is 'paradoxical', but the paradox stems from the two different contexts in which the words 'confrontation' and 'compromise' are used. An intuitively clearer description of the first ethical system (also given by Lefebvre) is as one where the principle 'the end justifies the means' is rejected and of the second as one where this principle is accepted. The opposite evaluations of confrontation and compromise between good and evil are introduced, because they are directly reflected in the mathematical formalism.

The first ethical system, in which justification of evil means to achieve good ends is rejected, is attributed by Lefebvre to Christian ethics and, by extension, to 'Western' ethical norms. The second ethical system is assumed to be the foundation of norms prevalent in totalitarian societies, specifically in the Soviet Union.

The identification of the first ethical system with the ideals of Christianity appears to be accurate if these ideals are understood as those expressed in the Sermon on the Mount and even more so in the light of Tolstoy's interpretation of Christianity. (The clearest example of this interpretation is found in Tolstoy's short story, 'The Godson', where a mortal, temporarily omnipotent tries to 'set things right' with disastrous results.) To what extent Christian ideals have guided the formation of social and interpersonal norms in the West ought to be regarded as an open question. Surely, the history of the West and of its relation to the rest of humanity raises grave doubts on this score. It is true that machinery for conflict resolution based on compromise has been developed in Western democracies. It is likely, however, that widespread and firmly institutionalized commercial practices have contributed to the formation of the ideology of compromise considerably more than Christian ideals.

Explicit acceptance of the principle 'the end justifies the means' is obvious and prominent in Soviet political practice. It appears especially clear in the emphasis on humanitarian ideals in Soviet ideological pronouncements. It is embodied in Stalin's answer to Lady Astor's question, "When will you stop killing people?" – "When it is no longer necessary."

Apologetics of this sort are less prominent in other totalitarian ideologies, to which Lefebvre ascribes his second ethical system. The Nazis glorified cruelty as such without justifying it as 'sacrifice' for future generations.

The importance of this book, however, does not depend on the degree of appropriateness of the characterizations of the social systems associated with liberal and totalitarian ideologies. It should be seen in the way Lefebvre builds his mathematical apparatus to formalize concepts that do

not lend themselves to operational definitions. He constructs mathematical expressions for concepts that pervade discussions of ethics (e.g., 'good' and 'evil') as well as those prominent in psychological novels, especially in Russian literature (e.g., 'sacrifice', 'guilt', 'suffering') and concepts that reflect human archetypes (e.g., 'hero', 'saint', 'philistine', 'dissembler'). The theoretical leverage of these expressions is not to be sought in the accuracy with which they capture the rich (and necessarily vague) intuitive meanings of the concepts but rather in the circumstance that they permit deduction of logical consequences not immediately apparent but nevertheless in accord with observations, especially consequences derived from multi-level expressions of awareness.

A similar apparatus, based on a different formalism was developed by Lefebvre (1977a) in an earlier book, *The Structure of Awareness*. The present formalism is a significant step forward in that it permits the calculation of 'ethical statuses' of individuals and situations that can be compared with observations. Most enlightening is the analysis of major characters in Dostoevsky's (1964) *Crime and Punishment*, who reflect Lefebvre's four archetypes. Also noteworthy is the agreement between Lefebvre's theoretically-deduced conclusion about the emergent dominance of the 'philistine' in Soviet society and a similar conclusion drawn from Vera Dunham's analysis of Soviet popular literature in her book, *In Stalin's Time*.

Further corroboration of the models is provided in moral judgments of persons and situations elicited from respondents with Soviet and American backgrounds. That at least ordinal comparisons between attributed ethical statuses agree with those deduced from Lefebvre's models, suggests that the approach developed in this book is more than an intellectual exercise. It seems to capture some important aspects of ethical conceptions and of character typology that can serve as the fundamental concepts of a logically rigorous theory of ethics.

Even more intriguing is the remarkable structural analogy between the 'algebra of conscience' and mathematical formulations of quantum mechanics. Heisenberg's Uncertainty Principle has been advanced as 'scientific' support for the existence for free will, and connections have been noted between Bohr's Complementarity Principle and the limitations of self-awareness. The narrator in Tolstoy's novel *Childhood and Adolescence* becomes confused when he asks himself "What I am thinking about?" and answers, "I am thinking, 'What am I thinking about?'" which only raises the question, "And now what am I thinking about?" Free-wheeling

speculations on these matters, usually called philosophical, which Lefebvre calls metaphorical, seldom, if ever, lead to enlightenment. Only the vigour of mathematical reasoning points to a way out of impasses engendered by thought entangled in words. I do not share Lefebvre's view that the paradoxes manifested in quantum phenomena, in undecidable propositions, discovered in mathematical logic, and in problems of self-awareness are manifestations of laws governing the universe. I do believe, however, that they are manifestations of some fundamental characteristics of human cognition and of human value judgments.

ANATOL RAPOPORT
Vienna, March, 1981

INTRODUCTION

In this book I try to demonstrate that the evaluation of the relationships between 'good' and 'evil', which is a norm for any given culture, predetermines the typology of individuals belonging to that culture. There is a certain connection between the ethical philosophy of an individual and his psychological type. This connection can be expressed formally in strict mathematical language.

We can distinguish two ethical philosophies. One is based on the principle 'the compromise of good and evil is evil'. The second is based on the principle 'the compromise of good and evil is good'. Further, we shall refer to these as the 'first' and 'second' *ethical systems*, and we include in the concept of ethical system both ethical philosophy and normative personalities. I shall offer experimental and empirical arguments demonstrating that the first ethical system is realized in Western culture and the second ethical system is realized in the culture of the Soviet Union. Thus, the difference between Western society and Soviet society is much deeper than is usually assumed: this difference touches upon the fundamental structures connecting the categories of good and evil.

The experimental and empirical data about the second ethical system relate only to Soviet culture. However, I am convinced that Soviet culture is not unique in this respect. With a high degree of certainty we can state that the second ethical system is realized also in continental China and in various countries of Indochina, the Middle East, and Africa. Very often, the second ethical system is connected with communist ideology. However, this connection is not necessary: the second ethical system may be tied in with fascism, for example, or with some extreme religious philosophies. The Soviet Union is the most developed society whose culture is based on the second ethical system. This permits an analysis of different phases in the development of such societies and an analysis of the catastrophic cataclysms that often occur during transition from one phase to another.

We shall also demonstrate briefly that Soviet culture, with its basis in the second ethical system, is not historically unique. For example, an analysis of Icelandic sagas suggests the hypothesis that pre-Christian Europe belonged to the second ethical system as well.

The main theoretical idea of this work consists in likening an individual to an abstract automaton. Such likening is commonly used in behavioristic investigations. In this work we shall demonstrate that *it is possible to construct an automaton model of a completely different, non-behavioristic type*. Such a model can be represented as an automaton with inputs, output, and an inner structure. This inner structure, in turn, can be represented by a mathematical formula containing various algorithms connecting inputs and output. But at the same time, together with the algorithmic meaning, the syntax of the formula contains a special description of the psychological state of an individual. In this way, we believe we have established a connection between two levels of modeling reality. On the one hand we have behavior which can be modeled in a stimulus-response scheme. On the other hand, we can read the state of the inner world of an individual in 'terms' of this inner world such as 'image of oneself', 'image of one's enemy', 'feeling guilty', 'suffering', 'sacrifice', etc. By changing a formula we can change both the behavior and the structure of the inner world of an individual. In this way we gain the possibility of connecting the peculiarities of the inner world with the peculiarities of the behavior of a given individual. This is the main technical idea of our investigation. Such automata, contrary to the traditional ones, reflect not only man's behavior, but also his 'semantics'.

How are stimuli and responses interpreted in this model? Stimuli are actions directed at an individual by the social environment. These 'directions' may be 'good' or 'bad' within the framework of the morality of the culture to which the individual belongs. An individual's responses may be of two types: he either agrees to accept the environmental demand or he refuses to do so.

We might call an individual 'better' the less frequently he performs 'bad' actions directed by the environment. Analysis of the structure of the 'inner world' of an automaton permits us to establish the connection between his 'psychological type' and his 'ethical status'.

The main result of the work may be described as follows. The theory predicts that an ideal individual of the first ethical system, who evaluates negatively the compromise between good and evil, nevertheless tends to establish relationships of compromise, or union, with another individual, even in a situation of conflict. The theory predicts also that an ideal individual of the second ethical system, who evaluates positively the compromise between good and evil, nevertheless, *a priori*, tends toward confrontation with his partner. Therefore, the theory predicts the existence

of a kind of paradox: ethical 'compromiselessness' is connected with a compromise in human relationships, and ethical compromise is connected with 'compromiselessness' in human relationships.

Our analysis demonstrates that 'positive' Americans evaluate compromise between good and evil negatively, but tend to compromise in situations of confrontation with a partner. 'Positive' Soviets evaluate compromise between good and evil positively, but they are uncompromising toward partners in confrontation. Thus, we see that the characteristics obtained from pure theoretical reasoning are realized in American and Soviet cultures; and we have to attribute to the United States the first ethical system and to the Soviet Union the second. We can draw the conclusion from this that our model correctly reflects the main laws of ethical systems. In addition, the model predicts several subtle differences in the normative personalities of the first and second ethical systems; some of these differences were found in experiments.

Our schematic description of the ideal American and the ideal Soviet man requires explanations. The author is far from the thought that everyone living in the United States and everyone in the Soviet union has the peculiarities described above. Neither the American nor the Soviet culture is a homogeneous unit generating individuals of strictly one type. Moreover, each of these societies today is a battle arena between the two ethical systems. But we can establish the predominant ethical philosophy and the predominant normative character of an individual, which is predetermined by a specific style of upbringing and education. Thus, when we say that in American society the first ethical system is realized and in Soviet society the second ethical system, we emphasize the fact of the predominance of one ethical system in a given society. In addition, in each culture there may exist subcultures belonging to another ethical system.

However, we did not attempt to conduct a broad empirical investigation. The empirical and experimental parts were limited by the goal of verifying the predictive ability of our model in relation to the predominant ethical tendencies in the United States and the Soviet Union. The data serve the purpose of demonstration rather than of strict empirical proof.

The author has experienced some difficulties in describing the results of his work, since he was aware of the necessity of simultaneously describing both theoretical results and applications. A separate exposition of the 'theory' and 'applications' would render many details of this work less understandable.

The book consists of seventeen chapters and ten appendices. The

appendices contain technically complicated material, unnecessary at first reading.

In Chapter I, we describe the results of experimentally investigating evaluations of the relationships between good and evil given by Americans and recent Soviet emigrés. The investigation was made with the help of a questionnaire containing four groups of two statements. The first statement in each group describes the possibility of using bad means to attain good ends (the compromise of good and evil). The second statement describes a prohibition against the use of bad means to attain good ends (the confrontation of good and evil). Subjects were asked to express their agreement or disagreement with each of these statements.

The answers of Americans and former Soviet citizens turned out to be completely different. Americans agree with the confrontational statements and disagree with the statements of the compromise type. Recent emigrés from the Soviet Union, in contrast, agree with the compromising statements and disagree with the statements of the confrontational type.

These results experimentally favor the argument that Americans evaluate ethical compromise negatively and ethical confrontation positively, but that Soviets evaluate ethical compromise positively and ethical confrontation negatively.

In Chapter II, we begin constructing a formal model. The two different axiomatic systems are presented. Their difference reflects the difference found in Chapter I. Elementary equations of Boolean algebra underlie the two sets of axioms. '1' is employed for the concept of 'good', and '0' for the concept of 'evil'. The difference between the two sets of axioms is that in the first set compromise is expressed as 'logical multiplication' and confrontation as 'logical addition'; in the second set, vice versa, 'logical multiplication' is used for confrontation and 'logical addition' for compromise. The first set of axioms is the basis for the first ethical system, and the second is the basis for the second ethical system. The common axiom for both ethical systems is the statement: 'evil aware of evil becomes good'. A formal analogue for this statement is 'logical implication' (a function which plays a key role in our construction).

In Chapter III, the axioms of Boolean algebra and the laws for constructing propositional formulae are introduced. In distinction to the method of writing formulae, implication is expressed as an exponent, and this allows us to express long chains of implications compactly.

In Chapter IV, formal definitions of an individual and of a situation are introduced. A formula with a special hierarchical structure is called an

'individual'. An individual has an 'inner world', which may consist of images of this individual, of his partner, and of the relationships between them (confrontation or compromise). These images, in their turn, may have their 'inner worlds', etc. Then we include a formal concept of the correctness of images: an individual has either a correct or an incorrect image of himself, his partner, and their relationships. A 'situation' is represented by a formula expressing two individuals and the relationships between them. We introduce a formal definition of semantic equality of formulae and demonstrate the difference between semantic equality and logical equality. After this the algorithm for reading formulae is given.

In Chapter V, we introduce the concept of a semantic automaton. The formulae representing individuals are considered as transformers of a current of the environment's demands into a current of individuals' agreements or disagreements. We analyze currents with various (but constant) frequencies of appearance of 'good' and 'bad' environmental demands. The main attention is given to a random case. We introduce the concept of the ethical status of an individual: this is a frequency of appearance of 'good' answers of the individual as the automaton output for a randomized current of the environment's demands. Then it is shown that the ethical status of the simplest two-tier individual who has a correct image of himself is higher than the ethical status of an individual similar to the first one, but with an incorrect image of himself. Note that the ethical statuses of an individual can have only three values: 1 , $\frac{3}{4}$, $\frac{1}{2}$. An individual with ethical status 1 performs a 'good' action in response to any environmental demand. An individual with the status $\frac{3}{4}$ performs 'good' or 'bad' actions with equal probability when he receives evil demands from the environment. An individual with ethical status $\frac{1}{2}$ agrees with any environmental demand.

In Chapter VI, we introduce formal analogues of human states such as doubt, feeling guilty, condemnation, suffering.

An individual may *doubt* the correctness of his images; he *feels guilty* when his image of himself is in a state of 'evil'; he *condemns* his partner when his image of this partner is in a state of 'evil'; an individual *suffers* when he evaluates the situation in which he is involved as 'evil'. These feelings change depending on the environmental demands at the input. Thus we use frequencies to measure them. Feeling guilty is measured by a parameter which shows how often an individual sees himself in a state of evil. Condemnation corresponds to the parameter of how often an individual sees his partner in a state of evil. Suffering is a measure of how often an individual sees a situation as being in a state of evil.

At the conclusion of this chapter we demonstrate that feeling guilty is formally related to the structure of doubts. Feeling guilty is greater in individuals who doubt the correctness of their images of themselves than in those individuals who do not have such doubts. This is true for both ethical systems.

In Chapter VII, we begin to compare the two ethical systems. In each of them we study the connections between the structure of doubt and that of feeling guilty and of condemnation when an individual is in a state of confrontation with his partner. It is shown that feeling guilty appears more often in the first ethical system than in the second and that the absence of doubts as to correctness of one's image of oneself leads to a loss of feeling guilty in the second ethical system but does not lead to it in the first.

In Chapter VIII, we continue to compare the ethical systems and consider individuals in confrontation with each other. We analyze the ethical hierarchy depending on the structure of doubts. To do this we construct tables which demonstrate the connection between the structure of doubts and the ethical status. The presence of doubt as to the correctness of an individual's image of himself promotes an increase of the ethical status of this individual in both ethical systems. However, the doubt of an opponent's image has a different influence on the value of ethical status in the two ethical systems. The doubts of one's image of one's opponent in the first ethical system may increase this individual's ethical status, but can never decrease it. In the second ethical system the opposite is true: doubts of one's image of one's opponent may decrease one's ethical status, but can never increase it.

At the end of Chapter VIII we describe individuals who, in each ethical system, have maximal ethical status (in a situation of confrontation).

We continue with an analysis of the ethical structures vested in the heroes of classical Western and Soviet pieces of literature (Chapter IX). Hamlet was chosen as an example of a hero in Western literature. His conflict with Claudius is analyzed by our formal method; Hamlet would receive higher ethical status than Claudius only if their conflict were considered within the framework of the first ethical system. As examples of Soviet literary character we chose Kozhukh from the novel *The Iron Flood* by Serafimovich and Pavel Korchagin from *How the Steel was Tempered* by Nikolay Ostrovsky. The formulae of interaction with an adversary were constructed for each of these heroes. It turned out that the ethical status of a hero of Soviet literature is higher *only* if we make calculations in the second ethical system.

This is an argument in favor of the hypothesis that the first ethical system is realized in Western culture and the second ethical system is realized in Soviet culture.

In Chapter X, the results of experimental analysis of the ethical systems are described. Americans and recent Soviet emigrés were asked to fill out a questionnaire that gives a basis for constructing formulae of interactions of two men. The ethical priority of one man was given in advance or had to be defined by the subject. The data were interpreted in both the first and second ethical systems. The model based on the first ethical system predicts the ethical priorities of Americans better than the model based on the second ethical system. For recent Soviets the reverse is true: the model based on the second ethical system predicts ethical priorities better than the model based on the first ethical system. This is a second argument in favor of the hypothesis that in the U.S. and U.S.S.R. different ethical systems are realized.

In Chapter XI, we begin a formal analysis of an individual's activity in a situation where he may choose his relationships with his partner: compromise or confrontation. We accept a postulate that by making a decision an individual tends to maximize the ethical status of his image of himself. We prove that, in the first ethical system, an individual would prefer compromise with his partner, and in the second ethical system he would prefer confrontation. This is one of the most important results in our work. It clarifies the difference in *a priori* relationships toward one's partner in the first and second ethical systems.

At the end of this chapter we analyze the ethical and strategical elements which obtain during the process of choosing one's relationships with one's partner.

In Chapter XII, we investigate the connection between the procedure of choosing one's relationships with one's partner and feeling guilty and suffering. We prove that *when an individual maximizes the ethical status of his image of himself, he correspondingly minimizes his feeling guilty, but maximizes his suffering.* We interpret this fact as *sacrificial behavior*: decreasing the feeling guilty at the price of increasing the suffering. In the first ethical system a sacrifice is made through compromise with a partner ('In spite of the fact that this is hard for me, I have to offer my hand to this man, otherwise I would degrade myself in my own eyes'.) In the second ethical system a sacrifice is made through conflict. ('I have to oppose myself to this man, even though this is hard for me, else I would degrade myself in my own eyes'.) A formal analysis demonstrates that an individual, by

maximizing the ethical status of his image of himself maximizes, on the average, his own ethical status (from an external observer's point of view).

An individual who tends to minimize feeling guilty at the price of maximizing suffering is called a '*hero*'. Another kind of individual, one who minimizes his suffering at the price of increasing guilt feeling is called a '*philistine*'. A hero's ethical status is higher than that of a philistine.

Two more types of individuals are defined: *saint* and *dissembler*. A saint is an individual who behaves sacrificially, but who does not evaluate his behavior as sacrificial; he is modest. A dissembler falsely evaluates his own behavior as sacrificial.

It is proved that the highest ethical status is possessed by the saint, and the lowest by the dissembler, with the hero and philistine in between.

The saint has a surprising formal peculiarity: the 'negative feeling' which he tries to minimize cannot be isolated. A hero, for example, tends to minimize feeling of guilt; a philistine minimizes suffering; a dissembler tries to minimize both guilt and suffering. A saint, unlike all of them, tends to maximize both feeling of guilt and suffering.

We give a table which demonstrates the difference between normative individuals of the first and second systems. Individuals of the sacrificial type (saint and hero) in the first ethical system would tend toward compromising behavior, while in the second ethical system they tend toward aggressiveness. Individuals of the non-sacrificial type (philistine and dissembler) in the first ethical system would realize aggressive behavior, while in the second ethical system they would tend toward compromise.

All the above types were established as a result of the analysis of three-tier formulae. As a basis for the analysis we choose the peculiarities of structures of individuals. Further, we demonstrate that for analysis of formulae of arbitrary complexity the value of the ethical statuses may be taken as a basis. We introduce the concept of the 'perfect individual', whose ethical status is equal to 1. Such an individual cannot 'see' himself as perfect. Thus, 'perfection' is always an outer characteristic of an individual and cannot be a fragment of his own self-description. Saints are perfect more often than other types of individuals.

At the end of the chapter we demonstrate that the standard description of Lenin in Soviet propagandistic literature has all the features of a saint of the second ethical system.

In Chapter XIII, the results obtained in Chapter XII are generalized for formulae of wider application.

In Chapter XIV, we investigate the process of interaction between

individuals of different types, for example, between a hero and a philistine. To describe such interaction the concept of 'activity of an individual in a given interaction' is introduced. A measure of activity is the frequency with which an individual realizes his choices of the relationships with his partner. Thus, each individual may be in two phases: active or passive. In an active phase an individual chooses the type of relationships between himself and his partner and then realizes them. In a passive phase, an individual reflects the relationships chosen and realized by his partner. Concepts of active and passive ethical statuses are introduced.

In Chapter XV, we describe the results of our analysis of the typology of characters in the novel *Crime and Punishment*, by Dostoevsky. We assume that as a great artist Dostoevsky truthfully reflected ethical reality in his novel. We wanted to predict the type of behavior of Dostoevsky's characters by using their proclaimed philosophies and inclination to sacrifice (sacrificial or not). The ability of our model to make a correct prediction of this type may be considered, in our opinion, one of the methods of its verification.

In the chosen novel all elements necessary for the analysis (philosophy of the individual, type of sacrifice, and type of behavior) could be obviously and non-ambiguously separated from the text of the novel for five main characters.

Our analysis demonstrated the following: the main hero of the novel, Raskolnikov, combines within himself two different personalities belonging to the two different ethical systems. One of his personalities believes that 'the end does not justify the means', that is, belongs to the first ethical system; it is sacrificial and chooses compromise with a partner; this corresponds to our model's prediction. The second personality believes that 'the end justifies the means', that is, it belongs to the second ethical system. This personality is also sacrificial, but chooses aggressive behavior; this corresponds to our model as well. There are four other characters whose behavior can be predicted by our model. Two of them belong to the first ethical system, and two others to the second. Sonia Marmeladova and Svidrigailov are sacrificial individuals. Sonia's philosophy is of the first ethical system, and Svidrigailov's philosophy is of the second. Accordingly Sonia realizes compromising behavior, and Svidrigailov is aggressive. Luzhin and Lebeziatnikov are non-sacrificial: Luzhin's philosophy belongs to the first ethical system, Lebeziatnikov's to the second. Their behavior is opposite to that of sacrificial individuals. Luzhin is aggressive, and Lebeziatnikov tends to compromise. Thus we see that our model is capable

of predicting the type of behavior of the main characters.

In Chapter XVI, the correlation between ideology, morality, and political organization is analyzed. The main formulations of Soviet 'ethical ideology' are contained in the so-called 'Moral Code of a Builder of Communism', which is a part of the Program of the Communist Party. We compared this Moral Code with the main principles of Christianity and found an interesting difference. At the basis of The Old Testament are the Laws of Moses, with their prohibition of evil: Thou shalt not kill and Thou shalt not steal. Such formulations of ethical principles lead to negative evaluation of ethical compromise and to positive evaluation of ethical confrontation.

Christian 'theory of personality' is contained in The New Testament. A personality is enjoined to unify with another personality, to compromise with him. We see that the ethical demands of Christianity are coordinated in the framework of the first ethical system: compromiselessness in ethical evaluations and compromise in human relationships.

The 'Moral Code of a Builder of Communism' contains the approaches of both The Old and The New Testament. It has two parts: ethical principles and a theory of personality. Instead of ethical prohibitions the Moral Code contains a declaration of good: a person is called to be 'honest, truthful, morally pure, simple, and modest'. Such formulations lead to ethical compromise, since evil is not prohibited and may be used if it is necessary for the triumph of good. In the other part of the Moral Code there is a requirement to be ruthless toward an enemy. We can see that the ethical demands of communism are also logically coordinated, but in the framework of the second ethical system.

Therefore, prohibition of evil leads to the morality of the first ethical system, and declaration of good leads to the morality of the second ethical system.

In the second part of Chapter XVI we analyze the peculiarities of functioning of a governmental system which has the second ethical system as its basis. The life of such a system has two phases: heroic and philistine. In the first, the heroic phase, the dominant role is played by sacrificial individuals of the second ethical system. This individual tends toward sacrificial conflict with other individuals. The life of the society where such an individual is dominant becomes a permanent civil war of everyone against everybody. The government has to suppress this heroic element. Stalin's terror may be considered as a fight against this element. The main strategy in this fight was to destroy the 'heroic man' of the second ethical system. Facts supporting this idea are given in Chapter XVI. As a result of mass terror the 'proud man' of the second ethical system was destroyed and remained only in the system of propaganda as an abstract and unachievable

ideal. Soviet society now is in the second, 'philistine phase', in which personalities have low self-respect and easily obey any strong opponent. Their heroic behavior may be found only in non-dangerous situations in the family or in the store toward a weaker opponent. It looks like a caricature of an original heroic model.

At the end of the chapter we conclude that the ability at self-organization of a society based on the second ethical system is the higher, the lower the ethical status of its members. For a society based on the first ethical system the picture is different. The ability at self-organization is the higher, the higher the ethical status of its members.

In Chapter XVII, we generalize our formal scheme. We analyze the case of interaction of an arbitrary number of individuals, where each two of them are either in conflict or in union. The theory easily lends itself to generalization for the case of three individuals. But the case of four or more interacting individuals is not trivial. For some configurations of relationships there is no way to put a Boolean function into correspondence. Thus, we conclude, there are some sets of relationships between individuals that are ethically non-measurable in principle. In some cases there is no procedure that would permit us to calculate the ethical state of a group of individuals as a whole, in spite of the fact that we have the requisite information about each individual and his relationships with everyone. A situation which cannot be represented as a superposition of binary relations is non-measurable. In this way we prove the existence of fundamental limitations, that appear when we attempt to give ethical evaluations of situations containing a large number of individuals. This result is true for both ethical systems.

In the Conclusion, we give the substantiation of our axioms. Since some of them were verified experimentally and empirically, and others were introduced formally, one can suppose that another model based on a different set of axioms may exist and may reflect ethical reality better. From a formal point of view, 448 different models with different definitions of initial relationships between good and evil may exist. To select acceptable models from the 448 existing ones, we introduce the criterion of 'appropriateness': a simple two-tier individual with a correct image of himself (i.e., correct reflexion of his own intention) has a higher ethical status than a similar individual with an incorrect image of himself, and a simple three-tier individual with the correct image of himself and doubting the correctness of it has a higher ethical status than a similar individual with an incorrect image of himself and no doubts about its correctness. This criterion may be written down as three inequalities. We have analyzed all 448 models in

terms of these criteria and have found that only one model is appropriate – the model in which the process of reflexion is expressed by logical implication, and relationships between individuals are expressed by the operations of Boolean algebra. In this way we have substantiated our set of axioms.

The general method for representing judgments made by individuals is described in Appendix 1. These judgements may be of two types: judgments about the correctness of images and judgments about the correctness of judgments. We introduce a recursive algorithm for making judgments and represent these judgments by a chain of symbols from the top down (beginning from some image). Thus, to describe the inner world of an individual we use three directions: the diagonal gives levels of reflexions, the horizontal – images of situations, and the vertical – hierarchy of judgments.

In Appendix 2, we consider the generalization of our model to the case where the scale ‘good-evil’ is divided on an arbitrary finite number of intermediate values and the case of a continuum of values. We obtain a model based on multi-valued logics. All the inequalities which were true for a scale of two elements remain true on a multi-valued scale. However, the equalities may break down when we move from a two-valued scale to a three or more-valued scale. A three-valued scale gives us the complete information about the relationships ‘equal to’ and ‘greater than’ between formulae of individuals. We must note that only two-letter formulae were investigated, and our statement that a three-valued scale gives complete information of multi-valued scales is true only for the model of interaction between two individuals. We must note also that for $n \geq 50$ we have proved analytically that ratios of modules do not depend on n , but for $3 \leq n < 50$ we do not have an analytical proof, and this statement was verified with the help of a computer. Analysis of the three-valued scale has permitted us to discover some fine details of formal connection between inner structure and ethical status. For example, it occurred that in the second system the mere fact of awareness of the presence of an image of an enemy leads to a decrease in one’s ethical status.

In Appendix 3, we consider the case where the environmental demands are caused by individuals’ activity. It is demonstrated that for some quite ‘natural’ conditions an individual is either in a balanced state or asymptotically approaches a balanced state.

In Appendix 4, we describe an analytical method for calculating mean ethical statuses. In particular, it is shown that the correlation between guilt

and suffering remains true for the limiting case, where the number of levels tends toward infinity.

In Appendix 5, the classification of various types of adequacy between an image and its original is given.

In Appendix 6, algorithms for constructing formulae according to subjects' answers to the questionnaires are described.

Appendix 7 contains tables of formulae and ethical statuses.

In Appendix 8, the substantiation of our model is generalized for the case of arbitrary environments.

In Appendix 9, a new method of representing individuals is introduced.

In Appendix 10, we demonstrate that some phenomena of quantum physics have their analogues in the Algebraic Model of Ethical Cognition. Fifty years ago, Niels Bohr formulated the principle of complementarity. The main idea of this principle is that some observations can never be made simultaneously. For example, one cannot see an electron as a particle and as a wave at the same time. Two different experimental situations are necessary, and they cannot be realized simultaneously.

Niels Bohr suggested that this principle reflects the general laws of human knowledge and can perhaps be extended beyond physical investigations into the area of human reflexion. However, until recently this suggestion has been only a metaphor.

Working with the algebraic model of ethical cognition, the author unexpectedly found that the relations of complementarity pointed out by Neils Bohr can be realized in this model. Specific quantum concepts such as amplitude of probability and the square of the module of an amplitude of probability have their analogues in the model of ethical cognition. For example, a complex amplitude in quantum mechanics has as an analogue a Boolean function in the ethical model. The classical experiment of the interference of light passing through two holes also has an analogue in the model of ethical cognition. It is known that the paradox of interference stems from the fact that when we register particles passing through the holes, interference disappears. The analogue of the holes is an individual's image of himself and of his partner; the analogue of the particles' passing through is impulses of 'guilt' and of 'condemnation'; the analogue of the statistical picture on the screen is an individual's suffering, which is the sum of 'guilt' and 'condemnation'. It turned out that if an individual does not have a correct registration of his images, his 'suffering' is related to 'guilt' and 'condemnation' by a formula similar to the formula of interference.

However, if an individual has at least one correct registration of his images, the picture of interference disappears, and the formula becomes a simple sum. Thus, we have an almost complete analogy. The fact that peculiarities of ethical cognition appear to be related to the peculiarities of the micro-world suggests to us the idea that the deep laws of the Universe are reflected in human ethics and that ethics cannot be considered as a conventional epiphenomenon.

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CHAPTER I

MORAL COGNITION

We associate moral cognition with categories like conscience, guilt feelings, condemnation, repentance, etc. We feel intuitively that moral cognition in some degree is opposed to 'pragmatic cognition'. Let us consider the latter first. We can assume that pragmatic cognition of an individual may consist of three elements: one registers a goal, another some means directed at reaching that goal; yet another is a 'calculating' device for finding the 'profit' of reaching the goal and evaluating possible losses associated with particular means. In the simplest cases these calculations can be made in monetary units; however, in general, more refined individual evaluations, may be used, called 'utilities'. A scheme of an individual with pragmatic cognition is as shown in Figure 1.1.

We express symbolically the relation between a goal and a means by an arrow with a switch. First, an individual calculates the utility of reaching his goal without using a certain means (the switch is off); then he calculates the utility using the means (the switch is on). He compares the results and makes the 'optimal' decision. Various types of human activity can be expressed by the above scheme, but not moral decision-making. *'Natural' human moral cognition does not follow this scheme.* Let us clarify the term 'natural'. Everyone would probably agree that a normal adult possesses a complex range of inner feelings related to his reflexion of himself, of other people, and of their relationships. Everyone has the ability to condemn himself if, in his opinion, he does something bad or, to condemn others, if in his opinion, they do something bad. Evaluations of what is good and what is evil are predetermined by the culture to which an individual belongs. The development of moral cognition recalls the process of mastering a native language by a child. L. Kohlberg (1969) distinguished stages of moral development similar to those stated by Jean Piaget for the genesis of intellectual functions. Thus, the process of internalizing cultural moral norms is quite definite, and the resulting scheme of moral cognition has a strict structure. In this sense our moral cognition is *natural*. It dominates over all other human essences and works automatically. A normal individual cannot leave the 'jurisdiction' of his moral cognition by 'volitional effort' just as he cannot force himself to stop understanding his native

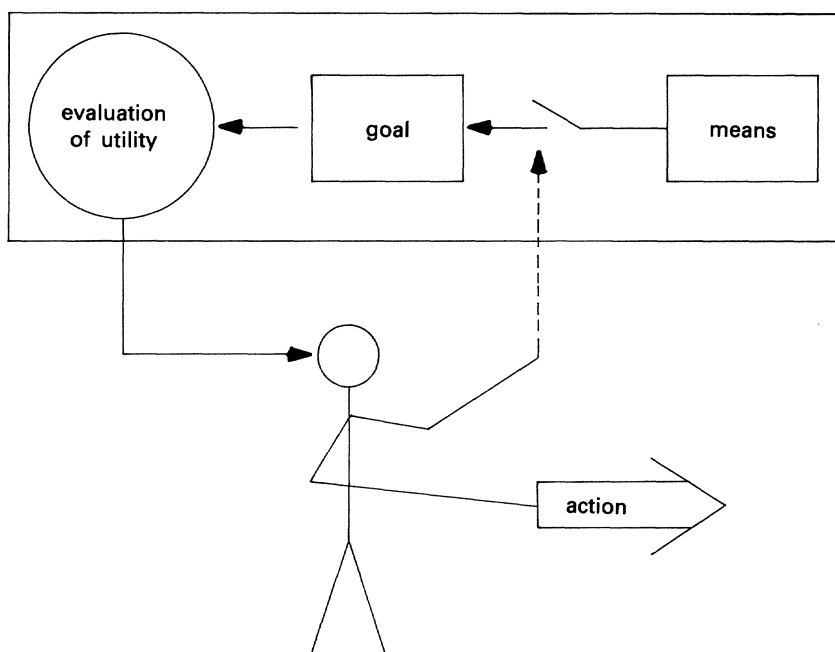


Fig. 1.1. Scheme of an individual with pragmatic cognition in a situation with goal and means.

language by 'volitional impulse'. To some degree, an individual is free in his actions, but not in his feelings about these actions. In this book we shall not touch upon the ontogenesis nor the phylogensis of ethical cognition. All our considerations are related to cognition which has already been developed.

In the initial construction of our model we suppose that the nucleus of ethical cognition of an individual is a binary structure 'good-evil'. Automatic, non-controlled by an individual, the functioning of this structure on 'higher tiers' of cognition generates psycho-moral phenomena of which we are aware as feeling guilty, condemnation, suffering, etc. The binary structure 'good-evil' can be in two states. The first state is polarization or confrontation and the second state is integration or compromise. These relations are fundamental and cannot be reduced to other, more simple relations. Moral cognition works by including various elements of a perceived situation into the evaluating mechanism 'good-evil'. A scheme of

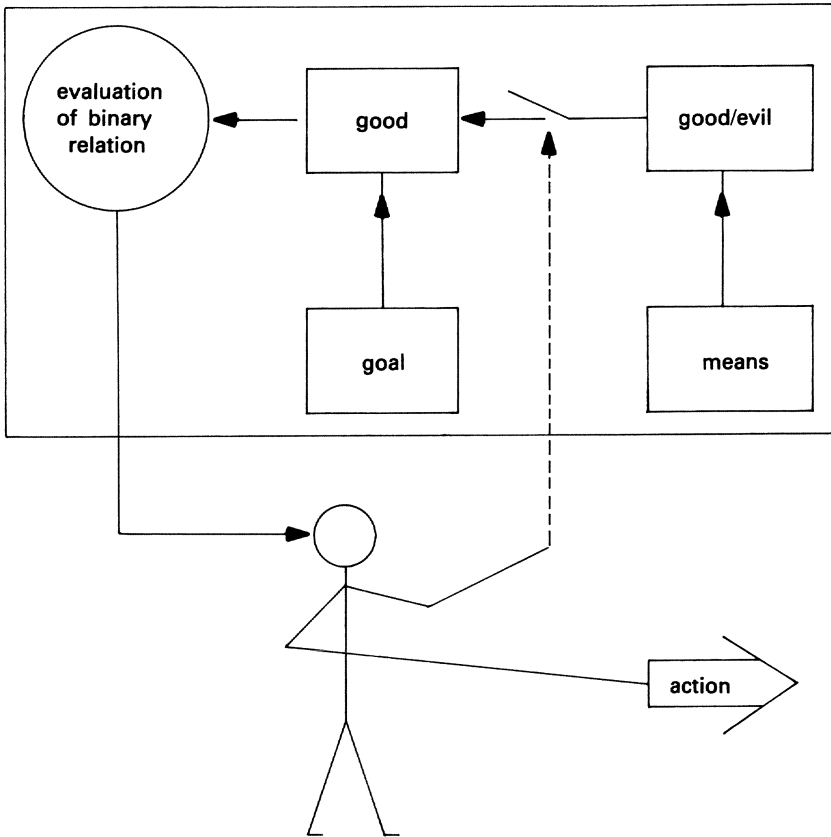


Fig. 1.2. Scheme of an individual with moral cognition in a situation with goal and means.

an individual with moral cognition in a situation with goal and means is shown in Figure 1.2.

The goal is replaced by 'abstract' good (if this is impossible, the goal is abandoned) and the means are replaced by good or evil. A conjunction of means and goal is transformed into the conjunction of the 'abstract categories' good and evil. Then a mechanism for evaluating the binary relations 'good-good' or 'good-evil' begins to operate without any connection to a concrete goal and concrete means. The goal may be 'the happiness of humanity' or the well-being of one person, and the means may be a 'child's tear' (Fedor Dostoevsky) or a murder. Good and evil in our

scheme do not have properties permitting us to measure their intensity and do not depend on the importance of a completed action. In Western culture a compromise of good and evil is evaluated as evil, and an individual who connects bad means with a good goal feels an impulse of 'moral suffering'.

A contemporary man consciously uses various scales of values. However, the binary scale 'good-evil' differs from all others. It is impossible to say that an individual uses this scale; it is an indescribable attribute of a human being who uses all other scales. Moral cognition is the oldest formation in human mentality; we may even suppose that the appearance of moral cognition based on the contraposition of good and evil, indicates the origin of a human being.

Consider now a situation where an individual has to choose his relationship with his partner. Suppose there can be either union or conflict. First, we consider an individual with 'pragmatic cognition' (Figure 1.3), which in this case may be called 'strategic'.

An individual calculates the utility for himself (or for society) if he chooses

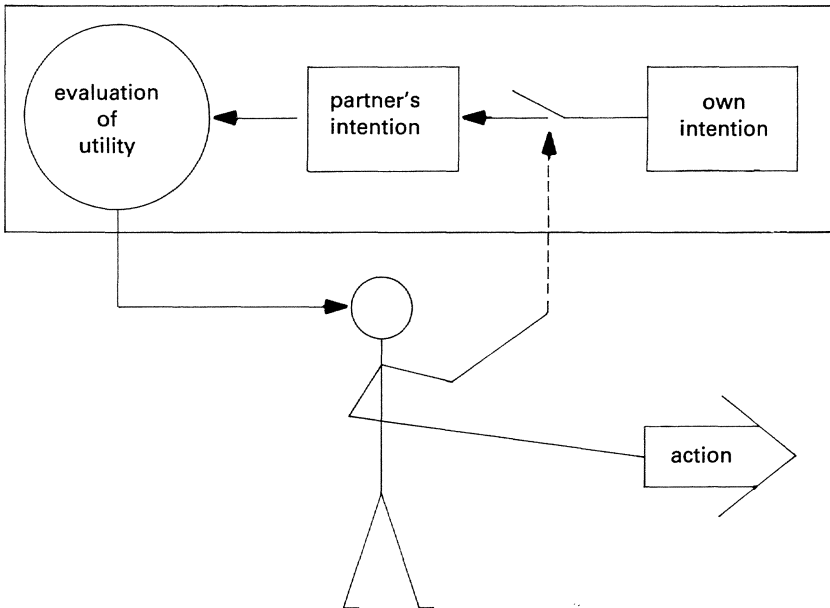


Fig. 1.3. Scheme of an individual with pragmatic cognition in a situation where he chooses his relationship with his partner.

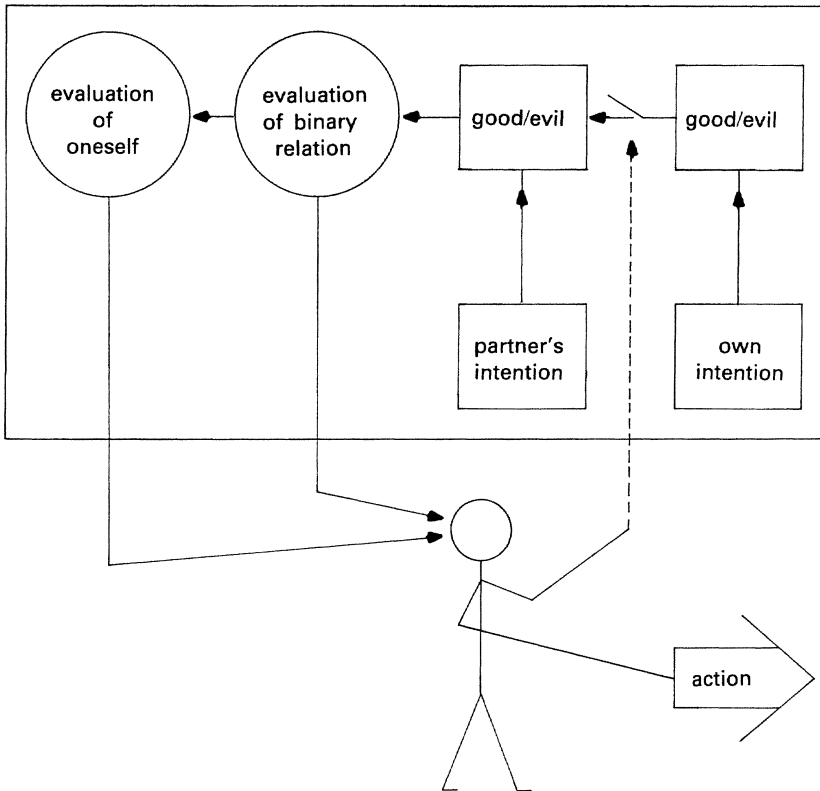


Fig. 1.4. Scheme of an individual with moral cognition in a situation where he chooses his relationship with his partner.

coalition with his partner (the switch is on), and then calculates the utility if he chooses confrontation with his partner (the switch is off). After comparing the results, an individual makes the optimal decision.

An individual with moral cognition in a situation of choice of relationship with his partner is expressed by the scheme in Figure 1.4. The individual substitutes 'good' or 'evil' for his own intention and his partner's intention; then he finds the relation between them and the mechanism of moral evaluation begins to operate. In addition to evaluating the binary relation of 'good-evil' (in the case of 'goal-means'), an individual evaluates his image of himself, as will be shown later. A negative evaluation of his own

image of himself generates an 'impulse of guilt', and a negative evaluation of the binary relation generates an 'impulse of suffering'. The most important peculiarities appear when we examine the statistical mean values of 'impulses of guilt' and 'impulses of suffering'. An individual can change these quantities by choosing different relationships with his partner. Thus, we can understand the psychological mechanism of inner motivation behind an individual's choice of relationship with his partner. The structure of moral cognition is, apparently, universal and does not depend on a given culture. However, there is one degree of freedom which allows the existence of two types of morality. This degree of freedom is connected with the possibility of evaluating the compromise between good and evil, either as evil or as good (and correspondingly the confrontation of good and evil can be evaluated either as good or as evil). As a result we obtain two *different ethical systems*, and we suggest the hypothesis that one ethical system is realized in American culture, and the other in Soviet culture.

The hypothesis about differences between Soviet and American individuals is based on personal observations by the author and his friends and colleagues. There are no serious works demonstrating the difference between normative types of these cultures. However, even superficial observation shows that in American culture a display of aggressiveness (with corresponding gestures and mimics) is a sign of bad manners and lack of restraint; in Soviet culture, this is evidence of 'having principles in the fight with life's defects' and is caused by objective conditions of upbringing. A smile or a formal benevolence toward an adversary is interpreted as a manifestation of weakness and cowardice.

The aggressiveness and compromiselessness of a Soviet man are not related to his personal peculiarities; they are normative features that are necessary for adequate functioning in Soviet society.

These standards, similar to those of language, do not belong to a single individual but rather are predetermined by the national culture of the country where the person was born and raised.

The differences in the evaluation of ethical compromise and ethical confrontation by Americans and Soviets can be shown by direct experiment. A special questionnaire was worked out and given to the two groups of subjects: recent emigrés from the Soviet Union (84 subjects ages from 19 to 66; 42 male and 42 female) and Americans (62 subjects ages from 17 to 67; 27 male and 35 female). The Soviet subjects were given this questionnaire in Russian. The questionnaire consists of four groups of two statements about situations where ends have to be correlated with means. The first of each

TABLE 1.1
Data from a study comparing Soviet and American attitudes toward ethical compromise and ethical confrontation.

Number of pair	Number of statement	Statements	Americans		Soviets	
			Percent in agreement	Two-tailed confidential interval for $p = 95\%$	Percent in agreement	Two-tailed confidential interval for $p = 95\%$
1	1	A doctor should conceal from a patient that he has cancer, in order to diminish his suffering.	8.0	± 6.8	89.0	± 6.7
	2	A doctor should not conceal from a patient that he has cancer, in order to diminish his suffering.	80.5	± 9.9	15.8	± 8.0
2	3	A malefactor can be punished more severely than the law requires, if this may serve as a deterrent for others.	11.5	± 8.0	84.5	± 7.8
	4	A malefactor cannot be punished more severely than the law requires, even if this may serve as a deterrent for others.	83.6	± 9.4	28.0	± 9.9
3	5	One may give false evidence in order to help an innocent person avoid jail.	19.9	± 8.4	65.0	± 10.0
	6	One must not give false evidence even in order to help an innocent person avoid jail.	82.25	± 9.6	42.5	± 10.7
4	7	One may send a cheat sheet during a competitive examination to a close friend.	8.0	± 6.8	62.0	± 10.1
	8	One must not send a cheat sheet during a competitive examination, even to a close friend.	90.3	± 7.4	37.5	± 10.3

group reflects compromise between 'good' and 'evil', the second confrontation between 'good' and 'evil'. The subject has to point out to what extent he agrees or disagrees with each statement. The results are shown in the Table 1.1.

We can see that Soviet and American subjects respond in dramatically different ways. Soviets evaluate positively statements reflecting compromise; Americans evaluate them negatively. On the other hand, Soviets evaluate negatively the statements reflecting confrontation, and Americans evaluate them positively.

We consider these differences to be the first argument in support of the hypothesis that different ethical systems are realized in the United States and in the Soviet Union. One might suppose that the reason for the above differences is that a Soviet individual in the described experiment uses not his moral cognition, but only his pragmatic orientation. However, analysis of the moral norms of Soviet culture (described in Chapter XVI), leads us to think that the choices of Soviet subjects are predetermined by their moral cognition.

CHAPTER II

ETHICAL SYSTEMS AND BOOLEAN ALGEBRA

Here are the conclusions drawn from the experiment: compromise between good and evil is evaluated positively in Soviet society and negatively in American society; confrontation between good and evil is evaluated negatively in Soviet society and positively in American society.

This difference may be taken as a point of departure for the author's model. In order to unify our concepts we say that, in Soviet society compromise between good and evil is rated as 'good' and confrontation as 'evil'; in American society, the evaluations are opposite.

Underlying the model is the idea of using classical propositional calculus for the representation of interaction of two abstract individuals. The category of 'good' corresponds to logical '1'; the category of 'evil' corresponds to logical '0'; and various relations between good and evil correspond to logical disjunction, conjunction, and implication.

We cannot give a formal definition of *good*, *evil*, *confrontation*, and *compromise*. Any attempt to define them by reducing them to other concepts only displaces the problem. For example, we can reduce 'good' to 'due' (which turns out to be as complicated as the category 'good'). The categories of 'confrontation' and 'compromise' may be reduced to the categories of 'conflict' and 'union' or 'incompatibility' and 'compatibility'. These categories are as complicated as the initial ones; however, such a reduction is helpful as a means of clarifying the connotation of the initial concepts.

In this work we shall construct abstract individuals who have inner models of their interactions with other individuals and who have the ability to evaluate these interactions. Our abstract individual has a recursive structure: models of individuals may also include models of their interactions with other individuals and of the ability to evaluate these interactions, and so on.

The core of the structure of an individual lies in the connection between the 'states' of the individual and his processes of awareness: *an individual in a state of evil and being aware of this state, transfers himself into a state of good*. We would like to consider this statement as a law and include it as an axiom in our theory. However, it is not a psychological law, but rather a

normative law of ethical evaluation. The 'state of evil' cannot be observed as a physical phenomenon. But we may observe that 'awareness of evil' by 'an evil person' is evaluated as good. Thus, when we say, 'This person has been aware that he was unfair', we change our initial evaluation of him from 'evil' to 'good' because of his awareness.

In all advanced cultures we see the phenomenon of an individual's preoccupation with concentrating attention on his own inner states and with the search for evidence of evil and imperfection in himself; oriental meditation, Christian repentance and modern group therapy serve this purpose.

Let us construct the basis of the two axiomatic ethical systems. These systems differ only in their first two axioms.

System I

- (1) Confrontation of good and evil is good.
- (2) Compromise of good and evil is evil.

System II

- (1) Confrontation of good and evil is evil.
- (2) Compromise of good and evil is good.

In both systems:

- (3) Confrontation of good and good as well as compromise of good and good is good.
- (4) Confrontation of evil and evil as well as compromise of evil and evil is evil.
- (5) Evil aware of evil becomes good.

The axioms above can be made to correspond in a natural way to the following Boolean equations.

I

- (1) $1 + 0 = 1$
- (2) $1 \cdot 0 = 0$

II

- (1) $1 \cdot 0 = 0$
- (2) $1 + 0 = 1$

- (3) $1 + 1 = 1$
 $1 \cdot 1 = 1$
- (4) $0 + 0 = 0$
 $0 \cdot 0 = 0$
- (5) $0 \rightarrow 0 = 1$

We see that in the first ethical system '+' is taken as the sign of confrontation, and '.' is the sign of compromise. In the second ethical system the opposite is true: '+' is the sign of compromise, and '.' is the sign of confrontation. Axioms (1) and (2) reflect the specific of interaction between good and evil. Axioms (3) (4) reflect the assumption that 'by itself' good does not turn into evil; neither does evil turn into good. Axiom (5) uses one of four possible equations of formal implication. We have to define three others;

- (6) $0 \rightarrow 1 = 1$ good aware of evil remains good;
- (7) $1 \rightarrow 1 = 1$ good aware of good remains good;
- (8) $1 \rightarrow 0 = 0$ evil aware of good remains evil.

Axioms (3) and (4), as well as (6), (7), and (8), cannot be verified by obvious empirical observations or experiments, but they do not contradict our intuition. The reason for choosing these particular axioms lies in the 'logic of consequences' and cannot be given before the corresponding analysis. The substantiation of these axioms is given in the Conclusion.

CHAPTER III

BOOLEAN ALGEBRA, EXPONENT, LOGARITHM

The traditional notation for the implication $a \rightarrow b$ is not convenient enough for our purposes. In this chapter, a new representation of implicational structures is described. While the current formulation and the usual one are equally adequate for doing Boolean algebra, the new formulation lends itself usefully to a semantic interpretation underlying the two structural models of this book.

We will call 'Boolean Algebra' an algebra with binary operation '+' and '.', unary operation 'bar above letter', elements 1, 0 and the following axioms:

- | | |
|---|---|
| <p>(1) $a + a = a$</p> <p>(3) $a + b = b + a$</p> <p>(5) $a + (b + c) = (a + b) + c$</p> <p>(7) $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$</p> <p>(9) $a + b = \overline{\overline{a} \cdot \overline{b}}$</p> <p>(11) $a + 1 = 1$</p> <p>(13) $\overline{1} = 0$</p> | <p>(2) $a \cdot a = a$</p> <p>(4) $a \cdot b = b \cdot a$</p> <p>(6) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$</p> <p>(8) $a + (b \cdot c) = (a + b) \cdot (a + c)$</p> <p>(10) $a + 0 = a$</p> <p>(12) $\overline{\overline{a}} = a$</p> |
|---|---|

In Boolean algebra, implication ($\rightarrow$) is defined in terms of disjunction, and negation as

$$b \rightarrow a \stackrel{\text{def.}}{=} \overline{b} + a.$$

We introduce, an alternative notation for implication as the exponential function:

$$b \rightarrow a \stackrel{\text{def.}}{=} a^b.$$

Thus, we postulate $1^1 = 1$; $1^0 = 1$; $0^0 = 1$; $0^1 = 0$.

The table form is given in Table 3.1.

The exponential representation of implication appeared in the context of

TABLE 3.1
Tabular representation of implication.

a	b	$b \rightarrow a$	a^b
1	0	1	1
0	1	0	0
1	1	1	1
0	0	1	1

the theory of functors and categories (McLane, 1971). However, such a representation may be efficient in a traditional logical context as well, for example, in dealing with long chains of implications (Lefebvre, 1977b).

Our motivation for introducing the new notation for implication is that the 'usual' properties of the exponential functions are easily verified for this function:

$$\begin{array}{ll}
 (1) & a^b \cdot a^c = a^{b+c} \\
 (2) & (a^b)^c = a^{b \cdot c} \\
 (3) & (a \cdot b)^c = a^c \cdot b^c \\
 (4) & a^0 = 1 \\
 (5) & a^1 = a \\
 (6) & 1^a = 1.
 \end{array}$$

Several additional statements are true (which do not have analogues in 'usual' algebra):

$$\begin{array}{ll}
 (7) & (a + b)^c = a^c + b^c \\
 (8) & (a + b)^c = a^c + b \\
 (9) & a^c + b = a + b^c \\
 (10) & 0^a = \bar{a}
 \end{array}$$

For Boolean algebra with two elements, 1 and 0, the following equations are also true:

$$a + \bar{a} = 1, \quad a \cdot \bar{a} = 0$$

as well as a few more:

$$\begin{array}{ll}
 (11) & a^a = 1 \\
 (12) & a^a \cdot b = 1 \\
 (13) & a^b + b^a = 1 \\
 (14) & a^{a+b} + b^{a+b} = 1 \\
 (15) & (a + b)^a = 1 \\
 (16) & a^{\bar{a}} = a
 \end{array}$$

The Equation (10) may be rewritten as $0^{\bar{a}} = a$. This allows us to include

the concept of logarithm into Boolean algebra (Lefebvre, 1977b). We define logarithm to the base 0 as

$$\log_0 a \stackrel{\text{def.}}{=} \bar{a}.$$

Thus, taking the logarithm corresponds to the unary operation of negation.

The next formulae follow from the definition of a logarithm:

- (1) $\log_0 (a \cdot b) = \log_0 a + \log_0 b$
- (2) $\log_0 b^a = a \cdot \log_0 b$
- (3) $\log_0 (\log_0 a) = a.$

In the traditional notation they correspond to the formulae:

- (1') $\overline{a \cdot b} = \bar{a} + \bar{b}$
- (2') $\overline{a \rightarrow b} = a\bar{b}$
- (3') $\bar{\bar{a}} = a.$

Any chains of implication could be written down in exponential form.

Examples:

- (a) $[a \rightarrow (b \rightarrow c)] = (b \rightarrow c)^a = (c^b)^a = c^{b \cdot a}$
- (b) $[(a \rightarrow b) \rightarrow c] = c^{a \rightarrow b} = c^{b^a}$
- (c) $[(a \rightarrow b) \rightarrow (c \rightarrow d)] = [b^a \rightarrow d^c] = (d^c)^{b^a} = d^{c \cdot b^a}$

We will write L for 'log₀' henceforth. The three preceding equations may be easily reduced to canonical form by double operations of logarithms:

- (a') $L(Lc^{b \cdot a}) = L(b \cdot a \cdot Lc) = Lb + La + c = \bar{b} + \bar{a} + c$
- (b') $L(Lc^{b^a}) = L(b^a \cdot Lc) = a \cdot Lb + c = a \cdot \bar{b} + c$
- (c') $L(Ld^{c \cdot b^a}) = L(c \cdot b^a \cdot Ld) = Lc + a \cdot Lb + d = \bar{c} + a \cdot \bar{b} + d.$

These representations of complex implicational structures simplify manipulations.

Let us next describe the concept of a propositional formula.

- (1) Latin letters a, b, c are called elementary formulae.
- (2) If A and B are formulae, then expressions $A + B$; $A \cdot B$; A^B ; $\bar{A}$ are also formulae.
- (3) The constants 1 and 0 are formulae.
- (4) If A is a formula, one can substitute any formula B and write $A = B$ if
 - (i) this expression is a formula or an axiom of Boolean algebra 1–12.
 - (ii) there is a finite chain of such substitutions transferring A into B , and each substitute formula agrees with (i)

CHAPTER IV

INDIVIDUALS, REFLECTION, AND INTERACTION

An individual will be designated by a particular propositional formula; however, logically equivalent formulae may not represent the same person's individual structure because the semantic interpretation may well be different. Each formula has two aspects: on the one hand it is a logical function, on the other hand it is a representation of an individual with his 'inner world' or of a situation of interaction between two individuals.

Let us introduce our *language* and *meta-language*. Elementary units of the language are symbols

$$a, \bar{a}, b, \bar{b}, +, \cdot.$$

Elementary units of the meta-language are symbols

$$\mathbf{a}, \mathbf{b}, A, B, W$$

as well as the same symbols with subscripts or superscripts. The semantics of the meta-language is defined as follows:

- ① $\mathbf{a}$ is a designation for both a and $\bar{a}$,
 $\mathbf{b}$ is a designation for both b and $\bar{b}$.
- ② A is a designation for both $\mathbf{a}$ and $\mathbf{a}^W$,
 B is a designation for both $\mathbf{b}$ and $\mathbf{b}^W$.
- ③ W is a designation for A , and for B , and for $A + B$, and for $A \cdot B$.

The formulae which can be designated as A or B will be called *individuals*.

The formulae which can be designated as $A + B$ or $A \cdot B$ will be called *situations*.

With the help of a chain of substitutions we can determine whether a given formulae is an individual or a situation or neither. For example, let us find what formula a^{a+b} is:

- (1) apply ① to formula a^{a+b} and obtain $\mathbf{a}^{a+b}$;
- (2) apply ② to formula $\mathbf{a}^{a+b}$ and obtain $\mathbf{a}^{A+B}$;
- (3) apply ③ to formula $\mathbf{a}^{A+B}$ and obtain $\mathbf{a}^W$;
- (4) apply ② to formula $\mathbf{a}^W$ and obtain A .

Thus the given formula is an individual designated by A . In this book only those formulae will be regarded which are either individuals or situations. In most cases the procedure for finding out what is a formula is trivial, and we will omit it in our treatment.

In the first ethical system the situation $A + B$ will be interpreted as a conflict or confrontation between individuals A and B , and the situation $A \cdot B$ will be interpreted as a union or compromise between individuals A and B . In the second ethical system the situation is reversed, i.e. $A + B$ will be interpreted as a union or compromise between individuals A and B , and the situation $A \cdot B$ will be interpreted as a conflict or confrontation between individuals A and B . A formula W in the formula a^W will be called the inner world of individual A . Sometimes we will say that individual A contains an inner world W , or that individual A *sees* W , *feels* W , *knows* W ; sometimes we will say that W is an *image of reality* which belongs to individual A , or reality from A 's point of view.

The symbol a in the formula $A = a^W$ will be called the *root of individual* A ; thus, individual A 'consists' of two parts: a root and an inner world. We will call the transformations of $A + B$ into $B + A$ and of $A \cdot B$ into $B \cdot A$ as well as the reverse of these *commutative transformations*. We will call two formulae *semantically equivalent* if they can be transformed one into the other only by commutative transformations of their parts. We will designate semantic equivalence by ' $\equiv$ ' and reserve the sign ' $=$ ' for Boolean or logical equivalence. Commutative transformation preserves logical equality, thus the semantic equivalence of two formulae implies their logical equivalence (but not conversely).

A_i, B_i , and W_i ($i = 1, 2, \dots$) will designate, respectively, concrete examples of formulas A, B , and W .

Let us look at the formula $A_1 \equiv a^{A_2 * B_2}$, where $*$ may be either ' $+$ ' or ' $\cdot$ '. We say that A_2 is A_1 's image of himself, and B_2 is A_1 's image of his partner, and $*$ is A_1 's image of his relationship with his partner.

Let us look at the formula $A_1 \equiv a^{A_2 * b^{W_2}}$. We have $B_2 \equiv b^{W_2}$. We say that W_2 is the inner world of A_1 's image of his partner, or W_2 is a partner's inner world from A_1 's point of view.

We say that an individual *has a correct* image of himself if the root of his image of himself is equal to the root of the individual, and we say that an individual *has an incorrect* image of himself if the root of his image of himself is not equal to the root of the individual. Thus, individual $a^{A \cdot B}$ has a correct image of himself if A is equal to a or to a^W , and this individual has an incorrect image of himself if A is equal to $\bar{a}$ or to $\bar{a}^W$.

Consider the situation.

$$a^{A_2 * B_2} * b^{W_1}$$

Let $A_1 \equiv a^{A_2 * B_2}$ and $B_1 \equiv b^{W_1}$. We say that individual A_1 has a correct image of individual B_1 if the root of A_1 's image of his partner is equal to the root of individual B_1 . Thus, A_1 has a correct image of B_1 if B_2 is b or b^{W_2} , and A_1 has an incorrect image of B_1 if B_2 is $\bar{b}$ or $\bar{b}^{W_2}$. If an individual is given without the context of his interaction with a partner, we cannot say anything about the correctness or incorrectness of his image of his partner.

Consider the situation

$$a^{A_2 * {}_2B_2} * {}_1b^{B_3 * {}_3A_3}$$

The signs $*_1, *_2, *_3$ have meanings '+' or '-' independently of each other. The meaning of sign $*_1$ we call *relationships* between partners. We say that individual A has a correct image of his relationships with his partner if $*_2$ has the same meaning as $*_1$. If $*_2$ differs from $*_1$, we say that A has an incorrect image of his relationships with his partner. And we say that individual B has a correct image of his relationships with his partner if $*_3$ has the same meaning as $*_1$. If $*_3$ differs from $*_1$, we say that individual B has an incorrect image of his relationships with his partner.

Let us interpret the following examples in more detail. (We assume that the individuals belong to the first ethical system.)

Examples

- | | |
|-------------------|--|
| (1) a^a | A has a correct image of himself. |
| (2) $a^{\bar{a}}$ | A has an incorrect image of himself. |

Next consider some other representations of an individual A now involving an interaction with another person B .

- | | |
|-------------------------|---|
| (3) a^{a+b} | A has an image of conflict between himself and B . |
| (4) $a^{a \cdot b}$ | A has an image of union between himself and B . |
| (5) a^{a+b^a} | A has an image of conflict between himself and B , who in A 's opinion has a correct image of A . |
| (6) $a^{a+b^{\bar{a}}}$ | A has an image of conflict between himself and B , who in A 's opinion has an incorrect image of A . A sees himself as a ; A thinks B sees him as $\bar{a}$; therefore A thinks that B is wrong. |

Now we present some joint possibilities representing both A and B .

- | | |
|--|--|
| (7) $a^b + b^a$ | A and B are in conflict. A has a correct image of B ; B has a correct image of A . |
| (8) $a^{\bar{b}} + b^{\bar{a}}$ | A and B are in conflict. Both of them have incorrect images of each other. |
| (9) $a^{a+b} + b^{b+a}$ | A and B are in conflict. Each of them has correct images of A and B and of their relationships. |
| (10) $a^{a+b} + b^{\bar{b}+\bar{a}}$ | A and B are in conflict. Each of them reflects this fact correctly. A has correct images of himself and of B . B has incorrect images of both himself and A . |
| (11) $a^{a+b} + b^{b \cdot a}$ | A and B are in conflict. A has a correct image of conflict, but B believes they are in union. |
| (12) $a^{\bar{a}+\bar{b}^a} + b^{\bar{b}+\bar{a}^b}$ | A and B are in conflict. Both of them reflect this fact correctly, and both have incorrect images of each other and of themselves. A sees his partner seeing a , but A himself sees $\bar{a}$, so he believes his partner is wrong. B sees his partner seeing $\bar{b}$, and B himself sees $\bar{b}$, so he thinks his partner has a correct image of him. |

A general rule for constructing judgments about the correctness of images and judgments is given in Appendix 1.

Let us introduce the concept of absolute adequacy of an image. In the formula

$$a^{B_2} * B_1$$

individual A has an absolutely adequate image of his partner if $B_2 \equiv B_1$. It is worth emphasizing the distinction between the concepts of absolute adequacy and correctness. Correctness here means the correct reflection of a root; adequacy is attained when the entire structure of an image is identical to the original. Thus, an absolutely adequate image is always correct, but the converse is not true. These problems are discussed in greater detail in Appendix 5.

CHAPTER V

AUTOMATA WITH SEMANTICS AND ETHICAL STATUS

Let us imagine an individual immersed in a social environment. This environment forces the individual to perform certain actions. In our schematic reasoning we will suppose that each action of an individual is normatively evaluated as either *good* or *bad*. An individual has either to perform or refuse to perform an action proposed by the environment.

We will now show that the formal individual introduced in the previous chapter may be considered as an automaton processing *the environment's demands* as his 'inputs'. Let us posit an abstract automaton with two inputs and one output (an example is given in Figure 5.1):

The input and output alphabets have two symbols: 1 and 0. The transformation is a Boolean function, which we will write in the exponential form representing an individual. Suppose each symbol (1 and 0) is fed with the probability $\frac{1}{2}$ to the first and second inputs independently. Depending on the two symbols at the inputs the transformer gives the corresponding value of its Boolean function as the output. We shall interpret 1 at the inputs as a demand of the environment to perform a good action, and 0 as a demand of the environment to perform a bad action. We shall interpret 1 at the output as a *consent* to perform a good action if there was 1 at the first input, and as a *refusal* to perform a bad action if there was 0 at the first input. We will interpret 0 at the output as a *consent* to perform a bad action if there was 0 at the first input, and as a *refusal* to perform a good action if there was 1 at the first input. (This last case is not analyzed in the given work; we have introduced it for formal completeness.)

The first input receives the demands of the environment for a given individual; the second input receives the demands of the environment which, *from the individual's point of view*, are given to his partner.

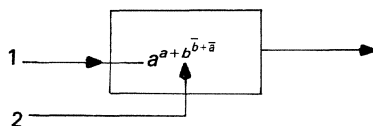


Fig. 5.1. An example of an automaton with semantics.

We shall assume that the root of an individual always accepts the value which is sent by the environment. Let us consider individual $\bar{a}^w$. The root of this individual is $\bar{a}$ (and not a !); therefore, the values of $\bar{a}$ are the environmental demands. In our model an investigator does not see the demand before it is accepted by a system. Thus, it will be more precise to say that we assume the particular demand was sent which was accepted by the system. The environmental demand accepted by the individual is a special intention of the individual to perform good or evil. So, when the root has the value 1, we consider that the individual has *the intention* of performing good; when the root is 0, we consider that the individual has *the intention* of performing evil.

When the root of an individual's image of a given person accepts the value 1, *the individual assumes* that the given person has the *intention* of performing good; when the root is 0, *the individual assumes* that the given person has the *intention* of perpetrating evil. If the individual has a correct image of himself, he models correctly his own intentions; if the individual has an incorrect image of himself, he models his own intentions incorrectly. Depending on whether the individual's image of his partner is correct or incorrect, he models correctly or incorrectly his partner's intention. In our model, an individual does not see the demand of the environment. He sees (correctly or incorrectly) only his intention and his partner's intention. Let us note that individuals always agree with good environmental demands, since $1^w = 1$.

Now we can introduce an evaluation of the automaton's work: it will be represented by the frequency of the appearance of 1 at the output. We will call this parameter *the ethical status of the individual*. It indicates the degree to which an individual resists the temptation proffered by the environment.

We will show that there are individuals who have only 1's at their outputs: they refuse all temptations. But there are also individuals who agree to any directions of the environment. The former have an ethical status equal to 1; the latter, to $\frac{1}{2}$. The ethical status of an individual is thus always located in the closed interval $[\frac{1}{2}, 1]$. Therefore, an individual may be



Fig. 5.2. Two automata corresponding to the simplest individuals; they have images of themselves but do not have images of their partners.

considered as a machine, improving the characteristics of some ‘stream of goodness’.

The ethical status of an individual A will be expressed as $|A|$. Sometimes we will use the concept ‘module of a formula’.

Let us consider the several examples.

The automaton 1.1 (shown in Figure 5.2) is an ‘absolute transformer’. Its output always gives 1. (Since $a^a = a + \bar{a} = 1$, so $|a^a| = 1$.) The formula a^a represents an individual who has a correct image of himself.

The automaton 1.2, unlike 1.1, does not improve the flow of symbols. It transfers the value from input to output. (Since $a^{\bar{a}} = a$, so $|a^{\bar{a}}| = \frac{1}{2}$.) The formula $a^{\bar{a}}$ represents an individual who has an incorrect image of himself. Thus, an individual with a correct image of himself has a higher ethical status.

In Figure 5.3, we will interpret the formulae in the first ethical system. The automaton 2.1 represents an individual who sees himself in conflict with his partner b and has a correct image of himself. The automaton 2.2 represents an individual who also sees himself in conflict with his partner b , but has an incorrect image of himself. Whose ethical status is higher? Consider Table 5.1.

In three cases out of four the individual a^{a+b} makes a ‘good’ response; but the individual $a^{\bar{a}+b}$ make a ‘good’ response only in half the cases. Thus, the first individual is better than the second one.

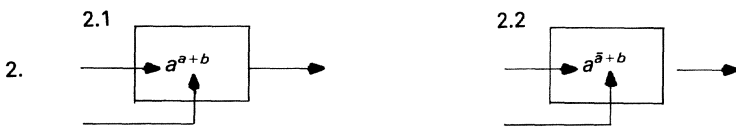


Fig. 5.3. The automata corresponding to individuals who have images of themselves and of their partner being in relationships ‘+’.

TABLE 5.1
An example of individuals’ reactions to various combinations of environmental demands.

a	b	a^{a+b}	$a^{\bar{a}+b}$
1	1	1	1
1	0	1	1
0	1	0	0
0	0	1	0

Let us now compare two individuals who see themselves in union with each other (Figure 5.4).

The individual 3.1 has a correct image of himself, and the individual 3.2 has an incorrect image of himself. Consider Table 5.2. We see that $|a^{a \cdot b}| = 1$, and $|a^{\bar{a} \cdot b}| = \frac{3}{4}$.

These results are shown in Table 5.3. We see that the ethical status of the individuals with simplest structures is higher if they have a correct image of themselves, compared with individuals differing only in that they have an incorrect image of themselves.

The above examination demonstrates the main principle of our in-

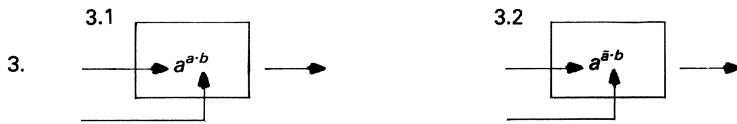


Fig. 5.4. The automata corresponding to individuals who have images of themselves and of their partner being in relationships '•'.

TABLE 5.2

Another example of individuals' reactions to various combinations of environmental demands.

a	b	$a^{a \cdot b}$	$a^{\bar{a} \cdot b}$
1	1	1	1
1	0	1	1
0	1	1	0
0	0	1	1

TABLE 5.3

A comparison of modules of two-tier individuals with correct *vs.* incorrect images of themselves.

Formula	Module	Formula	Module
a^a	1	$a^{\bar{a}}$	$\frac{1}{2}$
a^{a+b}	$\frac{3}{4}$	$a^{\bar{a}+b}$	$\frac{1}{2}$
$a^{a \cdot b}$	1	$a^{\bar{a} \cdot b}$	$\frac{3}{4}$

vestigation. We give a psychological interpretation of a formula, and examine its ethical status independently. By changing the structure of the formula and observing the change of status, we can establish the connection between the ethical status of an individual and those characteristics of the structure which may have a psychological interpretation. In this way we succeed in deriving the psychological peculiarities of 'heroes' and 'weak members' in each ethical system.

Not only an individual, but also a situation may be represented as an automaton. Let us consider the situation described in Example 10 of Chapter IV:

$$a^{a+b} + b^{\bar{b}+\bar{a}}.$$

The corresponding automaton appears in Figure 5.5.

This automaton consists of two interdependent automata. It has two inputs and three outputs. Inputs are the demands of the environment to the automata *A* and *B*. The two outputs of the automaton are the outputs of the automata *A* and *B*. The third output is the output of the situation, which is its evaluation calculated under the condition that the outputs of *A* and *B* are known. The evaluation is made in accordance with the evaluation rules governing the interaction between good and evil in each ethical system. The frequency of appearance of 1 in the output of the situation we will call the ethical status of the situation. The statuses of individuals and of the

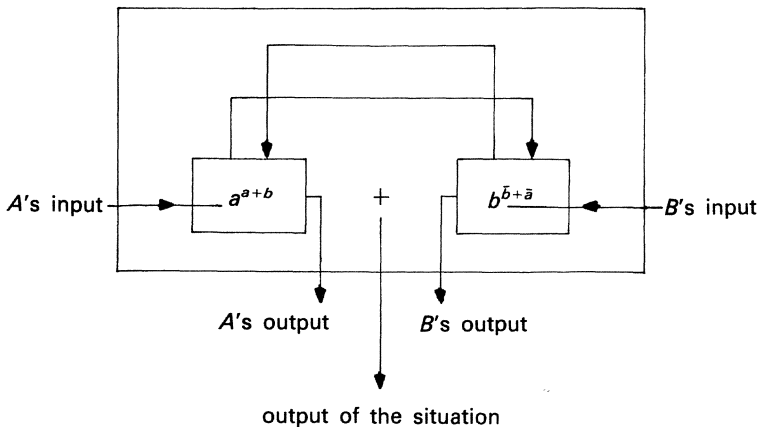
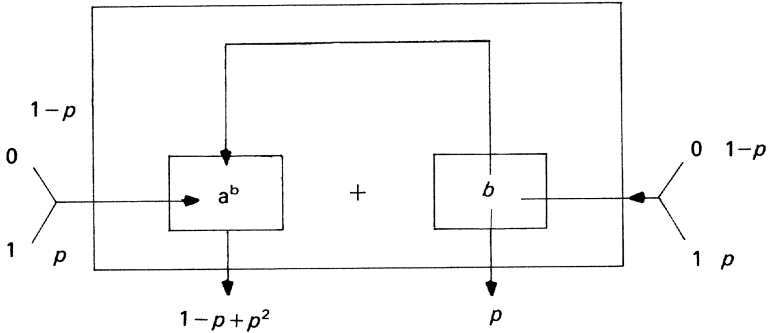


Fig. 5.5. A Structure of a situation represented as an automaton.

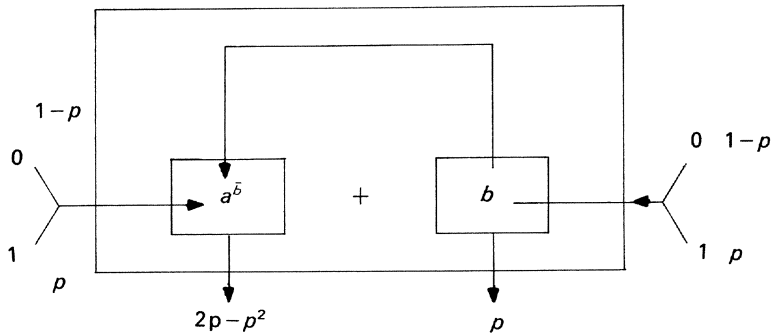
TABLE 5.4

An example of connection between individuals' reactions to certain environmental demands and corresponding assessment of situation.

a	b	A	B	$A + B$
1	1	1	1	1
0	1	0	1	1
1	0	1	0	1
0	0	1	0	1



(a)



(b)

Fig. 5.6. (a) Individual $A \equiv a^b$ has a correct image of his partner; (b) individual $A \equiv a^{\bar{b}}$ has an incorrect image of his partner.

situation is calculated from Table 5.4. Thus, $|A| = \frac{3}{4}$; $|B| = \frac{1}{2}$; $|A + B| = 1$.

Now we will briefly analyze the work of automata in environments with an unequal content of *good* and *evil*. Let us consider the automata given in Figure 5.6.

Let both work in the environment which offers 1 with the probability p and 0 with the probability $1 - p$. It is easy to see that the automaton a^b (Figure 5.6a) will generate 1 if it has 1 at the first input or zeroes at both inputs. Thus, the probability of obtaining 1 at the output is

$$\gamma_1 = 1 - p + p^2.$$

The automaton $a^{\bar{b}}$ (Figure 5.6b) will generate 1 if it has 1 at the first input, or if it has 0 at the first input and 1 at the second. Thus, the probability of obtaining 1 at the output is

$$\gamma_2 = p + p(1 - p) = 2p - p^2.$$

We can see that

- (1) $\gamma_1 = \gamma_2$ if $p = \frac{1}{2}$; or $p = 1$;
- (2) $\gamma_2 > \gamma_1$ if $p < \frac{1}{2}$;
- (3) $\gamma_1 < \gamma_2$ if $\frac{1}{2} < p < 1$.

It is possible to draw an important conclusion: The automaton a^b excels the automaton $a^{\bar{b}}$ in 'bad environments', where frequency of the appearance of 1 is less than $\frac{1}{2}$, and the automaton $a^{\bar{b}}$ excels the automaton a^b in 'good environments' where frequency of the appearance of 1 is higher than $\frac{1}{2}$. Therefore, the quality of the automaton's work may depend on the environment.

We can also easily obtain the mean value of the work of such automata in the entire range of various environments:

$$\gamma_1 = \int_0^1 \gamma_1 dp = \frac{5}{6},$$

$$\gamma_2 = \int_0^1 \gamma_2 dp = \frac{4}{6}.$$

Therefore, assuming that p is uniformly distributed in the interval $[0, 1]$, the automaton a^b works better.

Now let us recall the experiment mentioned at the outset. Two groups of subjects were invited to agree or disagree with a set of statements. Let us liken the subjects to automata and the statements to demands of the environment. The questionnaire was made in such a way that its odd statements were good demands in the second ethical system and bad ones in the first system, and its even statements were good demands in the first ethical system and bad ones in the second system.

Knowing the distribution of one's agreements and disagreements we can find out the ethical system an individual belongs to, *without any knowledge of his inner structure*. We used this type of reasoning when we discovered the significant differences between choices made by Americans and former Soviet citizens.

However, the theory we develop here suggests that the inner structures of individuals belonging to different ethical systems are also significantly different. The following chapters are devoted to the study of these structures.

CHAPTER VI

A FORMAL REPRESENTATION OF DOUBTS AND FEELINGS

Consider an individual

$$A_1 \equiv a^{a^{\bar{b}} + b}. \quad (6.1)$$

A_1 sees his partner $B_2 \equiv b$; in addition, A_1 sees himself seeing $B_3 \equiv \bar{b}$. Thus, a partner exists in A_1 's inner world in two roles: as 'reality' (B_2) and as an image or a model of this reality (B_3). A_1 can see that his image B_3 differs from the 'reality' B_2 . This is a structure where an individual can say, 'I am mistaken'. Of course, the structure itself does not show modality; we cannot fix the formal differences between situations where an individual 'is sure' that he is mistaken and 'supposes' that he is mistaken.

In this book we will not use a categorical interpretation such as '*being sure*'; we will consider that an individual *supposes* or *assumes* that his image is wrong. Using this interpretation we can say that an individual *doubts* the correctness of his image, and formula (6.1) will mean that individual A has

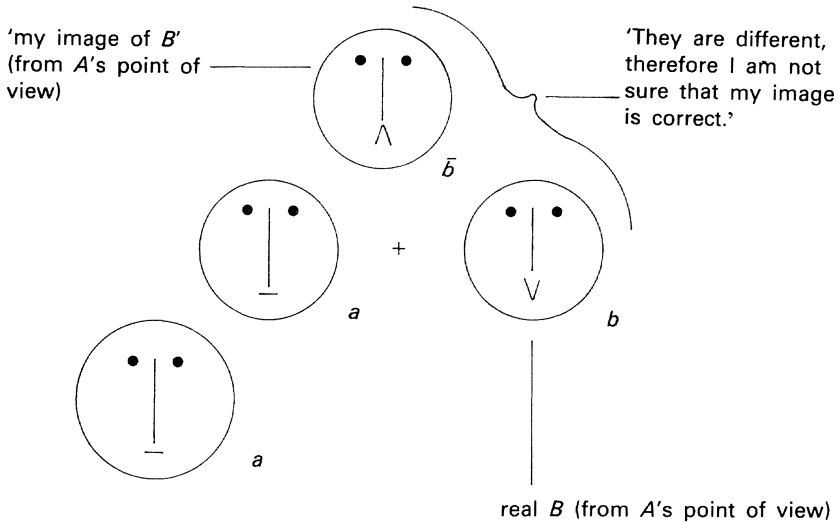


Fig. 6.1. Structure of an individual who doubts the correctness of his image of his partner.

doubts about the correctness of his image of B , i.e., about the correctness of $B_3 \equiv \bar{b}$.

Similarly let us consider an individual

$$a^{a^{\bar{a}}}. \quad (6.2)$$

Reasoning formally, we can say that A has a correct image of himself, but he sees himself as having an incorrect image of himself. Individual A sees that his 'image of A ' differs from the 'real A ', so A doubts the correctness of his image of himself.

Let us analyze an individual

$$a^{a^b + b}. \quad (6.3)$$

From A 's point of view the real B and A 's image of B coincide with each other. We will interpret this fact as an absence of doubts about the correctness of one's image of one's partner. Similarly an individual

$$a^{a^a} \quad (6.4)$$

does not doubt the correctness of his image of himself.

It is worth distinguishing the interpretation of structures (6.3) and (6.4) from that of structures

$$a^{a+b} \quad \text{and} \quad a^a.$$

In the latter structures an individual also does not doubt. But doubts cannot appear, because the individuals do not have elements of the third tier, the correctness of which they may doubt. This being the case, let us agree that when we speak of the absence of doubts, we assume that in the structure there are elements the correctness of which an individual does not doubt.

Which individual has a higher ethical status: a doubting or non-doubting one? A detailed analysis of this problem is given in the next chapter, but now we will only indicate that for the simplest (and most important) structures (6.2) and (6.4) the following relation is true:

$$|a^{a^{\bar{a}}}| > |a^{a^a}|$$

since

$$|a^{a\bar{a}}| = 1 \quad \text{and} \quad |a^{aa}| = \frac{1}{2}.$$

Now, let us analyze 'doubts in relationships': An individual

$$a^{a^a + b \cdot b} \tag{6.5}$$

sees the 'real' relationships between himself and his partner as $a \cdot b$; however, he sees himself seeing $a + b$. He can draw the conclusion that his image of the relationship with his partner is wrong (of course, in various modalities). Similarly to the previous analysis, we will say that A doubts the correctness of his image of his relationship with his partner, i.e., A doubts the correctness of $a + b$.

An individual

$$a^{a^a \cdot b + b} \tag{6.6}$$

will also be interpreted as doubting the correctness of his image of his relationships with his partner; in this case A doubts the correctness of $a \cdot b$.

Now let us introduce the concepts 'feeling guilty', 'suffering', and 'condemnation'. We will consider an individual

$$a^{A*B},$$

where A is an image of himself, B is an image of his partner, $*$ is an image of their relationship, and $A*B$ is an image of the situation. Let us introduce the concept *the state of an individual*. Let us fix the values of an automaton's inputs a and b and put them in the appropriate places of a formula. The resulting appearance of a formula (when each variable has its own particular value) we will call *the state of an individual*. For a particular state of an individual we can calculate the values of his substructures. We will say that an individual *feels guilty* when $A = 0$, *suffers* when $A*B = 0$, and *condemns his partner* when $B = 0$.

Finding evil in oneself is intuitively connected with the categories of 'feeling guilty' and 'repentance'. Feeling guilty is a kind of self-condemnation. This understanding of the feeling of guilt may be supported by numerous examples: any discussion of the problem of one's own guilt always contains an authentic or erroneous registration of evil in oneself in the past or in the present. Thus, in our context, *feeling guilty* is 'seeing evil' in oneself.

When $A * B = 0$, an individual sees a 'bad situation'; when $B = 0$, he sees a 'bad partner'. We have tried to distinguish feelings of the three types: one suffers because he sees *himself being bad*, another suffers because he sees *the situation being bad*, and yet a third suffers because he sees *his partner being bad*. The first feeling we call *feeling guilty*, the second one – *suffering*, and the third – *condemnation*.

Each pair of environmental demands corresponds to one state of an individual. Two currents of the demands correspond to a sequence of states. We can pose several questions: *how often A has value 0? How often A*B has value 0? How often B has value 0?* These frequencies are regarded as the *measure of feeling guilty*, the *measure of suffering*, and the *measure of condemnation*. It is obvious that the measure of feeling guilty is

$$u = 1 - |A|,$$

where $|A|$ is the module of one's own image: the measure of suffering is

$$v = 1 - |A * B|,$$

where $|A * B|$ is the module of one's image of the situation, or that of the 'inner world', and the measure of condemnation is

$$w = 1 - |B|,$$

TABLE 6.1
Connection between doubt and feeling of guilt.

Individual without doubts	Image of oneself	Feeling guilty	Individual with doubts	Image of oneself	Feeling guilty	Doubt about whose image
$a^{aa + b + b}$	$a^a + b$	$\frac{1}{4}$	$a^{a\bar{a} + b + b}$	$a^{\bar{a} + b}$	$\frac{1}{2}$	Oneself
			$a^{a\bar{a} + \bar{b} + b}$	$a^{\bar{a} + \bar{b}}$	$\frac{1}{2}$	Oneself and partner
			$a^{aa + \bar{b} + b}$	$a^a + \bar{b}$	$\frac{1}{4}$	Partner
$a^{a \cdot b \cdot b}$	$a^a \cdot b$	0	$a^{a\bar{a} \cdot b \cdot b}$	$a^{\bar{a} \cdot b}$	$\frac{1}{4}$	Oneself
			$a^{a\bar{a} \cdot \bar{b} \cdot b}$	$a^{\bar{a} \cdot \bar{b}}$	$\frac{1}{4}$	Oneself and partner
			$a^{aa \cdot \bar{b} \cdot b}$	$a^a \cdot \bar{b}$	0	Partner

where $|B|$ is the module of one's image of one's adversary. Therefore, the higher the module of one's image of oneself, the lower the feeling of guilt; the better one's image of the situation, the less the suffering; and the higher one's image of one's adversary, the less the condemnation.

Let us now see how *doubts* of the correctness of one's images of oneself are connected with *feeling guilty*. Images of oneself have two tiers, since the entire structures of individuals have three tiers (for the majority of real situations the interpretation using three tiers is sufficient). There are eight different images of oneself that at the same time contain both images of oneself and of one's partner. They are represented in Table 6.1.

It is easy to see that doubts about the correctness of one's image of oneself, compared with the same structure without doubts, always lead to an increase in feeling guilty.

Doubts only about the correctness of one's image of one's partner do not increase feeling guilty.

These results, in general, correspond to the centuries-old observation of mankind about the connection between feeling guilty and hesitations.

CHAPTER VII

A FORMAL COMPARISON OF ETHICAL SYSTEMS: FEELING GUILTY, CONDEMNATION, DOUBT

Recall that in the first ethical system, ' + ' corresponds to conflict, and ' . ' to union. In the second ethical system, the reverse holds: ' + ' corresponds to union and ' . ' corresponds to conflict.

Consider an individual in the first ethical system who has an image of his conflict with B :

$$a^{A+B},$$

where A and B are any structures.

We call the value of a 'intention'. Suppose $a = 0$. Let us pose the question: under what conditions can the individual transform a 'bad intention' into a 'good action?' To reach this it is necessary that

$$A + B = 0,$$

and this, in turn, is possible only if both $A = 0$ and $B = 0$. Therefore, an individual who is ordered by the environment to perform evil refuses to do so if he sees evil in himself and his adversary at the same time.

Consider the analogous structure in the second ethical system:

$$a^{A \cdot B}.$$

Suppose $a = 0$. The individual performs a good action if $A \cdot B = 0$, and this is possible when

- (1) $A = 0; B \neq 0$
- (2) $B = 0; A \neq 0$
- (3) $A = 0; B = 0$.

Therefore, in the second ethical system, an individual who is ordered by the environment to perform a bad action refuses to do this if

- (1) he sees evil in himself
- or
- (2) he sees evil in his adversary
- or
- (3) he sees evil both in himself and in his adversary.

If we identify 'evil seen in oneself' with feeling guilty and 'evil seen in one's adversary' with condemnation, then in the first ethical system a transformation of a bad intention into a good action is possible only if an individual feels guilty and condemns his enemy at the same time.

In the second ethical system such a transformation is possible if an individual

- (1) feels guilty
- or
- (2) condemns his enemy
- or
- (3) feels guilty and condemns his enemy at the same time.

We see that in the second ethical system the conditions for transformation are less stringent.

Let us now analyze the relation between guilt, condemnation, and doubt in conflict in the two ethical systems. We will consider four basic structures of doubt. To simplify our analysis we assume that an individual has a correct image of himself.

THE FIRST ETHICAL SYSTEM

Structure 1:

$$a^{A+B} \equiv a^{a^{\bar{a}}+b+b},$$

where $A \equiv a^{\bar{a}+b}$ and $B \equiv b$.

Substitute 0 for a :

$$0^{0^{\bar{0}}+b+b} = 0^{0^1+b+b} = 0^{0+b}.$$

Thus, $A = 0$ and $B = b$. Therefore, if A doubts the correctness of his image of himself and does not doubt that of his adversary, he always feels guilty when he has a bad intention.

Structure 2:

$$a^{A+B} \equiv a^{a^a+\bar{b}+b},$$

where $A \equiv a^a+\bar{b}$ and $B \equiv b$.

If $a = 0$,

$$0^{0^0 + \bar{b} + b} = 0^{0^{\bar{b}} + b} = 0^{b + b}.$$

Thus, $A = b$ and $B = b$. Therefore, if A doubts the correctness of his image of his adversary and does not doubt that of himself, he feels guilty when both he and his adversary have bad intentions.

Structure 3:

$$a^{A+B} \equiv a^{a^{\bar{a} + \bar{b}} + b},$$

where $A \equiv a^{\bar{a} + \bar{b}}$ and $B \equiv b$. This case is similar to that of Structure 1. Therefore, if A doubts the correctness of his images of himself and of his enemy, he always feels guilty when he has a bad intention.

Structure 4:

$$a^{A+B} \equiv a^{a^{a+b} + b},$$

where $A \equiv a^{a+b}$ and $B \equiv b$. If $a = 0$,

$$0^{0^{0+b} + b} = 0^{0^b + b} = 0^{\bar{b} + b}.$$

Thus $A = \bar{b}$ and $B = b$. Therefore, if A does not doubt at all, he feels guilty when he has a bad intention and his adversary has a good intention.

Let us now look at how guilt is related to condemnation.

Suppose $a = 0$ and $b = 0$ in Structures 1–4.

Structure 1, 2, 3:

$$A + B = 0.$$

Guilt and condemnation may appear at the same time.

Structure 4:

$$A + B \neq 0.$$

Guilt and condemnation cannot appear at the same time. In this structure doubt is absent. Therefore, *an absence of doubts implies the impossibility of feeling both guilt and condemnation at the same time*. In this case an individual's transformation of a bad intention into a good action is impossible by awareness of evil.

THE SECOND ETHICAL SYSTEM

Structure 1:

$$a^{A \cdot B} \equiv a^{a^{\bar{a} \cdot b} \cdot b},$$

where $A \equiv a^{\bar{a} \cdot b}$ and $B \equiv b$.

Suppose $a = 0$,

$$0^{0^1 \cdot b \cdot b} = 0^{\bar{b} \cdot b}.$$

Thus, $A = \bar{b}$ and $B = b$. Therefore if A doubts the correctness of his image of himself and does not doubt that of his adversary, he feels guilty when he has a bad intention and his adversary has a good intention.

Structure 2:

$$a^{A \cdot B} \equiv a^{a^a \cdot \bar{b} \cdot b},$$

where $A \equiv a^a \cdot \bar{b}$ and $B \equiv b$.

Suppose $a = 0$,

$$0^{0^0 \cdot \bar{b} \cdot b} = 0^{1 \cdot b}.$$

Thus, $A = 1$ and $B = b$. Therefore if A does not doubt the correctness of his image of himself and doubts that of his enemy, he never feels guilty.

Structure 3:

$$a^{A \cdot B} \equiv a^{a^{\bar{a} \cdot \bar{b}} \cdot b},$$

where $A \equiv a^{\bar{a} \cdot \bar{b}}$ and $B \equiv b$.

Suppose $a = 0$,

$$0^{0^1 \cdot b \cdot b} = 0^{b \cdot b}.$$

Thus, $A = b$ and $B = b$. Therefore if A doubts the correctness of his image of himself *and* that of his adversary, he feels guilty when both he and his adversary have bad intentions.

Structure 4:

$$a^{A \cdot B} \equiv a^{a^a \cdot b \cdot b}.$$

where $A \equiv a^a \cdot b$ and $B \equiv b$.

Suppose $a = 0$,

$$0^{0^0 \cdot b \cdot b} = 0^{1 \cdot b}.$$

Thus, $A = 1$ and $B = b$. Therefore if a does not have any doubts, he never feels guilty.

Let us see how guilt is related to condemnation (as we did for the first ethical system).

Structure 1:

$$A + B \neq 0.$$

Guilt and condemnation cannot appear at the same time.

Structure 2: No guilt at all.

Structure 3:

$$A + B = 0.$$

Guilt and condemnation may appear at the same time.

Structure 4: No guilt at all.

By comparing connections among good, evil, doubt, guilt and con-

demnation in situations of conflict in the first and second ethical systems, one can see:

- (1) Feeling guilty is more characteristic of an individual in the first ethical system than in the second (four times out of four in the first system and two times out of four in the second).
- (2) Simultaneous appearance of feeling guilty and condemnation is the necessary condition for transformation of evil into good in the first ethical system; in the second system only one of them is sufficient (feeling guilty or condemnation) for such a transformation.
- (3) The absence of doubts prevents the possibility of the transformation of evil into good in the first ethical system and does not prevent this in the second system.
- (4) If an individual doubts the correctness of his image of himself, and has a bad intention, then in the first ethical system he always feels guilty, but not always in the second system.
- (5) If an individual does not have any doubts, then in the first ethical system he feels guilty sometimes, but never in the second system.

CHAPTER VIII

A FORMAL COMPARISON OF ETHICAL SYSTEMS: DOUBTS AND ETHICAL STATUS

The ethical status of individuals whose expressions do not contain the signs ‘+’ and ‘.’ are equal in both systems. For example:

$$\begin{array}{ll} U_1 \equiv a & ; \quad |U_1| = \frac{1}{2}, \\ U_2 \equiv a^a & ; \quad |U_2| = 1, \\ U_3 \equiv a^{\bar{a}} & ; \quad |U_3| = \frac{1}{2}, \\ U_4 \equiv a^{aa} & ; \quad |U_4| = \frac{1}{2}, \\ U_5 \equiv a^{a\bar{a}} & ; \quad |U_5| = 1. \end{array}$$

Since ‘evaluations’ of such expressions do not depend on a concrete ethical system, we can say that these ‘evaluations’ form an ‘absolute ethics’. However, as soon as we begin to analyze an individual reflecting relationships with another person we leave the area of ‘absolute ethics’.

We will consider an individual *A*, who thinks he is in conflict with *B*, and his ‘reflexive structure’ does not contain ‘images from *B*’s point of view’. In Table 8.1, all possible structures of this kind are shown (up to the third level). In the left-hand column an individual and his ethical status are represented in the first ethical system; in the right-hand column the same individual, along with his ethical status, is shown in the second ethical system. In the first ethical system an individual of the chosen kind can have an ethical status of either $\frac{3}{4}$ or $\frac{1}{2}$. In the second ethical system we see also only two values of ethical status: 1 or $\frac{3}{4}$. We will compare ethical statuses inside each system, and the greatest one will be called maximum, and the lowest one, minimum. Maxima in both systems appear in the table with “√”. We can see that of the 18 structures of this kind, the maximum is reached 6 times in the first system, and 4 times in the second one.

Let us look now at the structures where ethical status is maximum in one of the systems and minimum in the other. There are four such ‘opposite’ structures in the table: 5, 6, 13 and 14.

Structure 5:

$$|a^{ab+b}| = \min; \quad |a^{a^b \cdot b}| = \max.$$

TABLE 8.1

All possible different two- and three-tier structures in either ethical system representing an individual *A* who thinks he is in conflict with *B* but where reflexive structure does not contain images from *B*'s point of view.

First ethical system		Second ethical system	
Formula	Ethical status	Formula	Ethical status
1. $a^a + b$	$\sqrt{\frac{3}{4}}$	$a^a \cdot b$	$\sqrt{1}$
2. $a^{\bar{a}} + b$	$\frac{1}{2}$	$a^{\bar{a}} \cdot b$	$\frac{3}{4}$
3. $a^{aa} + b$	$\frac{1}{2}$	$a^{aa} \cdot b$	$\frac{3}{4}$
4. $a^{\bar{a}\bar{a}} + b$	$\sqrt{\frac{3}{4}}$	$a^{\bar{a}\bar{a}} \cdot b$	$\sqrt{1}$
5.* $a^{ab} + b$	$\frac{1}{2}$	$a^{ab} \cdot b$	$\sqrt{1}$
6.* $a^{a\bar{b}} + b$	$\sqrt{\frac{3}{4}}$	$a^{a\bar{b}} \cdot b$	$\frac{3}{4}$
7. $a^{\bar{a}a} + b$	$\frac{1}{2}$	$a^{\bar{a}a} \cdot b$	$\frac{3}{4}$
8. $a^{\bar{a}\bar{b}} + b$	$\frac{1}{2}$	$a^{\bar{a}\bar{b}} \cdot b$	$\frac{3}{4}$
9. $a^{\bar{a}b} + b$	$\frac{1}{2}$	$a^{\bar{a}b} \cdot b$	$\frac{3}{4}$
10. $a^{a\bar{b}} + b$	$\frac{1}{2}$	$a^{a\bar{b}} \cdot b$	$\frac{3}{4}$
11. $a^{aa} + b + b$	$\frac{1}{2}$	$a^{aa} \cdot b \cdot b$	$\frac{4}{4}$
12. $a^{a\bar{a}} + b + b$	$\sqrt{\frac{3}{4}}$	$a^{a\bar{a}} \cdot b \cdot b$	$\sqrt{1}$
13.* $a^{aa} + \bar{b} + b$	$\sqrt{\frac{3}{4}}$	$a^{aa} \cdot \bar{b} \cdot b$	$\frac{3}{4}$
14.* $a^{a\bar{a}} + \bar{b} + b$	$\sqrt{\frac{3}{4}}$	$a^{a\bar{a}} \cdot \bar{b} \cdot b$	$\frac{3}{4}$
15. $a^{\bar{a}a} + b + b$	$\frac{1}{2}$	$a^{\bar{a}a} \cdot b \cdot b$	$\frac{3}{4}$
16. $a^{\bar{a}\bar{a}} + b + b$	$\frac{1}{2}$	$a^{\bar{a}\bar{a}} \cdot b \cdot b$	$\frac{3}{4}$
17. $a^{a\bar{a}} + \bar{b} + \bar{b}$	$\frac{1}{2}$	$a^{a\bar{a}} \cdot \bar{b} \cdot \bar{b}$	$\frac{3}{4}$
18. $a^{\bar{a}\bar{a}} + \bar{b} + \bar{b}$	$\frac{1}{2}$	$a^{\bar{a}\bar{a}} \cdot \bar{b} \cdot \bar{b}$	$\frac{3}{4}$

An individual *A* knows that he is in conflict with *B* and thinks that his image of *B* does not differ from the real *B* (from his point of view); that is, *he does not have doubts about the correctness of his image of his adversary*. In this case *A*'s ethical status is maximal in the second system and minimal in the first one.

Structure 6:

$$|a^{a^{\bar{b}} + b}| = \max; \quad |a^{a^{\bar{b}} \cdot b}| = \min.$$

An individual *A* knows that he is in conflict with *B*, and he thinks that his image of his opponent differs from that of the real opponent; that is, *he doubts the correctness of his image of his adversary*. In this case *A*'s ethical status is maximal in the first ethical system and minimal in the second one.

Structure 13:

$$|a^{a^a + \bar{b} + b}| = \max; \quad |a^{a^a \cdot \bar{b} \cdot b}| = \min.$$

Structure 14:

$$|a^{a^{\bar{a}} + \bar{b} + b}| = \max; \quad |a^{a^{\bar{a}} \cdot \bar{b} \cdot b}| = \min.$$

In both cases, *A* has doubts about his image of the opponent; in the first system *A*'s ethical status is maximal, in the second system it is minimal.

Thus, we see that a difference in the two ethical systems is connected with doubts about the correctness of one's images of his opponent. In the first ethical system such doubts cause maximization of ethical status; in the second ethical system they cause minimization of ethical status.

Absence of doubts about the correctness of an image of one's opponent might lead psychologically to an uncompromising fight with him. Presence of doubts about the correctness of an image of an opponent might lead one to a critical review of his opinion about his adversary and to a search for a possible compromise.

Let us now consider the full three-tier structures, including conflict with different patterns of negations on the second and third tier. For example:

$$a^{a^a + \bar{b} + b^b + \bar{a}}.$$

The total number of different structures for our example is $2^6 = 64$. But the number could be reduced to 32 by changing $\bar{b}$ into b .¹

¹ The complete set of these structures is given in Table A7.1.

There are two values of ethical status in the first ethical system ($\max = \frac{3}{4}$, and $\min = \frac{1}{2}$) and three in the second one ($\max = 1$, $\text{med} = \frac{3}{4}$, $\min = \frac{1}{2}$). Nine 'ideal' persons exist (whose ethical status is maximal) in the first ethical system. The general description of an 'ideal' person is as follows:

- (1) a person has a correct image of himself;
- (2) a person doubts the correctness of his image of himself or his opponent, or both;
- (3) a person thinks that his opponent has an incorrect image of himself, or of the person, or of both of them.

Only one 'ideal' person exists in the second ethical system:

$$a^{a\bar{a} \cdot b \cdot b\bar{b} \cdot \bar{a}}.$$

His description is as follows:

- (1) a person has a correct image of himself;
- (2) a person doubts the correctness of his own image, but he does not have any doubts about the correctness of his image of his opponent;
- (3) a person thinks that his opponent has incorrect images of both himself and of the person.

If this ideal person 'suddenly' begins to doubt the correctness of his image of his opponent, his ethical status will decrease to $\frac{3}{4}$.

There are 13 structures with 'medium' ethical status $\frac{3}{4}$. It is hard to make a general verbal description of these structures.² They may be regarded as 'ideals', but of a lower nature.

Let us examine one of them:

$$A \equiv a^{a\bar{a} \cdot b \cdot b\bar{b} \cdot \bar{a}}; \quad |A| = \frac{3}{4}.$$

This individual doubts neither his own image nor his image of his opponent. This might be viewed as a representation of a 'fanatic'. In the first ethical system, the ethical status of corresponding individual is equal to $\frac{1}{2}$;

² However, it is possible to use a special logical language, developed by B. Alpersen (1975), which can be interpreted by a meta-language for these structures. We use such a description in Appendix 2.

thus he cannot ethically excel his opponent. But in the second ethical system this is possible, for example,

$$\underbrace{a^{aa \cdot b} \cdot b^{\bar{a} \cdot \bar{b}}}_A \cdot \underbrace{b^{\bar{b} \cdot \bar{a}} \cdot a^{\bar{a} \cdot b}}_B$$

$$|A| = \frac{3}{4}; \quad |B| = \frac{1}{2}.$$

A description of person B is given in this example:

TABLE 8.2

A comparison of structures representing person A with an image of conflict with person B in terms of whether or not A doubts his image of conflict.

First ethical system			Second ethical system		
Formula	Ethical status	Change	Formula	Ethical status	Change
1. $a^{aa \cdot b + b}$	$\frac{1}{2}$	$\uparrow$	$a^{aa \cdot b \cdot b}$	$\frac{3}{4}$	$\downarrow$
$a^{aa \cdot b \cdot b}$	1		$a^{aa \cdot b + b}$	$\frac{1}{2}$	
2. $a^{a\bar{a} + b + b}$	$\frac{3}{4}$	$\uparrow$	$a^{a\bar{a} \cdot b \cdot b}$	1	$\downarrow$
$a^{a\bar{a} + b \cdot b}$	1		$a^{a\bar{a} \cdot b + b}$	$\frac{1}{2}$	
3. $a^{aa + \bar{b} + b}$	$\frac{3}{4}$	$=$	$a^{aa \cdot \bar{b} \cdot b}$	$\frac{3}{4}$	$\downarrow$
$a^{aa + \bar{b} \cdot b}$	$\frac{3}{4}$		$a^{aa \cdot \bar{b} + b}$	$\frac{1}{2}$	
4. $a^{a\bar{a} + \bar{b} + b}$	$\frac{3}{4}$	$\uparrow$	$a^{a\bar{a} \cdot \bar{b} \cdot b}$	$\frac{3}{4}$	$=$
$a^{a\bar{a} + \bar{b} \cdot b}$	1		$a^{a\bar{a} \cdot \bar{b} + b}$	$\frac{3}{4}$	
5. $a^{\bar{a}a + b + b}$	$\frac{1}{2}$	$\uparrow$	$a^{\bar{a}a \cdot b \cdot b}$	$\frac{3}{4}$	$\downarrow$
$a^{\bar{a}a + b \cdot b}$	$\frac{3}{4}$		$a^{\bar{a}a \cdot \bar{b} + b}$	$\frac{1}{2}$	
6. $a^{\bar{a}\bar{a} + b + b}$	$\frac{1}{2}$	$\uparrow$	$a^{\bar{a}\bar{a} \cdot b \cdot b}$	$\frac{3}{4}$	$\downarrow$
$a^{\bar{a}\bar{a} + b \cdot b}$	$\frac{3}{4}$		$a^{\bar{a}\bar{a} \cdot \bar{b} + b}$	$\frac{1}{2}$	
7. $a^{\bar{a}a + \bar{b} + b}$	$\frac{1}{2}$	$\uparrow$	$a^{\bar{a}a \cdot \bar{b} \cdot b}$	$\frac{3}{4}$	$\downarrow$
$a^{\bar{a}a + \bar{b} \cdot b}$	$\frac{3}{4}$		$a^{\bar{a}a \cdot \bar{b} + b}$	$\frac{1}{2}$	
8. $a^{\bar{a}\bar{a} + \bar{b} + b}$	$\frac{1}{2}$	$\uparrow$	$a^{\bar{a}\bar{a} \cdot \bar{b} \cdot b}$	$\frac{3}{4}$	$\downarrow$
$a^{\bar{a}\bar{a} + \bar{b} \cdot b}$	$\frac{3}{4}$		$a^{\bar{a}\bar{a} \cdot \bar{b} + b}$	$\frac{1}{2}$	

- (1) *B* has an incorrect image of himself;
- (2) *B* doubts the correctness of his images of himself and of his opponent;
- (3) *B* thinks that his opponent *A* has an incorrect image of himself and of *B*.

Consider now 'doubt about the correctness of assessing one's relation to an opponent as conflict'. Table 8.2 contains 8 blocks. Each block has two lines. The first line represents structures 'without doubts about correctness of assessment'; the structures in the second line differ from those in the first only by the presence of such doubts. The arrow pointing up shows increasing ethical status with transition from a structure 'without doubts about correctness of assessment' to a structure with such doubts. The arrow pointing down indicates a decrease in the ethical status with such a transition. We can see that in 7 of 8 cases this kind of transition causes an increase of ethical status in the first ethical system, and a decrease in the second ethical system. In one case in each ethical system the ethical status remains the same.

Therefore, 'doubt about the correctness of assessment of relationships with an opponent as conflict' increases ethical status in the first ethical system and decreases it in the second.

We will not consider in detail structures of union. Note only that structures of union in the first ethical system correspond to structures of conflict in the second ethical system, and structures of union in the second system correspond to structures of conflict in the first. Therefore, doubts in such a 'system of friendships' lead to decrease of ethical status of a doubting person in the first ethical system and to increase in the second.

CHAPTER IX

ETHICAL ANALYSIS OF ARTISTIC AND PROPAGANDISTIC LITERATURE

Let us now consider an anthropological aspect of the problem. Are the two ethical systems realized in real human cultures? To answer the question we need empirical data related to the two different societies. We can then examine whether the cultural peculiarities differ from each other in the way predicted by our theory (assuming that in these cultures the different ethical systems are realized).

The results obtained in the previous chapter suggest an experimental procedure. We know the connection between ethical status and doubts. By constructing a model of a *typical* conflict and by knowing which of the participants has moral priority (from a normative point of view), we can ask which of the two ethical systems best reflects this priority.

In particular, we can use the structure of 'fanatics'.¹ In the first ethical system a person described by such a structure cannot have an ethical status higher than that of his opponent. But this is possible in the second ethical system. Thus, if we found *normative* cases of ethical priority of a fanatic, this would corroborate the hypothesis that in a given culture the second ethical system is realized. In addition, we expect that an ideal person in the second ethical system is a person who doubts his image of himself but does not doubt his image of his opponent.

One possible method of empirical investigation is analysis of the classical literature of a given culture. We can attempt to represent in a reflexive structure a typical conflict between a 'positive hero' and a 'negative hero' to see which ethical system fits better, in the sense of attributing a higher

¹ The word 'fanatic' in English has a clear negative meaning. In nineteenth century Russian it also had the negative meaning. Vladimir Dal's Dictionary (1882) gives the following definition: "*Fanaticism* is barbarity, cruel, obstinate superstition instead of faith, persecution of differently minded. A *Fanatic* is a monster." However, in the modern Russian language this word has already two meanings. In Ozhegov's Dictionary (1953) one can read: "*Fanaticism* is excessive zeal. A *Fanatic* is (1) a person of frantic religiosity, a monster; (2) figurative meaning: a person fervently faithful to something. Example: fanatic of art." In Foreign Words Dictionary (Ed. Lehin, 1964) the second meaning is not considered figurative. In the modern Russian language the sentence 'A person fanatically devoted to Revolution' sounds extremely positive. Thus, the word 'fanatic's' positive or negative meaning depends on whether the idea he is faithful to is positive or negative.

ethical status to the positive hero. Of course, the literature chosen should be significant for a given culture. We have examined Shakespeare's *Hamlet* for Western culture and *How the Steel was Tempered* by Nikolay Ostrovsky and *The Iron Flood* by Alexander Serafimovich for Soviet culture.

Hamlet is obviously one of the most 'Western' pieces of literature involving confrontation between good and evil. This tragedy has motivated a great deal of critical and ethical commentary. Numerous scholars, scientists, and politicians (including Stalin) have expressed opinions about *Hamlet*.

The crux of the plot is the conflict between Hamlet and his uncle Claudius, who becomes King of Denmark after the death of the previous king, Hamlet's father. Claudius's immediate marriage to Hamlet's mother causes great trauma for Hamlet. The ghost of Hamlet's father appears to tell Hamlet a terrible secret: it was Claudius who poisoned him. It seems that Hamlet should avenge his father's death, but Hamlet hesitates over whether the ghost is actually his father's ghost. Hamlet enacts a complex experiment to eliminate his doubts about his image of his opponent; he invites actors to stage a scene recreating the old king's death, and together with his friend Horatio observes Claudius's reactions. Thus, Shakespeare employs a special scene in this tragedy to demonstrate Hamlet's doubts about the validity of his image of Claudius. With equal clarity, Shakespeare demonstrates Hamlet's doubts about himself in his monologue, "To be or not to be, that is the question."

Let us designate Hamlet by A and Claudius by B .² In the first ethical system, Hamlet would then be

$$H \equiv a^{\bar{a} + \bar{b} + b}; \quad |H| = \frac{3}{4} = \max.$$

Hamlet

- (1) has correct images of himself and of Claudius;
- (2) doubts the correctness of these images.

In the second ethical system, Hamlet would be

$$H \equiv a^{\bar{a} \cdot \bar{b} \cdot b}; \quad |H| = \frac{3}{4} = \min.$$

Claudius, on the other hand, has incorrect images of himself and of

² An algorithm of this analysis is described in Appendix 6.

Hamlet, and he doubts the correctness of his images of himself and of Hamlet. In the first ethical system, Claudius would be

$$C \equiv b^{\bar{b}b + a + \bar{a}}; \quad |C| = \frac{1}{2} = \min.$$

In the second ethical system, Claudius would be

$$C \equiv b^{\bar{b}b + a + \bar{a}}; \quad |C| = \frac{3}{4} = \min.$$

Hamlet's ethical status is higher than that of Claudius in the first ethical system; in the second they are equal. In the second ethical system Hamlet is a 'hero' of the second sort, unlike in the first one, where he is one of six 'ideals' according to the formal structure.

Shakespeare's creation coincided with a period of active formation of the European value system. The tremendous popularity of *Hamlet* suggests that this play might well correspond to an ethical norm of Western civilization. Apparently this Shakespearean tragedy has had normative significance.

Soviet culture is only 60 years old. Some people do not consider it a separate culture. However, the present author thinks that it is a separate culture, with its own ethical system different from the Western one.

It is easy to find normative heroes in Soviet culture. Very often an author's point of view coincides completely with a positive hero's position, which simplifies our analysis. One such work is the novel *The Iron Flood* by Serafimovich. The novel was written in 1924; its positive heroes are soldiers and commanders of a Red Army unit forcing its way and destroying everything in its path. Serafimovich sides with the 'iron flood'. The main hero, a Red commander Kozhukh, does not doubt at all, either the correctness of his image of himself or of his enemy. A standard of an enemy is Colonel Mikheladze.

In the first ethical system the conflict between the Red and White Armies may be expressed as follows:

$$\underbrace{a^{a + b + b\bar{b} + \bar{a}}}_{A} + \underbrace{b^{\bar{b}b + a + \bar{a} + b}}_{B},$$

$$|A| = \frac{1}{2}; \quad |B| = \frac{1}{2}.$$

A, Kozhukh (Red Army):

- (1) has correct images of himself and of his enemy;
- (2) does not have doubts concerning correctness of his images of himself and of his enemy;
- (3) thinks that his enemy has incorrect images of himself and of the Red Army (with which Kozhukh identifies himself). Kozhukh's structure is a fanatic's structure.

B, Mikheladze (White Army):

- (1) has incorrect images of himself and of his enemy;
- (2) does not have doubts concerning correctness of his images of himself and of his enemy;
- (3) thinks that his enemy has incorrect images of himself and of the White Army.

The first ethical system does not show that Kozhukh has a higher ethical status than Mikheladze.

In the second ethical system the conflict would be described by the structure:

$$\underbrace{a^{a^a \cdot b} \cdot b^{\bar{b}} \cdot \bar{a}}_A \cdot \underbrace{b^{\bar{b} \bar{b}} \cdot \bar{a} \cdot \bar{a}^a \cdot b}_B;$$

$$|A| = \frac{3}{4}; \quad |B| = \frac{1}{2}.$$

Thus, we see Kozhukh's ethical superiority to Mikheladze emerging in the second system, but not in the first. This is an argument favoring the idea that Serafimovich, while writing his novel, was within an ethical perspective related to the second ethical system.

Now we pose the question: is there a positive hero in Soviet literature whose ethical status reaches maximum?

An 'absolute ideal' in the second ethical system is not a fanatic, but a person who has doubts about the correctness of his own image but no doubts about the enemy's. One of the most significant normative works in Soviet literature is the novel *How the Steel Was Tempered* by Nikolay Ostrovsky. Nikolay Ostrovsky, as a very young boy, participated in the Russian Revolution of 1917 and in the Civil War. He believed in the idea of World Revolution. In 1930 he was blinded and paralyzed (he was 26 at that

time). For three years he dictated his autobiographical novel describing a revolutionary hero, Pavel Korchagin. The hero is critical of himself; he has many doubts about the correctness of his image of himself. When he becomes blind, he thinks about suicide; he banishes this thought with shame. However, Korchagin does not have any doubts about the correctness of his enemy's image. If he is in conflict with somebody, it is unalterable. His conflict is always social. He is conflicting not with a concrete person, but with a representative of a hostile social class. The world of people is dichotomized for him: comrades and enemies.

In the second ethical system Korchagin's structure may be represented as:

$$K \equiv a^{a\bar{a} \cdot b \cdot b}; \quad |K| = 1 = \max.$$

Korchagin

- (1) has correct images of himself and his enemy;
- (2) doubts the correctness of his own image, but does not doubt that of his enemy.

This individual reaches absolute maximum. Therefore, in Soviet literature we can see

- (1) 'positive' fanatics;
- (2) the 'ideal' of the second ethical system.

This compels us to the hypothesis that in Soviet culture the second ethical system is realized.

CHAPTER X

EXPERIMENTAL ANALYSIS OF NORMATIVE INDIVIDUALS

A series of experiments was performed for the purpose of examining the predictive power of the model. A subject is given a questionnaire about two conflicting people known to him. In some cases, the relation of ethical priority is given in the text, in others a subject has to indicate the ethical priority of one of the two people in the experimental questionnaire or their equality. The subject has to rely on his ethical intuition and transform it into a decision about ethical priority. When the subject has given an overall judgment about ethical priority, we ask him a programmed series of questions and use the answers to determine algorithmically a unique structure designed to represent the interaction as seen by the subject. The model predicts that the overall judgment will be consistent with the ethical status computed from the model if the correct ethical variant (of the two possibilities) is used to interpret the subject's answers. Thus, we can see which of the two ethical systems best represents each subject's choice. If one system consistently outperforms the other and also makes good predictions, its validity is argued. That is, according to the answers on the questionnaire, the formula of the interaction is constructed, and the relationships of ethical priority are found. Then we can establish whether the formula predicts that ethical priority which was indicated by a subject or given before the experiment began.

1. THE KENNEDY - NIXON EXPERIMENT

First we illustrate the method by examining the data from one subject. That subject was given the questionnaire illustrated by Table 10.1.¹ It shows a sample of actual data. According to these data we can construct formulae of the interaction in each ethical system.

The First Ethical System:

$$K^{K\bar{K} + N \cdot \bar{N}} + N^{\bar{N} + K + K}$$

$$|\text{Kennedy}| = 1; \quad |\text{Nixon}| = \frac{1}{2}.$$

¹ The general method of structuring formulae is given in Appendix 6.

TABLE 10.1

Questionnaire for the 'Kennedy-Nixon' experiment along with data from one subject indicated by checkmarks.

-
1. Do you think that Mr. J. Kennedy had a correct valuation of himself? ☒ yes ☐ no
 - Do you think that Mr. R. Nixon had a correct valuation of himself? ☐ yes ☒ no
 2. Did Mr. Kennedy have a correct valuation of Mr. Nixon? ☐ yes ☒ no
 - Did Mr. Nixon have a correct valuation of Mr. Kennedy? ☒ yes ☐ no
 3. Did Mr. Kennedy have any doubt about correctness of his valuation of himself? ☒ yes ☐ no
 - Did Mr. Nixon have any doubt about correctness of his valuation of himself? ☒ yes ☐ no
 4. Did Mr. Kennedy have any doubt about correctness of his valuation of Mr. Nixon? ☒ yes ☐ no
 - Did Mr. Nixon have any doubt about correctness of his valuation of Mr. Kennedy? ☐ yes ☒ no
 5. Did Mr. Kennedy doubt the correctness of his model of the relationships with Mr. Nixon as a conflict? ☒ yes ☐ no
 - Did Mr. Nixon doubt the correctness of his model of the relationships with Mr. Kennedy as a conflict? ☐ yes ☒ no
 6. What you think is true?
 - ☒ Mr. Kennedy ethically excelled Mr. Nixon.
 - ☐ Mr. Nixon ethically excelled Mr. Kennedy.
 - ☐ Mr. Kennedy and Mr. Nixon are ethically equal.
-

Kennedy's ethical status is higher than that of Nixon.

The Second Ethical System:

$$K^{K\bar{K} \cdot N + \bar{N}} + N^{\bar{N}N \cdot K \cdot K}$$

$$|\text{Kennedy}| = \frac{3}{4}; \quad |\text{Nixon}| = \frac{3}{4}.$$

The ethical statuses of Kennedy and Nixon are equal.

We can see that in the first ethical system the prognosis coincides with the subject's choice, but not in the second system. Similar analysis was made for data from all subjects.²

Twenty-eight American subjects (UCI volunteer students and faculty)

² The entire set of possible structures of one experimental person (Kennedy or Nixon) is given in Table A7.2.

participated in this experiment; the age ranged from 20 to 55; 17 male and 11 female. No explanation of meanings of terms on the questionnaire was offered to the subject.

For each of the two systems a 3×3 contingency table was constructed involving the prediction of the structure in that system and the actual response of the subject. The results were for the first system $\chi^2_4 = 14.99$, $p = 0.005$ and for the second $\chi^2_4 = 2.74$, $p = 0.50$. Thus we see that for this experiment system 1 gives statistically significant improvement while system 2 does not.

We would argue that if one of the ethical variants has a significant advantage over both the random model (null hypothesis) and the other ethical variant, then the following conclusions are justified: (a) subjects can make ethical comparisons that are meaningful; (b) our model and its computational and representational rules has some validity; and (c) the ethical system held by the respondent pool is characterized to some degree by the successful ethical variant. The significant results that were obtained for the first ethical system provide encouragement for the overall theoretical enterprise.

2. THE 'TOM AND JOHN' EXPERIMENT

In the previous experiment the relations of ethical priority were indicated by each subject. In the next experiment ethical priority was fixed in advance for all subjects. An experimental story was constructed (Table 10.2) with a conflict of Tom (a person with standard virtues) and John (an 'evildoer' with emphasized negative features).

Sixty-two American subjects participated in the experiment: ages from 17 to 60; 41 male and 21 female.

In contrast to the previous experiment, the present questionnaire allowed the construction of complete 'three-tier' structures.³ The probability of a correct prediction in random choices of structures is equal to:

the first ethical system – 0.20,
the second ethical system – 0.26.

The frequency of correct predictions calculated according to experimental data is equal to:

³ The entire set of 32 structures is given in Table A7.1.

TABLE 10.2
Questionnaire for the 'Tom and John' experiment.

John was a disgusting person. In his childhood he tortured cats. When he was sixteen he intended to kill his father, and the latter just barely escaped. John was a good student in the university, and after graduation he started to work for a furniture corporation. There he became acquainted with Tom.

Tom was an honest and kind person. He grew up in a family where everybody loved each other. He especially loved his father. He was sixteen when he went to the big town to the University, and he missed his home very much. Tom was also a good student, and after his graduation he started to work for the same furniture corporation.

Kathy was a young, blond, pretty girl, 20 years old, working in that corporation. Both John and Tom fell in love with her. For a short time they tried to keep their love a secret from each other, but soon they each understood that the other also loved Kathy. They could not be friends anymore.

On May 5, 1929, John killed Tom.

1. Do you think that John had an accurate image of himself? — yes — no
Do you think that Tom had an accurate image of himself? — yes — no
2. Did John have an accurate image of Tom? — yes — no
Did Tom have an accurate image of John? — yes — no
3. Did John have any doubt about his image of himself? — yes — no
Did Tom have any doubt about his image of himself? — yes — no
4. Did John have any doubt about his image of Tom? — yes — no
Did Tom have any doubt about his image of John? — yes — no
5. Did John think that Tom had an accurate image of himself? — yes — no
Did Tom think that John had an accurate image of himself? — yes — no
6. Did John think that Tom had an accurate image of John? — yes — no
Did Tom think that John had an accurate image of Tom? — yes — no

the first ethical system – 0.35,

the second ethical system – 0.24.

In the first ethical system we obtain a small but statistically significant improvement over chance ($\chi^2 = 8.5$; $df = 1$; $p < 0.01$). In the second system there is no improvement.

The two experiments are an argument in favor of the hypothesis that American subjects are best described by the first ethical system.

3. THE 'PAVLIK MOROZOV' EXPERIMENT

A similar experiment was used for the investigation of the normative ethical system of the Soviet Union. An experimental story was chosen from a

legend about Pavlik Morozov. Pavlik Morozov is a heroic person in Soviet propaganda. According to the official Soviet version, he was a son of a rich Siberian peasant who fought land collectivization (forcibly imposed by the Soviet government in the early 1930's). Pavlik joined the communist youth organization (Young Pioneers) and turned into a faithful helper of the local Soviet authorities. He informed on his father to the authorities; there was a trial in which Pavlik was a witness, and the father was sentenced to death. Afterwards, Pavlik was killed by his father's friends and relatives. Pavlik Morozov is a normative hero in the Soviet Union.

Since Soviet citizens are not available as subjects, recent emigrés from the Soviet Union were used. To eliminate the possible change in the subjects' mind under the influence of the struggle to emigrate and subsequent contact with Western culture, they were asked to answer the questions 'from the point of view of a typical Soviet person' and not from their own point of view. The experiment was performed as an interview; the same questions as in the experiment 'Tom and John' were asked orally. Altogether there were 14 subjects ranging in ages from 16 to 70, 8 male and 6 female.

The probability of a correct prediction in a case of random distribution is the same as in the earlier experiment involving 'Tom and John'. The frequencies of correct predictions obtained from experimental data are equal to:

the first ethical system—0.07

the second ethical system—0.86.

In the first ethical system there is a small statistically non-significant deterioration of prediction; in the second ethical system there is significant improvement ($\chi^2 = 20.3$; $df = 1$; $p < 0.001$).

This is an argument that the Soviet subjects in this experiment realized the second ethical system.

CHAPTER XI

THE PRINCIPLE OF MAXIMIZATION OF THE ETHICAL STATUS OF ONE'S IMAGE OF ONESELF

In the previous chapter, we analyzed given structures of individuals. In this chapter we will consider situations in which the individual *can choose* a type of relationship with his partner, and thus, can change his structure 'at will'.

We state the following postulate: *by making a decision an individual tries to maximize the ethical status of his own image*. The author thinks that this principle adequately reflects the ethical reality of decision-making by an 'ethical man'. An ethical self-evaluation and the decisions dependent upon it are important components of any real decision-making. This principle is stronger than utilitarian principles, because it encompasses the area of non-rational behavior.

Consider an individual

$$a^{\boxed{a^a * b}} * b$$

and let the sign '*' take on the meanings '+' or '·'. For convenience we frame the individual's image himself. Let us now put this individual in a situation where he has to choose the meaning of the sign; in other words, he himself has to establish the relation either of union or of conflict with his partner. At first glance it may seem that we (and the individual) do not have enough information to do so. However, this is not so if we assume that the individual can 'calculate', just as we can, the ethical statuses of individuals in his own inner world. The individual's image of himself is a^{a^b} . It is easy to see that

$$|a^{a \cdot b}| > |a^{a + b}|,$$

since

$$|a^{a \cdot b}| = 1 \quad \text{and} \quad |a^{a + b}| = \frac{3}{4}.$$

We can see that in all cases an individual will tend to choose the '·', not '+'. Therefore, a 'three-tier individual' belonging to the second ethical system would choose conflict, and one belonging to the first ethical system

would choose union. In the second ethical system an individual tries to be a fighter in any contact with another individual. In the first ethical system an individual always seeks 'collaboration'. This conclusion is a consequence of the formal theory.

In real situations an individual cannot always choose his relation to a partner. First, the relation depends on the partner; second, an individual is always included in a social context, which predetermines a concrete relation. His own choice degenerates to a tendency to compromiselessness or to a tendency to compromise. For example, an individual $a^{\overline{a^a + b}} \cdot b$ in the second ethical system would seek to stay in a state of conflict with his adversary, since in the case of a change of '·' to '+' the ethical status of his own image of himself will decrease. This is an example of compromiselessness toward an enemy.

An individual of the first ethical system $a^{\overline{a^a + b}} + b$ would have a tendency to change the '+' to '·', since in this case the ethical status of his own image of himself will increase. This is an example of a compromising relation to an enemy.

All previous conclusions were drawn by analyzing a single example, but the example completely reflects the general case, to be considered in Chapter XIII and Appendix 4.

Consider now a 'simple' individual, who does not have a third tier: $a^{\textcircled{a} + b}$, belonging to the second ethical system. From the point of view of an external observer, the ethical status of this individual is maximal when he is in conflict, and minimal when he is in union with his adversary. However, the individual 'does not know' this. His own image of himself is just a ; his ethical status is constant: $|a| = \frac{1}{2}$. In this example, the ethical status of one's image of oneself is not a function of a sign chosen by the individual, as it was in the previous examples. An individual does not have the ability of finding out the advantage of conflict over union. He is in an indefinite state. Therefore, this individual can transfer himself into a state with maximal ethical status only accidentally or else under external influence. (Propaganda of 'compromiselessness' is necessary!)

This example clarifies for us the functional role of the 'third tier'. An individual who has a 'third tier', i.e., 'a conscious individual', would choose sign by himself; he does not need an external compulsion. As will be shown in the next chapter, by choosing '·' instead of '+' the individual not only increases the ethical status of his own image of himself but also might increase his own ethical status from an external observer's point of view. Thus, *the third tier turns an individual into a self-maximizing system.*

The principle of maximization allows us to discover surprising connection between ethics and strategy. Consider a 'four-tier' individual:

$$a^{\boxed{a\bar{a} + b\bar{b} + a}} + b\bar{b} + a$$

in the second ethical system. The ethical status of the image of himself is equal to $\frac{1}{2}$. A change of relationship to the adversary alone would not change the individual's image of his own ethical status, since

$$|a^{\bar{a} \cdot b\bar{b} \cdot a}| = \frac{1}{2}.$$

The maximum of the module of the expression in the rectangle would be reached when the '+' is kept on the highest level and is changed to '.' on the level below:

$$|a^{\bar{a} \cdot b\bar{b} \cdot a}| = \frac{3}{4}.$$

What does this mean? The constant sign belongs to the adversary's image. If an individual changes his relationship with his adversary, this can be known to the adversary, who will reflect the fact of conflict, and the sign of the higher level would change. To keep this sign an individual has to hide the fact of the change of relationships from his adversary.

Let us imagine now that initially the individual was in conflict with his adversary:

$$a^{a\bar{a} \cdot b\bar{b} \cdot a} \cdot b\bar{b} \cdot a.$$

We have seen that an individual would have the ethical status of his image maximal if the sign on the upper level were to change from '.' to '+', while the other signs remained the same. How can this be done? The individual has to persuade his adversary that they are in union (although he must stay convinced they are in conflict). These examples demonstrate that secrecy and lying could be conditions for high self-evaluation.

Additionally, let us look at the expressions $|b^{\bar{b} \cdot a}| = \frac{1}{2}$; and $|b^{\bar{b} \cdot a}| = \frac{3}{4}$. By trying to force his adversary to believe that they are in union, rather than in conflict, an individual at the same time tries to diminish the adversary's ethical status. The reason is that in such cases, the condition of increasing the ethical status of one's own image is that the adversary's is decreased. Here we can see a way to understand the nature of moral conflict more deeply.

CHAPTER XII

FEELINGS AND SACRIFICES

There is a widespread opinion that willingness to sacrifice indicates altruism. In reality, altruism may be considered as a form of sacrifice. Examples of sacrificial thought show that only rarely are sacrifices made purely for the good of another person. Usually sacrifice is made in the name of an idea or for self-affirmation. Its meaning is almost always only symbolic. Furthermore, sacrifice does not bring any profit, but rather losses. Nevertheless, sacrifice has always been regarded as one of the better qualities of human beings.

The main feature of all real sacrifice is movement against the flow of adaptation and rationality. What moves Dostoevsky's heroes? Raskolnikov, suffering, kills an old woman; Sonia Marmeladova, suffering, sells herself. Here some mysterious mechanism of seeking suffering is at work.

In this chapter we shall demonstrate how this mechanism of sacrificial behavior may become clearer in the light of maximizing the ethical status of one's own image. For this purpose we relate the principle of maximization to 'feelings'.

As was shown in Chapter VI, for an individual

$$a^{A*B}$$

a measure of feeling guilty is

$$u = 1 - |A|,$$

and a measure of suffering is

$$v = 1 - |A*B|.$$

Consider an individual

$$a^{a^{a+b}+b}.$$

This individual has an image of himself $A \equiv a^{a+b}$, an image of his partner $B \equiv b$, and an image of the situation (an inner world) $W \equiv a^{a+b} + b$.

Let this individual change the sign of his relationship with his partner. First, the ethical status of his own image increases (as was shown in the previous chapter), since

$$|a^{a \cdot b}| > |a^{a+b}|.$$

Second, the feeling of guilt is decreased.

Let us look at suffering. Before the sign was changed suffering was

$$1 - |a^{a+b} + b| = 0.$$

After the change in sign *suffering increases*

$$1 - |a^{a \cdot b} \cdot b| = \frac{1}{2}.$$

Our analysis may be summarized with Table 12.1.

By changing '+' to '.' we obtain the following changes: (1) the module of one's image of oneself increases, and therefore feeling guilty decreases; (2) the module of one's inner world decreases, and therefore suffering increases; (3) the module of an individual increases. Thus, we see that while tending to maximize the ethical status of his own image, an individual *increases his suffering*. By changing '+' to '.' an individual transforms the image of himself from a^{a+b} to $a^{a \cdot b}$. Since $|a+b| > |a \cdot b|$, an individual *sees himself* suffering more after the change in sign than before. Thus, an individual not only increases his suffering, but also sees himself 'experiencing' this increase in suffering.

TABLE 12.1

The measures of feeling guilty, suffering, and ethical status of a particular individual as a function of the sign of his relationships with his partner.

Individuals	Module of individual's image of himself	Feeling guilty	Module of individual's inner world	Suffering	Module of individual
a^{a+b+b}	$\frac{3}{4}$	$\frac{1}{4}$	1	0	$\frac{1}{2}$
$a^{a \cdot b \cdot b}$	1	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$

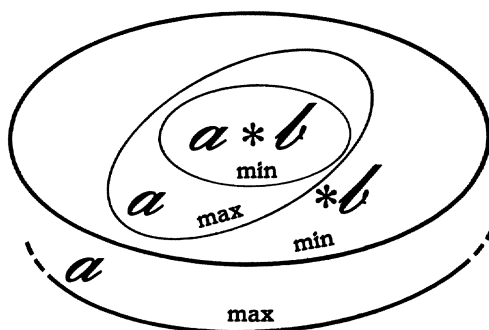


Fig. 12.1. Changes of modules of an individual and of his fragments while '+' is transforming into '•'.

Relations between the modules of various fragments of an individual a^{a*b*b} when sign '*' changes from '+' to '•' are shown in Figure 12.1. We may suppose that here we have succeeded in apprehending the psychological mechanism of sacrificial behavior in which an individual suffers deliberately in order to improve his image in his own eyes.

We may touch upon the mystery of the 'tendency to suffer', peculiar to Christian cognition and especially to Russian Orthodox cognition: while increasing his suffering and contemplating himself suffering, an individual ennoble himself in his own eyes. An external utilitarian setting for such behavior may be only an epiphenomenon. The individual tends to suffer because this is the only way to increase the ethical status of his own image, predetermined by the algebraic structure of cognition.

But we can see also the difference between sacrificial behavior in the first and second ethical systems. An individual of the first ethical system resorts to sacrificial compromise: 'I don't know this man, but I would degrade myself in my own eyes if I were to show my reservation or hostility. Despite its being hard for me, I have to extend my hand to him and smile. My compromise, my benevolence is a sacrifice, but I have to do so, in order to respect myself'. This is a quite simplified description of the ideology reflected in *the tendency to compromise with a partner* in the first ethical system.

An individual of the second ethical system resorts to sacrificial conflict. 'I don't know this man, but I would degrade myself in my eyes if I am benevolent and servile. Despite its being hard for me, I have to show that I

am ready to fight, to show this with my facial expression. My confrontation, my hostility is a sacrifice, but I have to do so, in order to respect myself'. This is a highly simplified description of the ideology reflected in *the tendency to conflict with a partner* in the second ethical system.

If an individual of the first ethical system is already in conflict with his partner, then his tendency toward dialogue and negotiations, which might decrease the strain, would increase the ethical status of his own image.

An individual of the second ethical system when in conflict with his partner, tries to dramatize the situation, since everything that decreases strain decreases also the ethical status of his own image.

A hero of Western culture is an individual with the tendency to compromise with his partner. A hero of Soviet culture is a person with a tendency to fight with his partner. The hero of the first ethical system is a Christian who loves his enemy; the hero of the second ethical system is a revolutionary preferring death to any dialogue with his adversary.

However, in contemporary Soviet mass cognition we do not find realizations of the heroic type. Heroic behavior appears toward a weak partner only; when dealing with a strong partner, a Soviet person resorts to compromise behavior, and he degrades himself in his eyes then.

Let us go back to Table 12.1. When the sign changes from '·' to '+', an individual's suffering decreases. Thus, if an individual tends to decrease his suffering (and not to increase the ethical status of his image of himself), his behavior is different. In the first ethical system he chooses conflict with his partner, and in the second ethical system such an individual chooses compromise with a partner. These individuals are not 'heroes': their ethical status is lower. We will call them 'philistines' (since philistines tend to minimize their suffering rather than to sacrifice). Philistines have a low ethical status and a high feeling of guilt.

Thus, we have found a formal correlation between suffering and feeling guilty. As the one grows, the other tends to diminish – and this is due to formal reasons. It is precisely these incompatible tendencies which separate individuals into the two types which we called 'heroes' and 'philistines'. But we can make further distinctions.

In the previous analysis we assumed that when an individual chooses a symbol for '*', he *sees himself* as having chosen that particular symbol. If we change this assumption and look at individuals whose reflections of the symbol of relationship are different in the second and third tiers, we see that the formal correlation between suffering and feeling guilty becomes

different. Such an individual may, at the same time, decrease both his feeling guilty and his suffering.

Let us analyze an individual

$$a^{a^a * b * b}, \quad (12.1)$$

who transforms himself into the structure

$$a^{a^a \cdot b + b}.$$

The ethical status of this individual is $\frac{1}{2}$, that is, minimal. This individual made a 'philistine' decision '+', but he sees himself as a person who made a heroic decision. He does not feel any guilt (since $|a^a \cdot b| = 1$), nor suffering (since $|a^a \cdot b + b| = 1$). This is an individual with excessively high self-evaluation; he is not modest. We will call him a *dissembler*. To simultaneously decrease both suffering and feeling guilty is possible only by violating the adequacy of reflection in the individual's inner world.

Let us analyze the same individual (12.1), who this time transforms himself into the structure

$$a^{a^a + b \cdot b}.$$

He has made a heroic decision, but sees himself as a person who made a philistine decision. The ethical status of this individual is 1, that is *maximal*. Feeling guilty and suffering simultaneously reach their possible maximum for this structure: feeling guilty is equal to $1 - |a^a + b| = \frac{1}{4}$, and suffering is equal to $1 - |a^a + b \cdot b| = \frac{3}{4}$. The suffering of this individual is the greatest possible within the framework of this structure, but he sees himself as 'experiencing' the least possible suffering. This individual has a low self-evaluation. He is modest. We will call him a *saint*.

There is an interesting peculiarity distinguishing a saint from a hero, a philistine, and a dissembler: a hero minimizes his feeling guilty; a philistine minimizes his suffering; a dissembler minimizes both suffering and feeling guilty. A saint does not minimize any kind of discomfort; he *tends* toward feeling guilty and suffering at the same time.

All types of individuals are represented in Table 12.2.

We will call an individual perfect if his ethical status is equal to 1.

Then the following statement is true:

THEOREM 12.1. *A perfect individual cannot consider himself perfect.*

TABLE 12.2
Descriptions of the types of normative individuals in both ethical systems.

Philosophy	First ethical system		Second ethical system	
	End does not justify means		End justifies means	
Sacrificial individual	Saint	Non-aggressive: tends toward compromise with a partner; has low self-evaluation	Aggressive: tends toward conflict with a partner; has low self-evaluation	
	Hero	Non-aggressive: tends toward compromise with a partner; has high self-evaluation	Aggressive: tends toward conflict with a partner; has high self-evaluation	
Non-sacrificial individual	Philistine	Aggressive: tends toward conflict with a partner; has low self-evaluation	Non-aggressive: tends toward compromise with a partner; has low self-evaluation	
	Dissembler	Aggressive: tends toward conflict with a partner; has high self-evaluation	Non-aggressive: tends toward compromise with a partner; has high self-evaluation	

Proof. Suppose the contrary, i.e., a perfect individual a^{A*B} considers himself perfect. This means that the following equations hold simultaneously:

$$|a^{A*B}| = 1, \quad \text{and} \quad |A| = 1.$$

Since $|A| = 1$, then $A = 1$ for any pair of values a and b , and therefore,

$$a^{A*B} = a^{1+B} = a \quad \text{if } * \text{ is } +,$$

and

$$a^{A*B} = a^{1 \cdot B} = a^B \quad \text{if } * \text{ is } \cdot.$$

In the first case the module of an individual is equal to $|a| = \frac{1}{2}$, which contradicts the definition of a perfect individual whose module is 1.

In the second case,

$$a^{A \cdot B} = a^B,$$

where B is either b or b^W .

If $B \equiv b$, then

$$|a^{A \cdot B}| = |a^b| = |\bar{b} + a| \neq 1.$$

If $B \equiv b^W$, then

$$|a^{A \cdot B}| = |a^{b^W}| = |W \cdot \bar{b} + a| \neq 1.$$

This also contradicts the condition that an individual's status is equal to 1.

For the simplest case a^A the individual has an image of himself but does not have one of his partner; it is clear that the two following equations

$$|a^A| = 1 \quad \text{and} \quad |A| = 1$$

cannot be true at the same time.

Therefore, our assumption that a perfect individual may consider himself perfect is wrong. $\square$

One of the consequences of the above theorem is that a perfect individual cannot consider himself a 'perfect saint', but can consider himself a dissembler. For example, an individual

$$a^{a^a \cdot b + b \cdot b}$$

has an ethical status equal to 1, but he sees himself as a dissembler:

$$a^{a^a \cdot b + b}.$$

This result quite well coincides with our intuitive picture of an ethically 'perfect person'. Such an individual is very modest, so he considers himself rather a dissembler than a saint; he is inclined to self-condemnation, rather than to self-admiration.

We see that perfection is an external characteristic of an individual: *a person is perfect from others' point of view and not from his own*. If a perfect individual occupies the place of an external observer and reflects himself as perfect, he would in this way destroy his perfection. One's own perfection is inaccessible to the 'inner sight' of an individual himself.

Let us look now at all formulae for saints, heroes, philistines, and dissemblers and find the number of perfect individuals in each group. We can consider only those individuals who have a correct image of themselves, *since if an individual has an incorrect image of himself, he cannot be perfect*. (We omit the simple proof of this statement.) Under the above conditions we obtain four types of individuals in each group:

$$(1) \quad a^{a^{a*2}b*1b},$$

$$(2) \quad a^{a^{\bar{a}*2}b*1b},$$

$$(3) \quad a^{a^{a*2}\bar{b}*1b},$$

$$(4) \quad a^{a^{\bar{a}*2}\bar{b}*1b}.$$

By calculating ethical statuses we found that among the four saints three are perfect; among the four heroes only one is perfect, and there are no perfect individuals at all among philistines and dissemblers.

To conclude this chapter we will analyze the peculiarities of saints of the first and second ethical systems. A saint of the first ethical system 'constructed' according to the framework of the formal model corresponds to the classical Christian saint: he is peaceful, modest and feels maximally repentant.

A saint of the second ethical system corresponds to the official portrait of the main Soviet saint—Lenin. The principal work which created Lenin's official portrait was Stalin's speech to the Kremlin Military School

graduates (1924) wherein he emphasizes Lenin's two main features: compromiselessness toward enemies of the Revolution and unusual modesty. Stalin used an oriental metaphor 'mountain eagle' to underscore Lenin's courage and irreconcilability:

Lenin was not just one of the leaders, but a leader of the highest rank, a mountain eagle, who knew no fear in the struggle, and who boldly led the Party forward along the unexplored paths of the Russian revolutionary movement (Stalin, 1953, p. 55).

The second part of this characterization is entitled 'modesty':

I first met Lenin in December 1905 at the Bolshevik conference in Tammerfors (Finland). I was hoping to see the mountain eagle of our Party, the great man, great not only politically, but, if you will, physically, because in my imagination I had pictured Lenin as a giant, stately and imposing. What, then was my disappointment to see a most ordinary-looking man, below average height, in no way, literally in no way, distinguishable from ordinary mortals....

It is accepted as the usual thing for a 'great man' to come late to meetings so that the assembly may await his appearance with bated breath; and then, just before the 'great man' enters, the warning whisper goes up: 'Hush!... Silence!... He's coming.' This ritual did not seem to me superfluous, because it creates an impression, inspires respect. What, then, was my disappointment to learn that Lenin had arrived at the conference before the delegates, had settled himself somewhere in a corner, and was unassumingly carrying on a conversation, a most ordinary conversation with the most ordinary delegates at the conference. I will not conceal from you that at that time this seemed to me to be something of a violation of certain essential rules.

Only later did I realize that this simplicity and modesty, this striving to remain unobserved, or, at least, not to make himself conspicuous and not to emphasize his high position, this feature was one of Lenin's strongest points as the new leader of the new masses, of the simple and ordinary masses of the 'rank and file' of humanity (Stalin, 1953, p. 56).

Compromiselessness and modesty are the first two features (among all eight mentioned by Stalin) in the canonical portrait of Lenin. These features coincide with those of our model prescribed for individuals of the second ethical system with the highest status – saints.

CHAPTER XIII

FORMAL CONNECTIONS BETWEEN MODULES OF INNER STRUCTURES AND INDIVIDUALS

In the previous chapter we analyzed only one structure:

$$a^{a^{a^{*b^{*b}}}}$$

Now we will show that the laws indicated there have a general character.

The simplest individuals who are able to maximize an image of themselves have the following structure:

$$S_1 \equiv \mathbf{a}^{\mathbf{a}^{\mathbf{a}^{*b^{*b}}}},$$

where $\mathbf{a}$ is either a or $\bar{a}$, $\mathbf{b}$ is either b or $\bar{b}$, and $*$ is either '+' or '·'. These individuals contain such important structures for us as an image of oneself:

$$S_2 \equiv \mathbf{a}^{\mathbf{a}^{*b}},$$

and an inner world:

$$S_3 \equiv \mathbf{a}^{\mathbf{a}^{*b^{*b}}}.$$

By substituting the letters $a, \bar{a}, b,$ and $\bar{b}$ for $\mathbf{a}$ and $\mathbf{b}$ and the symbols '+' and '·' for '*', we may generate concrete individuals for which we may find the modules $|S_1|, |S_2|, |S_3|$. Then we may find the mean for all individuals. The method which allows quite easy calculation of the mean ethical statuses (without calculating each of them individually) is given in Appendix 4. Later we shall study the connections between *the mean* ethical statuses.

By varying the symbols '+' and '·' we will obtain the following types of individuals:

- | | |
|----------------|--|
| (1) Saint | $\mathbf{a}^{\mathbf{a}^{\mathbf{a}^{*b^{*b}}}}$, |
| (2) Hero | $\mathbf{a}^{\mathbf{a}^{\mathbf{a}^{*b^{*b}}}}$, |
| (3) Philistine | $\mathbf{a}^{\mathbf{a}^{\mathbf{a}^{*b^{*b}}}}$, |
| (4) Dissembler | $\mathbf{a}^{\mathbf{a}^{\mathbf{a}^{*b^{*b}}}}$, |

where $\mathbf{a}$ and $\mathbf{b}$ are designations correspondingly for $a, \bar{a}$ and $b, \bar{b}$.

TABLE 13.1

Mean values of feeling guilty, suffering, and modules of individuals of different types

Individual	Modules					
	Module of individual's image of himself	Feeling guilty	Module of individual's inner world	Suffering	Module of individual $p = \frac{1}{2}$	Module of individual $0 \leq p \leq 1$
Saint	0.625	0.375	0.312	0.688	0.843	$\frac{11 + 5p}{16}$
Hero	0.875	0.125	0.437	0.563	0.781	$\frac{9 + 7p}{16}$
Philistine	0.625	0.375	0.813	0.187	0.593	$\frac{3 + 13p}{16}$
Dissembler	0.875	0.125	0.936	0.064	0.531	$\frac{1 + 15p}{16}$

The means of various modules, are given in Table 13.1.

We see that the maximal ethical status belongs to the saint, then to the hero, the philistine, and the dissembler. Feeling guilty is maximal in the saint and philistine and minimal in the hero and dissembler. Suffering is maximal in the saint, followed by the hero, the philistine, and the dissembler. When the sign changes from '+' (philistine) to '.' (hero), the mean suffering increases and the mean feeling of guilt decreases. This correlation between suffering and guilt is true not only for mean values, but for each concrete formula as well. Direct examination proves that

$$|\mathbf{a}^{\mathbf{a} \cdot \mathbf{b}} \cdot \mathbf{b}| < |\mathbf{a}^{\mathbf{a} + \mathbf{b}} + \mathbf{b}|$$

and

$$|\mathbf{a}^{\mathbf{a} \cdot \mathbf{b}}| > |\mathbf{a}^{\mathbf{a} + \mathbf{b}}|$$

are true for any substitutions of $\mathbf{a}$ and $\mathbf{b}$ by $a, \bar{a}, b$, and $\bar{b}$.

However, the ethical status of an individual increases only on the average (when '+' changes to '.'), since in some cases it does not change, for example,

$$|a^{a \cdot \bar{b} \cdot b}| = |a^{a \cdot \bar{b} + b}|.$$

But the ethical status of an individual never decreases when the sign ‘+’ changes to ‘.’.

We have analyzed a very important, but only a particular structure of individuals. What happens if the structure of an individual becomes complicated? A special theorem about this is proven in Appendix 4. Its significance consists in the fact that by increasing without limit the number of tiers in the structure of an individual, the limit of the mean modules of the sequence of structures with the symbol ‘+’ is equal to $2 - \sqrt{2} = 0.586$, and the limit for the sequence of structures with the symbol ‘.’ is equal to $\sqrt{3} - 1 = 0.732$.

Therefore, in the transition from the symbol ‘+’ to the symbol ‘.’ the ethical status of the individual increases; the ethical status of his image of himself increases also. But the module of the individual’s inner world decreases, as does the ethical status of the situation.

Direct calculations demonstrate that the mean values of the elements of the inner world given in Table 13.1 *do not depend* on the frequency of the environment’s positive demands. However, individual’s modules depend on this frequency (last column of Table 13.1). It is easy to see that

$$\frac{11 + 5p}{16} \geq \frac{9 + 7p}{16} \geq \frac{3 + 13p}{16} \geq \frac{1 + 15p}{16}$$

and the equality occurs only when $p = 1$.

CHAPTER XIV

INTERACTION. ACTIVITY AND ITS MEASURE

In this chapter we shall consider interactions between individuals of different kinds: heroes with philistines, heroes with saints, saints with philistines, etc. There are two possible cases:

The first case – an individual is involved in certain relationships with his partner apart from his own desire; he only reflects this reality.

The second case – an individual establishes his relationships with his partner by himself.

For example, let a hero be interacting with a philistine. We express their interaction by means of two arrows:

$$\underbrace{a^{a^a \cdot b \cdot b}}_{\text{hero}} \rightleftharpoons \underbrace{b^{b^b \cdot a \cdot a}}_{\text{philistine}}$$

As has been shown in previous chapters, a hero tends to establish the relationships ‘.’, and a philistine tends to establish ‘+’. Let us introduce discrete time: $t_1, t_2, t_3, \dots$. Each individual may act. A hero’s act is the establishment of ‘.’; a philistine’s act is the establishment of ‘+’. At each moment only one individual can act, and individuals act in turn: first a hero, then a philistine, then a hero, and so on.

At the moment t_1 a hero chooses the relationship ‘.’ and establishes it; a philistine reflects this:

$$\underbrace{a^{a^a \cdot b \cdot b}}_{\text{hero}} \rightarrow \underbrace{a^{b^b \cdot a \cdot a}}_{\text{philistine}}$$

At the moment t_2 a philistine chooses the relationship ‘+’ and establishes it; a hero reflects this:

$$\underbrace{a^{a^a \cdot b \cdot b}}_{\text{hero}} \leftarrow \underbrace{a^{b^b \cdot a \cdot a}}_{\text{philistine}}$$

Then a hero chooses ‘.’ and the cycle thus continues.

Thus, each individual may be in one of two phases: in an active phase, where he establishes the relationship with his partner, and in a passive phase, where the relationship is established by his partner.

In the above example the individuals acted in turn. However, we can consider a more general construction. At each moment an individual A may either be in an active phase with probability p , or in a passive phase with probability $1 - p$. Assume that the individuals cannot be in the same phase simultaneously; then an individual B is in an active phase with probability $1 - p$, and in a passive phase with probability p .

We call the probability of being in an active phase *a measure of the activity of an individual* in a given interaction. We call the *active ethical status* of an individual his ethical status in an active phase; we call the *passive ethical status* of an individual his ethical status in a passive phase. The *mean ethical status* of an individual is

$$p\gamma_1 + (1 - p)\gamma_2,$$

where γ_1 is his active ethical status, γ_2 is his passive ethical status, and p is his activity. The greater an individual's activity, the closer is his mean ethical status to his active ethical status; the less an individual's activity, the closer is his mean ethical status to his passive ethical status. It is easy to see that the ethical statuses of saints, heroes, philistines, and dissemblers, considered in previous chapters, are *active ethical statuses*.

Let us analyze an interaction between a saint and a dissembler:

$$\underbrace{a^{a^a + b \cdot b}}_{\text{saint}} \rightleftharpoons \underbrace{b^{b^b \cdot a + a}}_{\text{dissembler}}$$

Let a saint be in an active phase:

$$\underbrace{a^{a^a + b \cdot b}}_{\text{saint}} \rightarrow \underbrace{b^{b^b \cdot a \cdot a}}_{\text{dissembler}}$$

If a dissembler is in an active phase, then

$$\underbrace{a^{a^a + b + b}}_{\text{saint}} \leftarrow \underbrace{b^{b^b \cdot a + a}}_{\text{dissembler}}$$

We suppose that both a saint and a dissembler differ in signs on the second and third tiers *only* in their active phase. In the passive phase the signs on

these tiers are the same. In this kind of interaction the saint's and dissembler's images of themselves do not reflect changes of their relationships.

Note that the passive ethical status of a dissembler in his interaction with a saint is higher than his active ethical status. Therefore, if the measure of a dissembler's activity is equal to zero (absolute passivity), and the measure of a saint's activity is equal to one (absolute activity), then a saint 'increases' the mean ethical status of a dissembler up to the level of a hero. If a saint is absolutely passive, and a dissembler is absolutely active, then a dissembler 'decreases' the saint's mean ethical status to the level of philistine.

Other cases of interaction between individuals of different types can be analyzed analogously.

Our analysis of interaction was made under the assumption that the relationships between individuals depend only on these individuals. However, another kind of interaction is also possible, where relationships between individuals are imposed by some external agent, and the individuals do not act. In this case the measure of their activity is equal to zero.

We emphasize the important distinction between ethical response and ethical act. *Ethical response* is either agreement or refusal to perform an action proposed by the environment. It assumes that an individual has a fixed structure; his image of his relationship with his partner is determined (Chapter V). An *ethical act* is a choice of relationship made by an individual. An ethical act is related to a change of an individual's own structure. The initiator of an ethical act (but not of an ethical response) is always the individual himself.

Let us consider now the mean modules of situations. If there is a society of philistines only where various types of structures of doubt are represented equally and the probabilities of interaction of any two individuals are also equal, what then is the mean ethical status of situations that may appear in such a society and how will the status change if all individuals suddenly become heroes? An 'average' interaction of individuals may be represented as an automaton (Figure 14.1).

For each pair A_i and B_j we can find the module of the situation and then the mean module for all possible pairs. (Appendix 4 describes how to do this analytically.) We can see that when a society of philistines only transforms itself into a society of heroes only, the mean ethical status increases from 0.59 to 0.78, and the mean module of the situation decreases from 0.83 to 0.61. (We shall return to this problem in Chapter XVI.) The mean modules

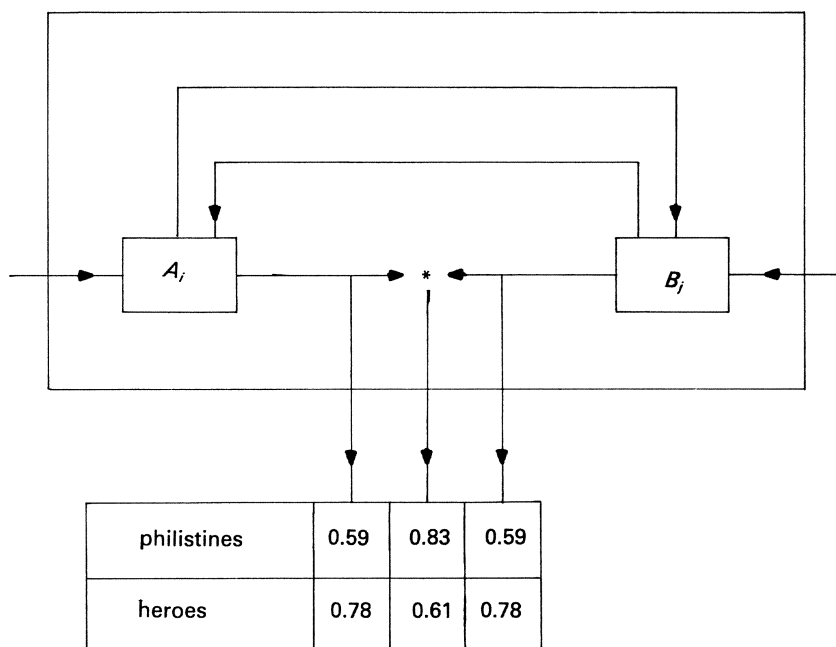


Fig. 14.1. Change of the mean modules of situations while a 'society of philistines' is transforming itself into a 'society of heroes'.

of situations in societies of only one type of individual are as follows (they do not depend on the activity of individuals):

philistines	0.83
dissemblers	0.81
saints	0.66
heroes	0.61

The best situation is that of a philistine society and the worst, of a society of heroes.

We may find also the mean modules of situations appearing in a 'mixed society', i.e., in a society where different types of individuals are in different proportions. The mean modules of a 'mixed society' lie between those of a society of philistines and a society of heroes.

CHAPTER XV

ETHICAL TYPOLOGY IN THE NOVEL *CRIME AND PUNISHMENT* BY DOSTOEVSKY

This novel is important for us, because it is structured around the collision of two ethical systems. The author brings these systems into contact in two ways. First, the two ethical systems collide in the soul of one person. Second, different personae representing the different ethical systems confront each other.

The scheme of our approach is as follows.

- (1) We assume that the laws connecting the various peculiarities of a normative character (as has been indicated in Chapter XII, see Table 12.2) are realized in a socio-cultural system.
- (2) Dostoevsky observes life and reflects it in his novel. Having keen insight into human nature, he *correctly* reflected, we assume, the *real* psychological peculiarities of human personalities.
- (3) We verify our assumption. If it is true, we expect to find in Dostoevsky's novel the laws *we* have stated.

As an analytic convenience we will simplify Table 12.2 by singling out three peculiarities of a character:

1. *Philosophy*. We are interested in how a persona evaluates compromise between ends and means *on the cognitive declarative level*, that is, how he explains his actions. All personae will be separated into two types: some of them evaluate compromise between ends and means as *evil* (I), others as *good* (II).

2. *Sacrifice*. We are interested in the readiness of a persona to violate his own inner comfort in the name of some person, idea, or of self-realization. A persona may be (1) *sacrificial* or (2) *nonsacrificial*.

3. *Behavior*. We will distinguish between personae *inclined toward conflict* and those *inclined toward compromise* with others. Thus, personae can be *aggressive* (*A*) or *nonaggressive* (*$\bar{A}$*).

The new table is shown in Table 15.1.

The number at the bottom right-hand corner of each square indicates the average ethical status for all personae belonging to this square. Note that we can compare statuses only within one ethical system. The ethical status

TABLE 15.1
Connection between 'philosophy' and type of behavior.

Type of sacrifice	Philosophy	
	Ends do not justify means (I)	Ends justify means (II)
Sacrificial (1)	(a) Non-aggressive $\bar{A}$ 0.812	(c) Aggressive A 0.812
Non-sacrificial (2)	(b) Aggressive A 0.562	(d) Non-aggressive $\bar{A}$ 0.562

of (a) is higher than that of (b), and the ethical status of (c) is higher than that of (d), but we cannot compare the status of (a) with that of (d), nor the status of (b) with that of (c).

The relations among philosophy, sacrifice, and behavior, shown in Table 15.1, may also be expressed by the following formulae:

- (a) $(I, 1) \rightarrow \bar{A}$
- (b) $(I, 2) \rightarrow A$
- (c) $(II, 1) \rightarrow A$
- (d) $(II, 2) \rightarrow \bar{A}$

From the formal point of view eight different expressions of this type can exist. Our theory points out that only four of them (those given above) can be realized. *The realization of any one not included in the four given above must be regarded as a disconfirmation of the theory.*

There are five personae in the novel, whose philosophy, inclination to sacrifice, and type of behavior are apparent. They are Raskolnikov, Sonia Marmeladova, Svidrigailov, Luzhin, and Lebeziatnikov.

RASKOLNIKOV

Raskolnikov is 'possessed' by two philosophies. The basis of the novel is the struggle between his two personalities. Each of these personalities has not

only different philosophies, but also different types of behavior. *They are united by the sacrificial nature of the character.* The content of one of the philosophies is described in Raskolnikov's article, and he retells it in his conversation with Porfiry:

In my opinion, if the discoveries of Kepler and Newton, by some combination of circumstances, could not have become known to the world in any other way than by sacrificing the lives of one, or ten, or a hundred or more people, who might have hampered or in some way been obstacles in the path of those discoveries, then Newton would have had the right, or might even have been under an obligation . . . to *remove* those ten or hundred people, so that his discoveries might be revealed to all mankind . . . (Dostoevsky, 1964, p. 249).

. . . . In a word, I deduce that all of them, not only the great ones, but also those who diverge ever so slightly from the beaten track, those, that is, who are just barely capable of saying something new, must by their nature, inevitably be criminals – in a greater or less degree, naturally (Dostoevsky, 1964, p. 250).

Here we see a clear formulation of the principle 'the ends justify the means'.

Raskolnikov's other philosophy is not given in the novel in as straightforward a manner. The key to it may be found in the following dialogue with Porfiry:

'You do believe in the New Jerusalem, then?'

'Yes, I do,' answered Raskolnikov firmly; he said this with his eyes fixed on one spot on the carpet, as they had been all through his long tirade.

'A-and you believe in God? Forgive me for being so inquisitive.'

'Y-yes. Why are you asking all this?'

'You believe in it literally?'

'Yes' (Dostoevsky, 1964, p. 251).

Therefore, it becomes clear that Raskolnikov is a Christian and cannot accept the principle 'ends justify means'.

Each philosophy in Raskolnikov is connected with the corresponding type of behavior. The compromising philosophy leads to an uncompromising action: murder of an old woman.

He pulled the axe out, swung it up with both hands, hardly conscious of what he was doing, and almost mechanically, without putting any force behind it, let the butt-end fall on her head (Dostoevsky, 1964, p. 73).

The uncompromising philosophy leads to a compromising action: to kneel before people.

He knelt in the middle of the square, bowed to the ground, and kissed its filth with pleasure and joy. He raised himself and then bowed down a second time (Dostoevsky, 1964, p. 505).

Both personalities of Raskolnikov are sacrificial. The murder of an old woman is a special *sacrifice* of a person belonging to the second ethical system. There was no utilitarian goal behind this murder.

I longed to kill without casuistry, to kill for my own benefit, and for that alone! I would not lie about it even to myself! I did not commit murder to help my mother – that's rubbish! (Dostoevsky, 1964, p. 401).

I did not commit murder in order to use the profit and power I gained to make myself a benefactor to humanity. Rubbish! I simply murdered; I murdered for myself, for myself alone, and whether I become a benefactor to anybody else, or, like a spider, spend the rest of my life catching everybody in my web and sucking the life-blood out of them, should have been a matter of complete indifference to me at that moment! . . . And, most important, it was not money, so much as something else . . . I know all this now . . . Understand me: perhaps, were I to pursue the same course, I should not commit murder again. I needed to experience something different, something else was pushing me along: what I needed to find out then, and find out as soon as possible, was whether I was a louse like everybody else or a man, whether I was capable of stepping over the barriers or not. Dared I stoop and take power or not? Was I a trembling creature or had I the *right* . . . (Dostoevsky, 1964, p. 402).

The second personality sacrificially repents on bended knee.

He had suddenly remembered Sonia's words: 'Go to the crossroads; bow down before the people, and kiss the ground, because you are guilty before them, and say aloud to all the world "I am a murderer!"' A shudder shook his whole body at the remembrance. He was so crushed by the weight of all the unescapable misery and anxiety of all this time, and especially of these last hours, that he almost flung himself on the possibility of this new, complete, integral sensation. It had come down on him like a clap of thunder; a single spark was kindled in his spirit and suddenly, like a fire, enveloped his whole being. Everything in him softened on the instant and the tears gushed out. He fell to the ground where he stood . . . (Dostoevsky, 1964, p. 505).

Sometimes one, sometimes the other personality seizes Raskolnikov, and their struggle turns into his dashing about between Sonia (personifying the first ethical system) and Svidrigailov (personifying the second).

Meanwhile he was still hurrying to see Svidrigailov; could it mean that he expected something *new* from him, some information, some way of escape? A drowning man clutches at a straw! Could destiny, or some instinct, be bringing them together? Perhaps it was only weariness and despair; perhaps he needed not Svidrigailov but somebody else, and Svidrigailov had merely happened to turn up. Sonia? But what should take him to Sonia now? Should he plead for her tears again? But the thought of Sonia was terrible to him now. She represented an irrevocable sentence, an unchangeable resolution. He must choose between her way and his own.

Especially at this moment he was in no condition to see her. No, would it not be better to try Svidrigailov, and find out what he meant? And he could not help acknowledging to himself that for a long time he really had felt a kind of need of Svidrigailov for some reason (Dostoevsky, 1964, p. 444).

And the entire novel is sustained by these frantic sojourns. The first personality of Raskolnikov can be expressed as

$$(a) \quad (I, 1) \rightarrow \bar{A}$$

the second one as

$$(c) \quad (II, 1) \rightarrow A.$$

SONIA

Sonia can be thought of as the conscience of the first personality of Raskolnikov. Her character is schematic and artistically not very interesting. One can feel the auxiliary function of this persona. She is religious, thus she belongs to the first ethical system; she is sacrificial – sells herself for the sake of her small sisters and her brother. She is an uncomplaining creature, tending to please everyone, i.e., she is compromising in relationships with others. Sonia realizes the expression

$$(a) \quad (I, 1) \rightarrow \bar{A}$$

SVIDRIGAILOV

Svidrigailov is a mysterious villain. Perhaps he is a murderer and seducer, but, he has some largesse and nobility. He had been a benefactor of Sonia and of his young fiancée. Spiritually devastated, he commits suicide. His philosophy is contained in his words:

'It is a long story, Avdotya Romanovna. Here you have... how can I put it?... a kind of theory, the same sort of thing which makes me, for example, consider that a single piece of wrongdoing is allowable, if the chief aim is good. One single evil and a hundred good deeds! (Dostoevsky, 1964, p. 471).

He is aggressive and ready to sacrifice his life, but not to compromise:

Dunya raised the revolver... He took a step forward and a shot rang out. The bullet grazed his scalp and lodged in the wall behind him. He stood still and laughed softly:

'The wasp has stung me! She aims straight at the head . . . What's this? Blood!' He took out his handkerchief to wipe away the blood which flowed in a thin trickle down his right temple. The bullet seemed barely to have grazed the skin . . .

'Well, you missed! Fire again, I will wait,' said Svidrigailov softly, still laughing, if rather sourly. 'If you behave like this, I shall have time to seize you before you have got the revolver cocked!'

Dunechka shuddered, hastily cocked the gun, and again levelled it.

'Leave me alone!' she said desperately. 'I swear I will shoot again . . . I . . . will kill you! . . .'

'Well . . . at three paces it is impossible not to kill. But if you don't kill me . . . then . . . ' His eyes gleamed, and he took two more paces forward.

Dunechka pulled the trigger, and the gun misfired.

'You didn't load properly. It doesn't matter! You have another cap there. Put it right. I'll wait.'

He stood in front of her, two steps away, waiting and gazing at her with wild determination in this sombre eyes, inflamed with passion. Dunya realized that he would rather die than let her go. And . . . she would certainly kill him now, at two paces! . . . (Dostoevsky, 1964 pp. 476–477).

Svidrigailov is similar to the second personality of Raskolnikov: compromising philosophy, uncompromising aggressiveness toward a partner, and ability to sacrifice:

(c) (II, 1) $\rightarrow$ A.

LEBEZIATNIKOV AND LUZHIN

These two personae are presented in a different manner. Here are irony, and grotesqueness,

Naturally, he had soon recognized in Andrey Semenovitch [Lebeziatnikov] a singularly commonplace and silly person (Dostoevsky, 1964, p. 348).

As he [Luzhin] stooped to go through the door, he lifted his hat with exaggerated care to shoulder level, and the very curve of his back seemed to express resentment at being so shockingly insulted (Dostoevsky, 1964, p. 147).

The name Lebeziatnikov itself comes from the verb "lebezit," meaning to cringe or fawn. The name is a characternym, a symbol of mockery. The character and philosophy of Lebeziatnikov are almost exhaustively given in the author's comment:

In spite of these good qualities, Andrey Semenovitch really was rather stupid. He had joined the forces of progress and 'our younger generation' out of conviction. He was one of that countless and multifarious legion of nondescripts, putrescent abortions, and uninformed obstinate fools who instantly and infallibly attach themselves to the most fashionable current idea, with the

immediate effect of vulgarizing it and of turning into a ridiculous caricature any cause they serve, however sincerely (Dostoevsky, 1964, pp. 348–349).

Lebeziatnikov's goal is to construct a new society, for which he is ready to encourage a certain Dunechka

if she thought of taking a lover in the first month of their marriage, or not to have his future children christened, and so on – all in the same strain (Dostoevsky, 1964, p. 349).

In other words good social goals justify the necessary means.

The philosophy of Luzhin shows itself in antipathy to the philosophy of Lebeziatnikov. Luzhin has traditional views on morality. In his own cognition he is formally respectable and therefore has to believe that 'ends do not justify means': 'But what about morals? And, so to say, principles?' (Dostoevsky, 1964, p. 145).

In spite of his mean actions, they do not realize the principle 'ends justify means':

'What are you making so much fuss about?' broke in Raskolnikov unexpectedly. 'It has worked out in accordance with your own theory.'

'How do you mean, in accordance with my theory?'

'Carry to its logical conclusion what you were preaching just now, and it emerges that you can cut people's throats...'

'Oh, come, that's nonsense!' exclaimed Luzhin (Dostoevsky, 1964, p. 145).

Unlike Raskolnikov's, Luzhin's morality and behavior are separated.

Luzhin is aggressive and wicked:

'My dear sir!' exclaimed Luzhin, in furious irritation. He was blushing and confused. 'My dear sir... such a distortion of my meaning! (Dostoevsky, 1964, p. 146).

His annoyance redoubled when he suddenly realized that he ought never to have told Andrey Semenovitch of the result of yesterday's happenings. That was another mistake he had made yesterday, one made in the heat of the moment, when irritation had made him unduly expansive... (Dostoevsky, 1964, p. 345).

And Lebeziatnikov is cowardly and wicked:

'That is all rubbish, and it is slanderous!' flared up Lebeziatnikov, who always flinched from reminders of this incident. 'It wasn't like that at all!' It was different (Dostoevsky, 1964, p. 351).

Lebeziatnikov, in spite of his great kindness of heart, was also finding it difficult to tolerate his fellow lodger and former guardian, Peter Petrovich (Dostoevsky, 1964, p. 349).

Neither is inclined to sacrifice. Here is Luzhin's point of view:

If you love yourself alone, you will conduct your affairs properly, and your cloak will remain whole. Economic truth adds that the more private enterprises are established and the more, so to say, whole cloaks there are in a society, the firmer will be its foundations and the more will be undertaken for the common good (Dostoevsky, 1964, pp. 142–143).

And here is Lebeziatnikov's point of view:

'What does honourable mean? I don't understand such expressions as used to define human activities. "More honourable". "nobler" – that's all rubbish; those are absurdities, antiquated prejudices which I reject! Everything that is *useful* to humanity is honourable. I understand only one word, *useful*! You may snigger if you like, but it is true!' (Dostoevsky, 1964, p. 355).

Neither Luzhin, nor Lebeziatnikov perform sacrificial actions. (The only *honest* action of Lebeziatnikov was when he saved Sonia by unmasking Luzhin, but he *did not risk*.) Therefore, we see that the first ethical system is realized in Luzhin, the second in Lebeziatnikov. Luzhin is aggressive, but Lebeziatnikov is a coward and inclined to compromise, and neither is sacrificial. Luzhin corresponds to the expression

$$(b) \quad (I, 2) \rightarrow A,$$

and Lebeziatnikov to the expression

$$(d) \quad (II, 2) \rightarrow \bar{A}.$$

TABLE 15.2
Typology of the main characters in the novel
Crime and Punishment by Dostoevsky.

Type of sacrifice	Philosophy	
	Ends do not justify means (I)	Ends justify means (II)
Sacrificial (1)	(a) Raskolnikov Sonia	(c) Raskolnikov Svidrigailov
Non-sacrificial (2)	(b) Luzhin	(d) Lebeziatnikov

The general picture of ethical personae realized in the novel is represented in Table 15.2.

That is, only the structures predicted by our theory are realized in the novel.

We have already pointed out differences in Dostoevsky's intonation when he describes Luzhin and Lebeziatnikov. We can also see different intonations in describing other personae. Why does Dostoevsky disdain Lebeziatnikov? And why does respect appear when he speaks about the villain Svidrigailov? (By the way, the illusion of mystery in Svidrigailov is attained by the contrast between the blackness of his soul and the author's evident sympathy, which reaches its culmination in the suicide scene.) Why does the author disdain Luzhin and if he does not respect, at least sympathize with Raskolnikov? We may suppose that the author's (Dostoevsky's) intonation is an *indication of the character's ethical status*. And this indicator does not depend on the type of ethical system. The status of Svidrigailov in the framework of the second ethical system is high: so he should be respected (although he could be hated at the same time). The ethical status of Lebeziatnikov is low; so he should be disdained. The relation respect-disdain is the scale on which ethical status is projected. Thus, the difference between ethical statuses and their comparison may show itself as a special evaluating intonational layer of the novel.

CHAPTER XVI

IDEOLOGY, MORALITY, AND POLITICAL ORGANIZATION

When analyzing Soviet ideology one does not find any advocacy of evil. On the contrary, Soviet ideology is imbued with the *declaration of good*. A man is called upon to be honest, humane, and truthful, to love the Motherland, to be a good family man, to love children. Numerous posters demanding high morals are hung everywhere in schools and universities. On the declaration of good the entire Soviet upbringing and educational process is built.

Let us see how this leads to the realization of the second ethical system. How does Soviet ideology differ from Christian ideology? The answer is: Christian ideology is based on *prohibition of evil*; Soviet ideology is based on *declaration of good*. There is a great difference between the principles 'do not lie' and 'be truthful'.

The fundamental postulates of Christianity are contained in the Old Testament. The postulates regulating human behavior sound as prohibitions.

5:17 Thou shalt not kill.

5:18 Thou shalt not commit adultery.

5:19 Thou shalt not steal.

5:20 Thou shalt not bear false witness against thy neighbor.

5:21 Thou shalt not covet thy neighbor's house; thou shalt not covet thy neighbor's wife, nor his manservant, nor his maidservant, nor his ox, nor his ass, nor anything that is thy neighbor's (*Deuteronomy 5:17–21*).

These are prohibitions against certain means for accomplishing any end. A goal may be anything, but the prohibition is *absolute*. Therefore, any compromise between good and evil, for example, a *murder for the good* of mankind, is evaluated as *evil*. Any confrontation between good and evil, for example, *the abstention from a murder*, even though it may bring good to mankind, is evaluated as *good*. Such an ideological structure hinders a person from perpetrating evil by seeking good. The New Testament, in contrast to the Old Testament, may be considered as a *normative theory of personality*, which contains instructions on right human relationships. Whereas the postulates of the Old Testament through *prohibitions* lead to a

negative evaluation of the compromise between good and evil, the New Testament calls for *compromise relationships* among people:

But I say unto you, Love your enemies, bless them that curse you, do good to them that hate you, and pray for them who despitefully use you, and persecute you (*Matthew 6:44*).

As has been shown in previous chapters this correlation is formally coordinated in the first ethical system.

Now let us analyze the structure of the dictates of communist ideology. The canonical formulation is in the so-called 'Moral Code of a Builder of Communism', which is a part of 'The Program of the Communist Party of the USSR', approved by the XXII Party Congress in 1962. Here is its text:

Devotion to the communist cause; love of the socialist motherland and of the other socialist countries. (1)

Conscientious labor for the good of society – he who does not work, neither shall he eat. (2)

Concern on the part of everyone for the preservation and growth of public wealth. (3)

A high sense of public duty; intolerance of actions harmful to the public interest. (4)

Collectivism and comradely mutual assistance: one for all and all for one. (5)

Humane relations and mutual respect between individuals – man is to man a friend, comrade and brother. (6)

Honesty and truthfulness, moral purity, modesty, and unpretentiousness in social and private life. (7)

Mutual respect in the family, and concern for the upbringing of children. (8)

An uncompromising attitude to injustice, parasitism, dishonesty, careerism, and money-grubbing. (9)

Friendship and brotherhood among all peoples of the U.S.S.R.; intolerance of national and racial hatred. (10)

An uncompromising attitude to the enemies of communism, peace, and the freedom of nations. (11)

Fraternal solidarity with the working people of all countries, and with all peoples. (12), (p. 122).

This code contains both ethical dictates and a 'theory of personality'. Thus, it may be considered as a combined Old and New Testament of Communism. For the sake of convenience we have numbered the statements of the Code.

The statements numbered (1), (2), (3), and (7) give positive features, which 'a builder of communism' has to possess. There are dictates on *what kind of man one should be*:

—faithful to the Motherland (1);

—honest, truthful, morally pure, simple, and modest (7);

and on *what one should do*:

- work honestly for the good of society (2);
- concern about preserving and increasing national welfare (3).

We see that there are prescriptions in the Code on *what to be* and *what to do*, but that there is nothing at all on *what one should not be* and *what one should not do*. There is the declaration of good, but there is no prohibition against perpetrating evil. Therefore, a compromise (between good and evil) as reflected in 'the devotion to communism at the price of renouncing one's father' is evaluated positively, and the confrontation 'a refusal to renounce one's father in order to demonstrate one's faith to communism' is evaluated negatively.

Thus, an ideology based on the propagation of good leads to a positive evaluation of the compromise between good and evil and to a negative evaluation of the confrontation between the two.

In statements (4), (5), (6), (8), (9), (10), (11), and (12) we find the principles of human relationships. These principles subdivide into the principles behind relationships with 'comrades' and 'enemies'. The principles behind relationships with 'comrades' are:

- collectivism and comradely mutual assistance; one is responsible for all and all are responsible for one; (5)
- humane relations and mutual respect (6);
- concern for children (8);
- friendship and brotherhood among all peoples of the U.S.S.R. (10);
- fraternal solidarity with the working people of all countries, and with all peoples (12).

And here are the principles of the relationships with 'enemies':

- intolerance toward *those who* violate public interest (4);
- irreconcilability with *those who* are unfair, *who* tend to live at the expense of others, and *who* have as a main life goal their own career (9);
- intolerance to *those who* show national and racial hatred (10);
- uncompromising attitude to the enemies of communism (11).

The Code openly and obviously requires that one be intolerant toward and irreconcilable with one's enemy.

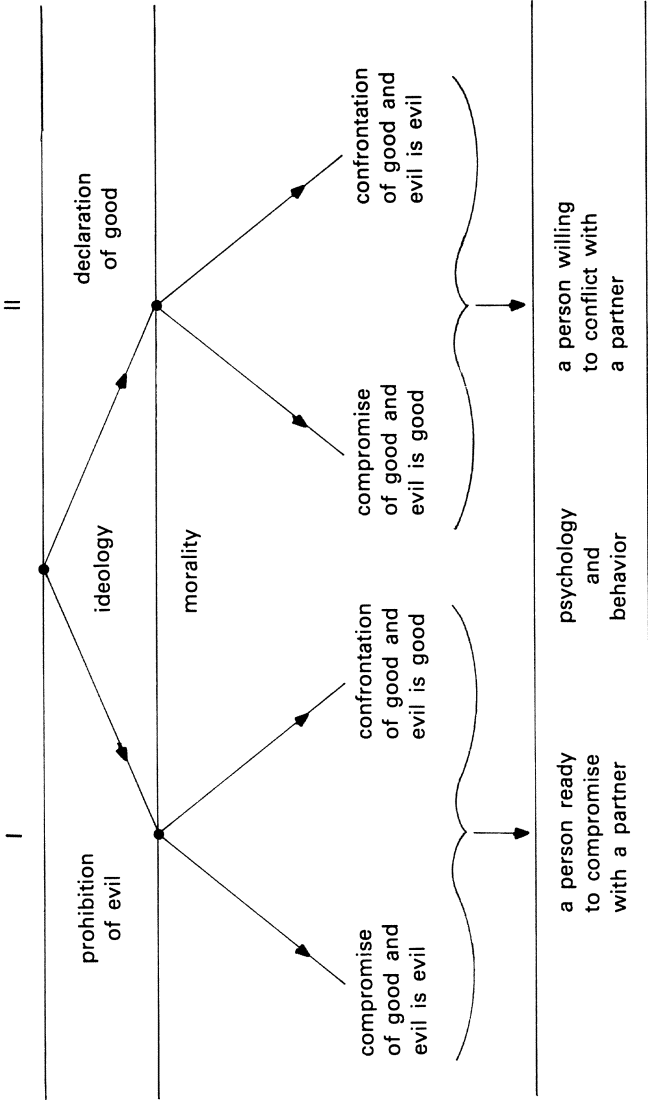


Fig. 16.1. The general scheme of connections between ideology, morality, and behavior.

We can see that the ideology reflected in the 'Moral Code of a Builder of Communism' is formally consistent in the framework of the second ethical system: on the one hand – ethical compromise, on the other hand – compromiseless relationships toward an adversary. The general structure of the relations among ideology, morality, psychology, and behavior can be represented in the scheme shown in Figure 16.1.

Initially an ideology can be formulated in two ways: through the prohibition of evil or through the declaration of good. It would seem that they are just two different formulations of the triumph of good. But this turns out not to be the case:

The prohibition of evil leads to the morality of the first ethical system (compromise between good and evil is evil; confrontation between good and evil is good).

The declaration of good leads to the morality of the second ethical system (compromise between good and evil is good; confrontation between good and evil is evil).

Now we can consider the connection between an ethical system and the political organization to which it leads. Let us start with the first ethical system. The idealized case is that in which all individuals belong to the heroic type. In this instance, every member is willing to compromise with every other. This system is optimal for the appearance of various internal cooperative subsystems. We can suppose that Western culture and economics flourished through the realization of the potential of compromise inherent in the first ethical system. However, additional research is needed here.

But the first ethical system displays itself not only in its heroes but also in its philistines (see Table 12.2). The philistines, in contrast to the heroes, tend toward conflict in their relationships with each other, since in this way they reach inner comfort. Western civilization can be considered as the first ethical system torn by the two polar types. In its different parts sometimes heroic principles prevail, sometimes philistine principles prevail.

Let us consider now the second ethical system. Let us start from the idealized case, where all the individuals belong to the heroic type. We see a system, all members of which are in uncompromising conflict with each other. In this way we gain the possibility of a new view of the causes of such social catastrophies as the Russian revolution and Stalin's terror.

When considering the reasons behind such upheavals, some are of the opinion that people simply lost their moral values. But now we have a special criterion of 'ethical man' and we can verify the above opinion. This

criterion is the ability of an 'ideal' of a particular culture to sacrifice. The ability to sacrifice is incompatible with the absence of morals. In Russia of the 1920–1930's, we would find normative fearless fighters capable of bloody sacrifice. This is an ideal of the second ethical system. Thus, morals were not lost; rather they changed.

We may suppose that in every great culture, both ethical systems are realized, but one of them is dominant, while the other is suppressed. We may suppose also that in Russia at the beginning of our century, the 'official' normative ethical system was the first, while the second, belonged to the 'criminal world' or 'lumpen-proletariat'. The strategy of Bolsheviks in the October Revolution of 1917 consisted in attracting these groups to their side. Thus, it is natural that after the Bolsheviks came to power, the second ethical system became dominant, and further development of Soviet culture can be thought of as arising on its basis. The Russian Revolution expressed itself not only in a change in the socio-economic structures of society, but to a greater extent in the change in ethics.

Let us go back to the hypothetical case where all members of the society are in constant conflict with one another: a picture of a permanent civil war. Something similar could be observed in the Soviet Union in the 1920's and early 1930's. Such universal conflict destroys economic and social institutions. The government can survive only if its structure is able to resist destructive forces. In the Soviet Union the reaction of the government was an unprecedented punitive system, based on the second ethical system. As a result, twenty million people were killed. Thus, the second ethical system contains a terrible trap. The triumph of a hero of the second ethical system leads to bloody anarchy. To resist this anarchy, a government creates a repressive agency which 'without any doubts' crushes the most active part of the population. Stalin wanted to crush the 'proud man of the revolution' (especially inside the Communist Party). Evident affiliation with the heroic type of the second ethical system was often tantamount to the death sentence. Here are Khrushchev's memoirs about the death of Svanidze, Stalin's brother-in-law:

The fate of the brother of Stalin's first wife, Alyosha Svanidze, who was less known to the rank and file of our Party, was also a tragic one. He was a veteran Bolshevik, but Beria, by means of various machinations, made a case to the effect that Svanidze had been placed near Stalin by the German secret service, although he was Stalin's very close friend. And Svanidze was shot. Before he was shot he was told that Stalin had said that if he asked forgiveness he would be pardoned. When Stalin's words were repeated to Svanidze he asked: 'Why should I ask forgiveness? I have not committed any crime.' He was shot. After Svanidze's death, Stalin said:

'See how proud he is, he died but wouldn't ask forgiveness.' It never occurred to Stalin that Svanidze was, above all, an honest man.

And that is how many innocent people died (Khrushchev, 1961, p. 229).

Almost everyone of any significance in Stalin's regime was executed. Blood was running throughout the country. We can see here a strange phenomenon. The main task of interrogators was to obtain a confession of guilt from the 'defendant'. In the overwhelming majority of cases, the defendant did not have anything to confess, so the investigators *themselves* invented the guilt for him, and he only had to accept their version. The Russian writer R. Ivanov-Razumnik (1953), one of the first serious investigators of the Soviet punitive system, described a prison at that time as follows:

During all the time I spent in prison I met only twelve people who had the courage 'not to confess' even after the heaviest interrogations under torture, 'Not to confess' would not have been an act of heroism if torture were not applied. It is a completely different matter 'not to confess' when as a result of an interrogation one had to be sent to the hospital in a dead faint. Such courageous people, I met only twelve in a thousand. The majority of the 'fully confessed' treated these heroes with obvious hostility, although, maybe, with a secret respect. But hostility dominated. 'Ah, in spite of the tortures you still did not confess, and I could not stand it and "confessed". You want to be better than I?' In a forgotten short story, 'Darkness' by Leonid Andreev, such a psychology is expressed briefly in the words addressed by a prostitute to a revolutionary: 'How do you dare to be good, when I am bad?' To tell the truth, however, hostility was never expressed widely. But according to what we heard, in other prisons it reached incredible limits.

Towards the middle of 1938 in our cell, No. 79, there arrived, coming from Tseliabinsk and Sverdlovsk, a 'saboteur', who had spent 3 months in prisons of these two towns. He was, of course, enchanted with the heavenly conditions of our life at Butyrka; he told terrible things about life in the provincial prisons of those towns, where some new prison barracks had been rapidly built. But these barracks were intended only for those who 'already confessed'. Those who still had not done so had remained in prison, and were treated with the provincial methods of coercion. If all these coercive methods did not give satisfactory results, they used to tell the obstinate prisoner: 'Well, tomorrow we will transfer you to barracks No. 1.' This was the barracks of the 'already confessed', famous all around the prison. Its headman was a ferocious Georgian, who fully acknowledged the psychological approach of Andreev's prostitute. The resisting prisoner was transferred to the barracks and it was announced: 'This one does not wish to confess!' – 'Ah, you don't want to confess, and I did confess? You want to be better than I? How dare you be good, when I am bad? Just wait a minute!' – And such ferocious tortures were applied, that those used in the prison seemed mild in comparison. The Georgian started with putting the resisting prisoner up to the neck into a barrel full of urine and faeces and kept him there for 24 hours. If this still did not help – the tortures which I prefer not to remember, started again . . . The fame of barracks No. 1 was so well known that many resisting prisoners preferred 'to confess' after the first threat of being sent to this barracks. The Georgian was a beast and a monster. But the whole barracks of hundreds of people looked at

it, saw it, some maybe helped, some rejoiced at others' misfortune.... This shows to what moral degradation the brutal Soviet regime can lead (Ivanov-Razumnik, 1953, p. 386).

We see that the main goal of this infamous enterprise was to destroy human dignity, to transform the population of the country into philistines of the second ethical system. It is noteworthy that the interrogators destroyed each other also. Here is a quote from V. Bukovsky (1979):

Later I met a man whose story excellently illustrates the mechanics of this accelerating process. It happened in 1947. A colonel in the tank corps, he was arrested on a false denunciation and accused of high treason. There was no evidence of his guilt and none was sought. All that his interrogators wanted from him was new names and new victims. They demanded that he name the people who had recruited him for foreign intelligence, and they tortured him cruelly. He was prepared to sign any idiocy against himself, but not to incriminate his totally innocent friends. At last, feeling that he could no longer hold out against the torture, and fearing to write a false denunciation of someone while he was delirious, he did something that took even himself by surprise.

The interrogation and torture were being carried out by three MGB investigators – one chief investigator and two assistants. One day when they were demanding that he name the enemies who had recruited him, he suddenly pointed his finger at the chief investigator and said: 'You! It was you, you bastard, who recruited me! Don't you remember? On maneuvers, outside Minsk, in 1933, in the birch grove!' 'He's raving, he's gone mad, take him away!' said the chief investigator. 'No, no, why do that?' said the other two with sudden interest. 'It's very interesting, let him go on.' He never saw that chief investigator again – he was probably shot. One of the assistants took his place, the case was quickly brought to an end, and my acquaintance was dispatched to a camp with a 'quarter' (twenty-five years) (Bukovsky, 1979, p. 135).

After the purges of the late 1930's at the head of the Soviet Union were completely obedient representatives of the second ethical system. Here are the memoirs of Marshal A. Vasilevsky, Chief of General Headquarters during the Second World War. Marshal Vasilevsky's father was a Russian orthodox priest. After the revolution both priests and their families were subject to persecution. The reasons and peculiarities of Vasilevsky's successful career became comprehensible from the following dialogue with Stalin:

The conversation did not stop. 'Tell me please', he went on, 'why you and your brothers do not bring material help to your father? As far as I know, one of your brothers is a medical doctor, the second one an agronomist, the third one a commander, a pilot – a person very well off. I think all of you could be of some help to your parents; so that your father would have abandoned the church, not at present, but a long time ago. He needed this as a means of surviving.'

I replied that since 1936 I have severed all connection with my parents. Otherwise, I not only

would not have been accepted as a member of our party, but could not take part in the Labour-Peasant Army, especially in the Army Headquarters. To prove this I quoted the following.

Some time ago for the first time after many years, I received a letter from my father. (In all the official questionnaires which I had filled out, it was pointed out that I had no connection with my parents.) I immediately reported the fact to the Secretary of the Party, who asked me to go on having the same kind of relations with my parents as in the past.

Stalin and the members of the Politburo participating in the lunch were surprised. Stalin told me to get immediately in touch with my parents, to help them, and to inform the party organization of the Headquarters about this permission (Vasilevsky, 1976, p. 104).

For his career's sake, Vasilevsky accepted bad means to reach his ends, i.e., he belongs to the second ethical system. Such behavior was considered both by him and by those around him as right.

We can draw the following conclusions:

- (1) The society where the first ethical system is realized has the more ability for self-organization and unity *the higher* the ethical status of its members.
- (2) The society where the second ethical system is realized has the more ability for self-organization and unity *the lower* the ethical status of its members.

In this case we see the contradiction between the tendency toward the ethical ideal and the processes of social self-organization. The society based on the second ethical system reaches its stability when it destroys the live realization of its own heroic ideals.

CHAPTER XVII

GENERALIZATION. PROOF OF EXISTENCE OF ETHICALLY NONMEASURABLE SITUATIONS

In previous chapters we constructed a theory of ethical interaction of two individuals. The question arises: can this theory be generalized to the case of interaction of three, four, or any finite number of individuals?

We assume that any two individuals in a given group are either in conflict or in union with each other, and the configuration of these relationships can be arbitrary.

Therefore, a situation of interaction may be represented as a graph, the nodes of which are individuals, and the sides of which are the relationships between two individuals (conflict or union). The next task is to represent this graph with the help of a Boolean formula.

Consider the case of interaction between three individuals. The graph of their relationships is a triangle. Union and conflict are marked as '·' and ' + ' on the sides of the triangle. At least two sides are always marked by the same sign. Thus, there is always at least one individual whose relationships with each of his partners are the same. *We will consider these common relationships of an individual with each of the partners as a relationship with a group of two individuals, independent of the relationship between the other two.*

Thus, we *postulate* the possibility of transition from relationships between individuals to relationships between groups: if the relationship between any two individuals belonging to two different groups is the same, then the relationship between their respective groups is the same. We call this *the postulate of enlargement*. The following reasoning is based on this postulate.

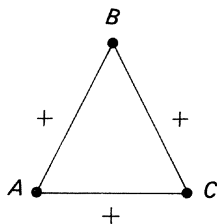


Fig. 17.1 Triangle structure where all relationships are ' + '.

If all three relationships between any two individuals in the group of three are the same, each individual may be considered as being in relationships with a system of two others. Let all sides of a triangle (Figure 17.1) be marked by '+'.

Three different descriptions of the relationships correspond to this triangle:

$$\begin{aligned} A + (B + C), \\ B + (A + C), \\ C + (A + B). \end{aligned}$$

In view of the commutativity and associativity of '+', we can write $A + B + C$ for all three.

For triangle (Figure 17.2) the description is

$$A \cdot B \cdot C.$$

Consider the triangle in Figure 17.3.

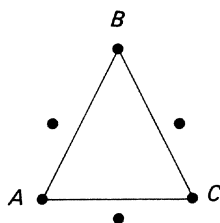


Fig. 17.2. Triangle structure where all relationships are '•'.

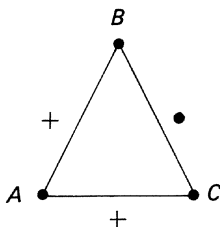


Fig. 17.3. Triangle structure with two '+' and one '•'.

A has the same relationship with his partners, so the description is

$$A + (B \cdot C).$$

For the triangle in Figure 17.4

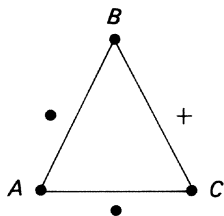
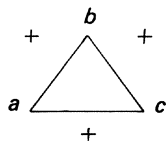
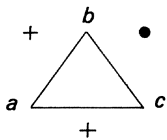


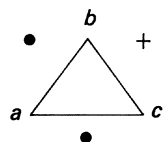
Fig. 17.4. Triangle structure with two '•' and one '+'.
 $S_1 = a + b + c, \quad |S_1| = \frac{7}{8}.$



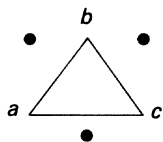
$$S_1 = a + b + c, \quad |S_1| = \frac{7}{8}.$$



$$S_2 = a + (b \cdot c), \quad |S_2| = \frac{5}{8}.$$



$$S_3 = a \cdot (b + c), \quad |S_3| = \frac{3}{8}.$$



$$S_4 = a \cdot b \cdot c, \quad |S_4| = \frac{1}{8}.$$

Fig. 17.5. Triangle structures, simple situations corresponding to them, and their modules.

we can write the formula

$$A \cdot (B + C).$$

These four formulae exhaust all possible configurations of relationships between three individuals. They may be regarded as representations of Boolean functions of three independent variables, a , b , and c . By varying the values of these variables and calculating the value of the given function, we can find the proportion of 1's in all values of the function, that is, we can calculate the ethical status of the situation.

Therefore, for each triangle with marked sides there is a corresponding number. For example, for the simplest individuals a , b , and c there are the ethical statuses shown in Figure 17.5.

The theory of relationships between three individuals may be constructed in the same way as was done for two individuals, but the formula will contain three letters. For example,

$$a^a \cdot (b + c) \cdot (b^a \cdot (b + c) + c^a \cdot (b + c)).$$

The rule for reading such formulae may be constructed similarly to that for the case of two individual interactions, but we will not now consider this problem in greater detail.

Let us consider the case of interaction between four individuals. We would like to use the same idea as for three individuals: to find an individual whose relations with each of the others are the same. Sometimes this is possible.

Individual B on the graph in Figure 17.6 is connected with all others by the sign '+'. We know how to describe the relationships of the other three (A , C , D)

$$D \cdot (A + C).$$

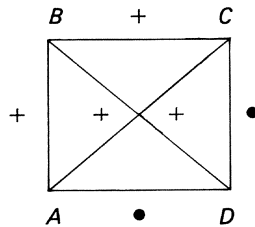


Fig. 17.6. An example of a four-node graph that can be put into correspondence with a formula.

Thus the entire graph may be represented as

$$B + D \cdot (A + C).$$

But a graph may not contain an element connected by the same sign with each of the other three. In this case we try to separate two groups of two elements in such a way that the elements of one group are connected with each of the elements of the other group by the same sign. For example, in the graph in Figure 17.7 each element of the group (A, D) is connected with each element of the group (B, C) by the sign $\cdot$, and between themselves the elements of each group are connected by the sign $+$. Therefore, we may represent this system with the formula

$$(A + D) \cdot (B + C).$$

However, there might exist cases where neither elements nor groups of elements are connected with each of the others with the same sign. Figure 17.8 is an example.

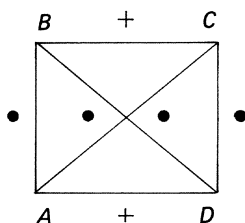


Fig. 17.7. Another example of a four-node graph that can be put into correspondence with a formula.

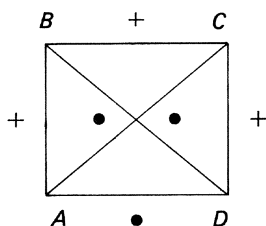


Fig. 17.8. An example of a four-node graph that cannot be put into correspondence with a formula.

Here there is no subset of the elements, each one of which is connected with each of the others with the same sign. *And we are not able to put this graph into correspondence with a Boolean formula, using the postulate of enlargement and therefore, with a certain number.*

What is the interpretation of the fact that some graphs cannot be represented by Boolean functions? (And no other principle of representation of a graph with the help of Boolean functions is apparent.) The most natural answer is that *the situations represented by such graphs are ethically non-measurable*. These situations cannot be evaluated in a binary system of relationships; they are too complex for this. We cannot say, for example, that a given situation is more or less difficult than another. The binary technique of our ethical evaluations is limited. And if we cannot find a Boolean function for such situations, enabling us to evaluate them, an individual involved in such situations cannot evaluate them either.

Therefore, such situations cannot, in principle, be ethically reflected by individuals involved in them.

Let us consider now the problem of ethical measurability for a situation with n individuals, where $n \geq 4$. Each situation can be rendered as a set of n elements with given relationships of two types. We exclude reflexive relationships, i.e., our graphs do not contain loops. Moreover, we assume that the relationships are symmetrical, i.e., $aRb \leftrightarrow bRa$. If each element of a subset of nodes is connected with each element of another subset with a relationship R , we consider these two subsets as being in relationships R . In view of associativity,

$$AR(BRC) = (ARB)RC;$$

therefore, we may write

$$ARBRC.$$

Henceforth, we will analyze graphs, any two elements of which are in one of two different relationships: R or $\bar{R}$.

The expression 'graph G ' we will understand as 'graph with a set of apices G ', and the expression 'a set G ' would mean a set of apices of a graph.

We shall call a nucleus of graph G a subset H such that $HR(G - H)$, where $G - H$ is a complementary nonempty subset. It is clear that $G - H$ is also a nucleus. We call a nucleus minimal if there is no such subset $H_1 \subset H$ which would be a nucleus of G . We call a graph simple in relation to R if it does not have a nucleus in this relation.

It is clear that if graph G has a nucleus H (in relation to R), then there is a subset $H_{\min} \subseteq H$ which is a minimal nucleus of G (in relation to R), because the number of elements in G is finite.

THEOREM 17.1. *A minimal nucleus H of graph G in relation to R is a simple graph in this relation.*

Proof. We shall call an element external in relation to subset H of set G if this element belongs to G and does not belong to H . Let h be a nucleus of H in relation to R ; let $\alpha \in h$ and $\beta \in G - h$. Two cases are possible: first — $\beta \in H - h$, then $\alpha R \beta$, since α belongs to a nucleus in H ; second — $\beta \in G - H$; then $\alpha R \beta$, since α belongs to a nucleus in G . Therefore, $\alpha R \beta$ for any external element β , and h is a nucleus of G and at the same time it is a subset H . But this is impossible, since H is a minimal nucleus. Therefore, H is a simple graph in relation to R . $\square$

THEOREM 17.2. *The intersection of two different minimal nuclei H and P in relation to R of set G is empty.*

Proof. Let us consider any element α belonging to the intersection and any element β out of it. If $\beta \in H$, then β is an element external to all elements belonging to P , and therefore, $\alpha R \beta$. If $\beta \in P$, then β is an element external to all elements belonging to H , and therefore, $\alpha R \beta$. If β belongs to the complementary set for joint H and P , then $\alpha R \beta$, since α belongs to at least one minimal nucleus. Therefore, for any element external to the intersection of H and P $\alpha R \beta$ is true. Thus, the intersection of H and P is a nucleus. But minimal nuclei are different, therefore their intersection is a subset of at least one of them and at the same time it is a nucleus of G . This is impossible according to the definition of a minimal nucleus. Therefore minimal nuclei do not intersect. $\square$

THEOREM 17.3. *If graph G is not simple in relation to R , each of its nodes belongs to some minimal nucleus.*

Proof. Let us prove that graph G may be represented as a group of minimal nuclei. Since G is not simple in relation to R , it contains a finite number K of minimal nuclei. The number K is finite because the number of G 's nodes is finite and the minimal nuclei do not intersect.

Let us suppose now that graph G cannot be represented as a group of minimal nuclei. Then graph G may be represented as a group of K minimal nuclei and a set S which is not a minimal nucleus and does not intersect with any minimal nucleus. Set S is a nonminimal nucleus, since S is a

supplement to the group of minimal nuclei which are a nucleus. But a non-minimal nucleus has a minimal nucleus among its subsets. Thus graph G contains $K + 1$ minimal nuclei, and this contradicts the statement that graph G contains exactly K minimal nuclei. Therefore, our assumption that graph G cannot be represented as a group of minimal nuclei is wrong. The minimal nuclei do not intersect, and therefore, each node of graph G belongs to one and only one minimal nucleus. $\square$

THEOREM 17.4. *If graph G is not simple in relation to R , it is simple in relation to $\bar{R}$.*

Proof. Let us suppose that a graph is not simple in relation to both R and $\bar{R}$. Then, there are two true expressions: $HR(G - H)$ and $\bar{P}\bar{R}(G - P)$, where H is a nucleus in relation to R , and P is a nucleus in relation to $\bar{R}$. Let $\alpha \in H$. This element belongs either to P or to $G - P$. We shall consider that it belongs to P . Let $\beta \in G - H$. Then $\alpha R \beta$. Element β also belongs either to P or to $G - P$. Let $\beta \in G - P$, then $\alpha \bar{R} \beta$. Thus $\alpha \bar{R} \beta$ and $\alpha R \beta$ are true at the same time. This is impossible. Therefore all elements $\beta \in G - H$ belongs to P , and each element $\alpha_1 \in G - P$ must belong to H . Let us pick an arbitrary $\alpha_1 \in G - P$ and obtain $\alpha_1 \bar{R} \beta$. Since $\alpha_1 \in H$ and $\beta \in G - H$, then $\alpha_1 R \beta$, $\alpha_1 \bar{R} \beta$ and $\alpha_1 R \beta$ cannot be true at the same time; therefore, our initial assumption is incorrect. $\square$

It follows from the theorems above that each graph that is non-simple in relation to R may be represented as a juncture of nonintersecting minimal nuclei

$$H_1 R H_2 R \dots R H_k \quad k \geq 2, \quad (17.1)$$

and such a representation is unique except for permutations of $H_i (i = 1, 2, \dots, K)$. Each H_i is simple in relation to R , but may not be simple in relation to $\bar{R}$. We will call a graph absolutely simple if it is simple in relation to R and to $\bar{R}$ at the same time.

For each H_i which is not simple in relation to $\bar{R}$ we find an expansion (17.1), and for each of the elements of the obtained expansions we reiterate this procedure until all elements are absolutely simple graphs. Let us note that each node of graph G is an absolutely simple graph. Since the initial graph G is finite, this procedure is finite (Figure. 17.9).

We will call graph G ethically measurable if *all* absolutely simple graphs

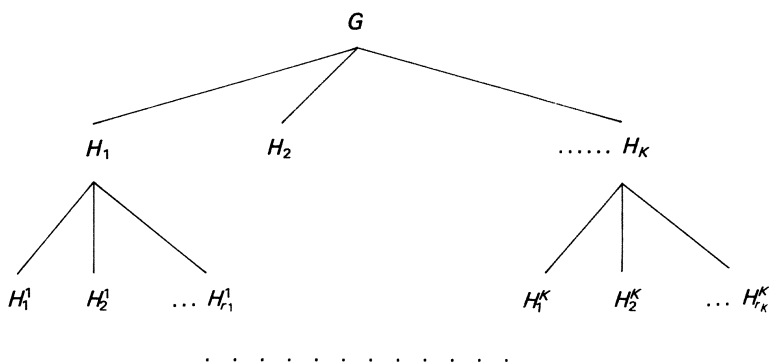


Fig. 17.9. Scheme of a procedure of decomposition of a graph.

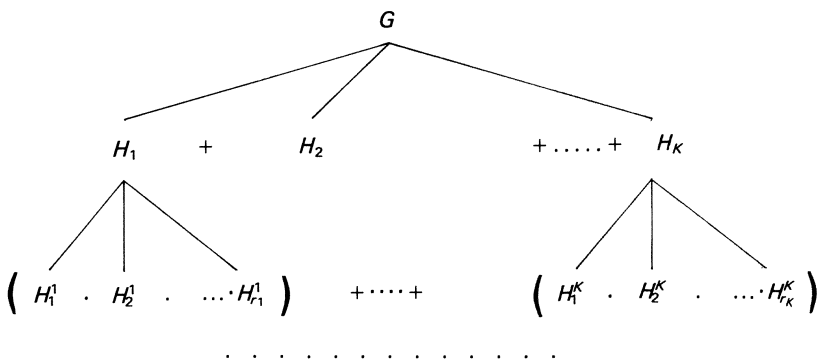


Fig. 17.10. Scheme of a procedure for generating a formula.

appearing in the above procedure are the elements of G . In the opposite case we will call graph G ethically non-measurable. It is easy to see that this definition is a generalization of a procedure for finding Boolean functions, the procedure which we have considered for the case of $n = 4$. If, for example, instead of the symbol R we write the sign '+', then instead of the symbol $\bar{R}$ we have to write the sign ' $\cdot$ ', and the above procedure would generate a formula shown in Figure 17.10.

And the procedure can be completed only if all absolutely simple graphs appearing in it are the nodes of graph G . According to the proven

uniqueness of the expansion, such a formula (if it exists for a given graph G) is unique. It is easy to see that a necessary and sufficient condition for a graph ($n \geq 4$) to be absolutely simple is its connection in relation to both R and $\bar{R}$ at the same time. The existence of an infinite number of non-measurable graphs follows from this. It is easy to demonstrate that the number of measurable graphs is also infinite. In the case of the graph with one type of relationship (for example, '+') the generating procedure generates the formula already at the very first level (Figure 17.11), where a_i is a node of the graph ($i = 1, 2, \dots, K$).

In sociology there is the concept of a 'balanced graph'. A graph is said to be balanced if the set of all its nodes may be divided into two non-intersecting sets in such a way that any 'positive' side connects nodes of the same set, and any 'negative' side connects nodes of the different sets. These kinds of graphs, along with corresponding situations, were studied by Heider (1946) and Cartwright and Harary (1956).

We call a graph complete if each two nodes of it are connected by a side. It is easy to see that every complete balanced graph is ethically measurable. The converse, however, is not true. Consider the graph shown in Figure 17.12.

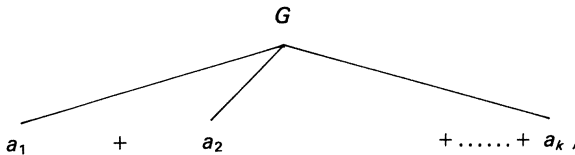


Fig. 17.11. Scheme of a procedure for generating a formula in the case where all graph connections are '+'.

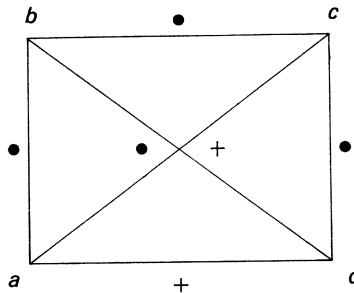


Fig. 17.12. Example of an ethically measurable unbalanced graph.

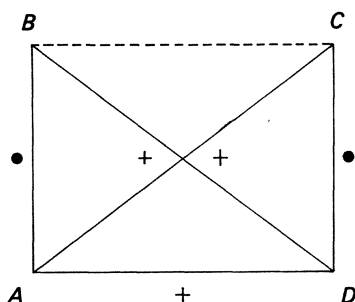


Fig. 17.13. Example of a graph that is ethically measurable if BC is '+' and ethically nonmeasurable if BC is '•'.

This graph is not balanced (independently of interpretation of signs '•' and '+'); however, it is ethically measurable, and its Boolean function is:

$$b \cdot (a + (c \cdot d)).$$

The concept of balance can be generalized. A graph is called 'grouped' if the set of its nodes can be divided into several subsets (called groups) in such a way that every 'positive' side connects nodes inside the same subset and every 'negative' side connects nodes of different subsets (Davis, 1967, Kemeny and Snell, 1962). Each complete grouped graph is ethically measurable; the converse is not true.

The existence of ethically non-measurable situations must radically change the character of ethical decisions in groups of more than three members, in comparison with the character of ethical decisions in groups of two or three members. Consider the situation indicated in Figure 17.13.

Suppose individual B can himself establish his relationship with C . What kind of relationship would he choose? In Chapter XI we demonstrated that an individual may be guided by the principle of the maximization of the ethical status of his own image, by choosing a relationship from the two alternatives: '+' and '•'. In the situation above individual B apparently does not have alternatives: he has to choose '+', since only in this case would he be able to 'calculate' his ethical status. If he chooses '•', the situation is ethically non-measurable, and thus his ethical status is non-calculable.

Consider the following situation of five individuals (Figure 17.14).

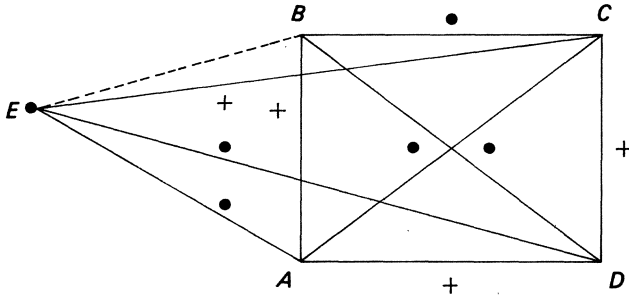


Fig. 17.14. Example of an ethically non-measurable graph (independently of relationship EB).

Suppose individual E has to establish his relationships with B . Both when E chooses '+' and '-', the situations become ethically non-measurable, therefore, E cannot choose his relationship with B on the basis of ethical advantage. Perhaps in similar cases an individual tends to change relationships among other members of the group only with the purpose of reaching an ethically measurable situation that allows him a base for ethical orientation.

While analyzing ethical measurability, we found limitations similar to the well-known fact in mathematical logic that in formal calculi there may exist 'correctly' formulated propositions which are not decidable within the framework of a given calculus. In other words, we cannot say whether these theorems are true or false.

In our case we have 'correctly' constructed situations of ethical interaction. We can make an ethical evaluation of the actions of each individual (he behaves 'well' or 'poorly') but, in principle, we cannot ethically evaluate the whole situation: we cannot answer the question of whether it is 'good' or 'bad'.

CONCLUSION

THE PROBLEM OF SUBSTANTIATING THE INITIAL AXIOMS

The model described in this book is constructed on axioms. First we formulated several fundamental statements: then we deduced their consequences. However, it is important to note that our axioms represent a logical, rather than an empirical, basis.

It is necessary to make a distinction between logical substantiation of the model and its experimental and empirical verification. It is natural to choose those statements as axioms, from which consequences may be deduced with the greatest logical simplicity. But on the other hand, among these statements, some may turn out to be difficult to verify or even wrong. In such a case we have to choose consequences which conveniently lend themselves to verification, to verify these consequences, and then to prove that the given consequences logically lead to the given set of axioms. For example, the axioms of geometry, it might be noted, were chosen by Euclid for their functional significance in constructing his theory, and not for their ease of verification. For example, Euclid's famous fifth postulate, on the non-convergence of parallel lines, cannot be directly verified empirically, but it is necessary in the deductive scheme of his theory.

Let us consider the axioms described in Chapter II. Some of them can be verified directly, but others are defined in such a way as to obtain a mathematical object convenient for analysis and development of the theory. Here an intriguing question arises: why were those particular definitions chosen? We also have empirical arguments in favor of our fifth axiom, evil aware of evil becomes good. Systems of religious thought have special procedures for searching out and recognizing evil in oneself and others. Such recognition is a precondition for overcoming evil in one's own activity.

However, Axioms (6), (7) and (8) cannot be grounded in empirical arguments. Maybe it is possible to find a function more convenient for modelling the process of awareness than the formal implication

$$a^b \stackrel{\text{def}}{=} b \rightarrow a?$$

To support this particular expression we analyze the consequences that result if we reject them. We shall demonstrate that by any substitutions for

this expression we obtain consequences empirically unacceptable.

Let us analyze the set of all possible non-commutative algebras that have as elements

- (1) the symbols 1 and 0;
- (2) a binary operation which puts any pair of symbols (1, 1), (0, 1), (1, 0), (0, 0) into correspondence with 1 or 0;
- (3) the unary operation 'bar': $\bar{1} = 0, \bar{0} = 1$.

Suppose that a pair (a, b) corresponds to c . We write this as $a^b = c$. Let us designate the module of this expression as the proportion of times c is equal to 1, when a and b independently have values 1 and 0.

We call an algebra 'appropriate' if the two following expressions are true at the same time:

$$|a^a| > |a^{\bar{a}}|, \quad (C1)$$

$$|a^{a\bar{a}}| > |a^{\bar{a}a}|. \quad (C2)$$

Inequality (C1) reflects the statement that an individual with a correct evaluation of himself has a higher ethical status than one with an incorrect evaluation of himself. Inequality (C2) reflects the statement that an individual with a correct evaluation of himself and who doubts the correctness of his own image of himself has a higher ethical status than one with an incorrect evaluation of himself and no doubts about the correctness of his own image of himself. It seems natural to require that the theory have these as true statements.

Now we will look for appropriate algebras. The entire set of such algebras are given in Table C.1.

TABLE C.1
Values of combinations depending on number of algebra.

Combination	Number of algebra															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1^1	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0
0^0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0
1^0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0^1	1	1	1	1	0	0	0	0	1	1	1	1	0	0	0	0

TABLE C.2
Values of modules of formulae depending on number of algebra in
Table C.1.

Formulae	Number of algebra															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
a^a	1	$\frac{1}{2}$	$\frac{1}{2}$	0	1	$\frac{1}{2}$	$\frac{1}{2}$	0	1	$\frac{1}{2}$	$\frac{1}{2}$	0	1	$\frac{1}{2}$	$\frac{1}{2}$	0
$a^{\bar{a}}$	1	1	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	0
$a^{a\bar{a}}$	1	1	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	0
$a^{\bar{a}\bar{a}}$	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	0	$\frac{1}{2}$	0

The table for corresponding values of

$$|a^a|, |a^{\bar{a}}|, |a^{a\bar{a}}| \text{ and } |a^{\bar{a}\bar{a}}|$$

in each algebra is shown in Table C.2.

We can see that only in Algebra 5 both (C1) and (C2) are true at the same time. This algebra is defined by the expression $1^1 = 1$, $0^0 = 1$, $1^0 = 1$, $0^1 = 0$. This is the very interpretation which was used in our work and corresponds to the Boolean operation of implication. Thus, we have proved that only that implication is appropriate to construct the theory in which both (C1) and (C2) are true and, thus, we have grounded the choice of Axioms (6), (7), (8), which cannot be verified directly.

Let us now analyze other axioms. The different formulations of the first two axioms led us to distinguish two different ethical systems. We have empirical and experimental arguments justifying the use of the axioms.

However, Axioms (3) and (4) cannot be grounded in empirical arguments. Why do we consider the following statements true?

- (3) The confrontation of good and good as well as the compromise of good and good is good.
- (4) The confrontation of evil and evil as well as the compromise of evil and evil is evil.

To support these statement we have to analyze the consequences that result if we reject them. We shall demonstrate now that by any substitutions for these statements we receive consequences empirically unacceptable.

We add to Algebra 5 (introduced above) a commutative binary operation

$*$, which puts any pair of symbols (1, 1), (1, 0), (0, 0) into correspondence with 1 or 0.

It seems natural that an individual with a correct evaluation of himself should have a higher ethical status than another individual equal to him in all respects, but with an incorrect evaluation of himself. Let us designate by ' $*$ ' relationships between individuals. Then, if the expression

$$|a^{a*b}| > |a^{\bar{a}*b}| \quad (C3)$$

is true we will call a theory acceptable, if it is untrue, the theory is unacceptable.

The entire set of operations can be represented by Table C.3.

Now for each algebra we will calculate modules of the expressions

$$a^{a*b} \quad \text{and} \quad a^{\bar{a}*b}$$

and find among them those where expression (C3) is true. All results are summarized in Table C.4.

TABLE C.3
Values of combinations depending on number of operation.

Combination	Number of operation							
	1	2	3	4	5	6	7	8
1*1	1	1	1	0	0	0	1	0
1*0	1	1	0	1	0	1	0	0
0*0	1	0	1	1	1	0	0	0

TABLE C.4
Values of modules of formulae depending on number of operation in Table C.3.

Formulae	Number of operation							
	1	2	3	4	5	6	7	8
a^{a*b}	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	1	1
$a^{\bar{a}*b}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	1	$\frac{3}{4}$	$\frac{3}{4}$	1

Expression (C3) is true only for two operations: (2) and (7). We will analyze them.

$$\begin{array}{ll}
 (2) & 1 * 1 = 1 \\
 & 1 * 0 = 1 \\
 & 0 * 0 = 0
 \end{array}
 \quad
 \begin{array}{ll}
 (7) & 1 * 1 = 1 \\
 & 1 * 0 = 0 \\
 & 0 * 0 = 0.
 \end{array}$$

It is easy to see that in these algebras operation ‘*’ coincides with the operations ‘+’ and ‘.’ of Boolean algebra.

$$\begin{array}{ll}
 (2) & 1 + 1 = 1 \\
 & 1 + 0 = 1 \\
 & 0 + 0 = 0.
 \end{array}
 \quad
 \begin{array}{ll}
 (7) & 1 \cdot 1 = 1 \\
 & 1 \cdot 0 = 0 \\
 & 0 \cdot 0 = 0.
 \end{array}$$

Therefore, we have proven that only for operations of *Boolean* algebra is expression (C3) true. And in the framework of our considerations the acceptable theory is a theory based on Boolean algebra. Thus, striving to make a theory acceptable leads us to the necessity of using Axioms (3) and (4), which are not directly verifiable.

Thus we have proved that the three inequalities (C1), (C2), and (C3) are true only if the process of reflexion is modeled with implication, and the process of individuals’ interaction is modeled with operations of Boolean algebra. This is the substantiation of our initial axioms.

EPILOGUE

The results of this work indicate that two different ethical systems exist, and that one of them is realized on the whole in contemporary Western society and the second one in contemporary Soviet society.

We have demonstrated that the first ethical system is quite definitely connected with Christianity, so we have good reasons for the assumption that the dominant ethical system in any Christian society is the first ethical system.

But what about the second ethical system? Is its appearance in Soviet society a unique phenomenon? Has it existed in history before? Let us analyze a European of pre-Christian times as he is described in the epos of Northern peoples. His character has very little in common with the Christian examples which followed. A true hero of that time is imprudently courageous and cruel. The more unparalleled the hero's death, the more terrible, unprecedented, and unusual its circumstances, the more significant the hero. There one may find a man who demands the cruel death of his brother and then is killed himself; a father who in his enemy recognizes his son but nevertheless kills him; a woman who kills her lover, then herself (A. Gurevich, 1979). All these personages are heroes.

Here is the plot of an ancient heroic story 'Lay of Greenland About Atly' (Atakvida): Two brothers own an enormous hidden treasure. A neighboring chief, Atly, wants to seize this treasure and invites the brothers to his castle. They are warned about Atly by his wife, who happens to be their sister, but they nevertheless go to Atly, since true heroes cannot intentionally avoid danger. Atly captures the brothers and threatens to kill them if they do not tell the whereabouts of the treasure. Then the older brother promises to reveal the secret place if he receives his brother's heart on a plate, and when he sees it, he says:

Atly, you will not find
happiness,
for you will not see
our treasure.
Only I alone,
since my brother is dead,

know where
the treasure of Nifflungue is hidden.

Atly orders the brother thrown into a pit of poisonous snakes, where he dies.

Here one sees a hero who does not compromise at all; moreover, he tends to make the situation as difficult as possible. This is an heroic individual of the second ethical system who performs a ritual of sacrificial conflict. We have here an example of a European pre-Christian society in which the second ethical system appears to be realized.

Gurevich, who analyzed the 'Lay of Greenland about Atly', points out that contemporary investigators are confused in evaluating the older brother's cruel act as an act of a positive hero. Gurevich offers to explain this act from the point of view of the psychology of ancient man, which first has to be reconstructed. However, such psychology exists in modern times.

In 1936 a famous Soviet poet, Konstantin Simonov, wrote 'The Story About Hidden Arms'. He indicated that the plot was borrowed from one R. H. Sander. The poem immediately became a Soviet classic; youths throughout the Soviet Union learned it by heart. This is its plot:

Two Red fighters, who know where arms are hidden, are captured by enemies. They are tortured, but keep the secret. Then the older of the two promises to reveal the hiding place if he witnesses the death of the younger man. A boss by phone orders the younger fighter to be shot, and having seen the execution, the older man says:

You want to hear about the arms,
Do you not senor?
We hid them long ago;
Only we two knew where they are.
My friend could give out
Under torture.
I bound forever the tongue
Of him who might betray the place.
He is dead and cannot speak
I am alive and I am silent.

This is the same 'Lay About Atly' returned to life a thousand years later in another place among another people – after these people lost Christianity.

The simultaneous existence of two ethical systems is behind the present ethical split of the world. A man of the first ethical system countervails to an archaic man of the second ethical system, as it was a thousand years ago.

APPENDIX 1

CONSTRUCTION OF JUDGMENTS ABOUT THE CORRECTNESS OF IMAGES AND JUDGMENTS

1. RECURSIVE STRUCTURE OF AN INDIVIDUAL

Consider a formal individual $A \equiv a^W$, where W is equal to A_1 or to B_1 or to $A_1 * B_1$. Suppose we have interpreted W , then individual A can also interpret W . This formal individual is a 'reality for an external observer'. But he can be linked to an external observer, since individuals in his inner world are a 'reality for him'; this reality may differ from an external observer's reality. In their turn, individuals who are the images of a given individual may have their own inner world with other individuals as images of images. Such a hierarchy can be of arbitrary depth.

2. IMAGES AND JUDGMENTS

Our model is constructed in such a way that an external observer can make judgments about the correctness or incorrectness of individuals' images. We analyze a case of interaction between two individuals: A and B . To make a judgment about the correctness of 'an image of A ' belonging to some individual, the external observer has to compare 'an image of A ' with an A who is 'real for the external observer'. If their roots are the same, the external observer makes the judgment: 'the image is correct'. If the roots are different, the external observer makes the judgment: 'the image is incorrect'. To make a judgment about the correctness of an image of the relationship between individuals, the external observer compares this image with the relationship between 'real' (from his point of view) individuals.

The judgment 'the image is correct' we shall designate by t (true); we insert this letter into a formula under a symbol standing for a given image. The judgment 'the image is incorrect' we shall designate by f (false). Consider the following situation:

$$W \equiv a^{\bar{a}+b} + b^b \cdot \bar{a}$$

From an external observer's point of view, A has an incorrect image of himself, a correct image of his partner, and a correct image of their relationship; B has a correct image of himself, an incorrect image of his

partner, and an incorrect image of their relationships. These judgments by an external observer may be expressed as:

$$W \equiv a \begin{array}{cc} \bar{a} + b \\ f \quad t \quad t \end{array} + b \begin{array}{cc} b \cdot \bar{a} \\ t \quad f \quad f \end{array}$$

Let the external observer regard individual A 'outside the context' of his relationship with B :

$$a^{\bar{a}+b}.$$

In this case the external observer can make the judgment that A has an incorrect image of himself, but the external observer cannot make a judgment about the correctness of A 's image of his partner or of A 's image of his relationships with his partner, because he has no 'reality' with which to compare these images. Thus, we have an expression:

$$\begin{array}{c} \bar{a} + b \\ a \quad f \end{array}$$

3. JUDGMENT ABOUT JUDGMENTS

It is natural to assume that an individual can also make judgments about the correctness of images. Let A see the situation W analyzed above:

$$W_1 \equiv a \begin{array}{cc} \bar{a} + b \\ a \quad f \quad t \quad t \end{array} + b \begin{array}{cc} b \cdot \bar{a} \\ t \quad f \quad f \end{array} + b \begin{array}{cc} b \cdot \bar{a} \\ b \cdot \bar{a} \end{array}.$$

In this formula, judgments about correctness are made by A . Now, the external observer has to make a judgment about the correctness of A 's images and *in addition* about the correctness of A 's judgments. To do so, the external observer has to compare the images of the third tier with the 'real individuals' whose roots are in the first tier and judge their correctness from his point of view; this allows him to agree or disagree with A 's judgments. If the external observer agrees with A 's judgment, we write t underneath; if the external observer disagrees with A 's judgment, we write f . In the above example the external observer agrees with all A 's judgments:

$$W_1 \equiv a \begin{array}{cc} \bar{a} + b \\ a \quad f \quad t \quad t \\ t \quad t \quad t \quad t \end{array} + b \begin{array}{cc} b \cdot \bar{a} \\ t \quad f \quad f \\ t \quad t \quad t \end{array} + b \begin{array}{cc} b \cdot \bar{a} \\ b \cdot \bar{a} \\ t \quad t \end{array}.$$

Another example:

$$W_2 \equiv a^{a^b + \bar{b}} + b.$$

First we write down the judgments made by A . A 's inner world is

$$a^b + \bar{b}.$$

Individual $B \equiv \bar{b}$ is a reality for A , and A sees that his image of B differs from 'real B ', thus A makes the judgment:

$$\begin{array}{c} b \\ a \ f + \bar{b}. \end{array}$$

The external observer sees this judgment and disagrees with it (since for him 'real B ' is $B \equiv b$):

$$\begin{array}{c} b \\ a \ f + \bar{b} \\ a \ f \quad + b. \end{array}$$

If individual A is given 'outside the context' of his relationship with B :

$$a^{a^b + \bar{b}},$$

then he makes the judgment:

$$\begin{array}{c} b \\ a \ f + \bar{b} \\ a \end{array}$$

but the external observer cannot make a judgment about this judgment, because of the absence of 'reality'.

4. THE STEPS IN THE GENERAL RULE FOR MAKING JUDGMENTS (FOR THREE-TIER FORMULAE)

1. Let A (or B) be an individual with a root on the second tier. If we can find on the first tier a root of the same name, we shall write t underneath (in the

APPENDIX 2

ETHICAL SYSTEMS AND MULTIVALUED LOGICS

In previous chapters we have considered modules of functions in Boolean algebra of two elements: 0 and 1. In this appendix we are going to consider the peculiarities of modules of functions in algebras of an arbitrary number of elements (for a class of functions corresponding to formulae written in a two-letter alphabet).

The main result is that the relation of order for modules can be found exhaustively in an algebra of three elements. This means that a scale in Figure A2.1 gives a final evaluation for the equality or inequality of ethical statuses of individuals. We would not obtain any new information by dividing the scale into smaller parts.

It is also shown that the relation 'greater than' connecting modules of two functions (corresponding to two fixed formulae) which is true in an algebra of two elements is also true in algebras of an arbitrary number of elements.

Therefore, all inequalities of modules used in previous chapters are definitive. However, equalities may be violated.

1. ALGEBRA OF MAXIMA AND MINIMA. SCALES

Consider a set of numbers in a closed interval $[0, 1]$. Let us define 'addition' and 'multiplication' of these numbers as follows:

$$a + b = \max(a, b)$$

$$a \cdot b = \min(a, b).$$

Additionally, we will introduce an operation 'bar above':

$$\bar{a} = 1 - a.$$

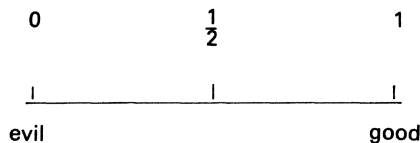


Fig. A2.1. Scale for the algebra of the three elements 0, $\frac{1}{2}$, 1.

It is well known that a set of numbers with the three operations given above is a Boolean algebra (Novikov, 1964).

The equations

$$a + \bar{a} = 1 \quad \text{and} \quad a \cdot \bar{a} = 0,$$

true for Boolean algebra of two elements, are not true in the general case; sometimes the term quasi-Boolean algebra is used.

In this appendix we shall consider sets of numbers of the type:

$$0, \frac{1}{n-1}, \frac{2}{n-1}, \dots, \frac{n-2}{n-1}, 1.$$

We shall call them ' n scales'. In addition, the continuous scale consisting of all numbers in the closed interval $[0, 1]$ will be considered.

2. PROPOSITIONAL FORMULAE

An alphabet of symbols consists of the following main signs: $a, b, +, \cdot, -, (,)$. The expressions $a, b, a + b, b + a, a \cdot b, b \cdot a, b^a, a^b$ are called formulae. For operational convenience, formulae may be taken in parentheses. The rules for factoring are the same as in standard formal logic.

Let f_1 and f_2 be formulae; then the expressions $f_1 + f_2, f_1 \cdot f_2, f_1^2, \bar{f}_1$ are also called formulae.

Two formulae are called logically equal if one of them can be transformed into a second one with the help of Axioms (1)–(12) of Boolean algebra (Chapter III). Each propositional formula is a formula connecting elements of Boolean algebra and can be used for any scale.

3. FORMULAE AND THEIR MODULES

Let an arbitrary scale n and a propositional formula F be given. Each letter a and b can independently have n different values. The general number of different pairs of values is n^2 . Thus, the function has n^2 values corresponding to different pairs. It is obvious that all values of the function belong to the set

$$0, \frac{1}{n-1}, \frac{2}{n-1}, \dots, \frac{n-2}{n-1}, 1.$$

Let us find the arithmetical sum of all values of the function and divide it by n^2 . The number obtained will be called a 'module of formula F in scale n '.

4. GRAPHICAL INTERPRETATION OF MODULE OF FORMULA

Suppose there is given a formula $F(a, b)$ and a scale n . Let us take a cube, with edges equal to 1 (Figure A2.2), and consider three of its edges meeting in one vertex.

Let one edge correspond to variable a , the second to variable b , and the third to the value of the function. We divide edges a and b into n parts, draw perpendiculars through points of division, and obtain squares. Each interval on the axes a and b corresponds to the number $(i - 1)/(n - 1)$, where i is its ordinal number from the vertex. Thus, each square on the base corresponds to a pair of numbers. Let us calculate the value of the function

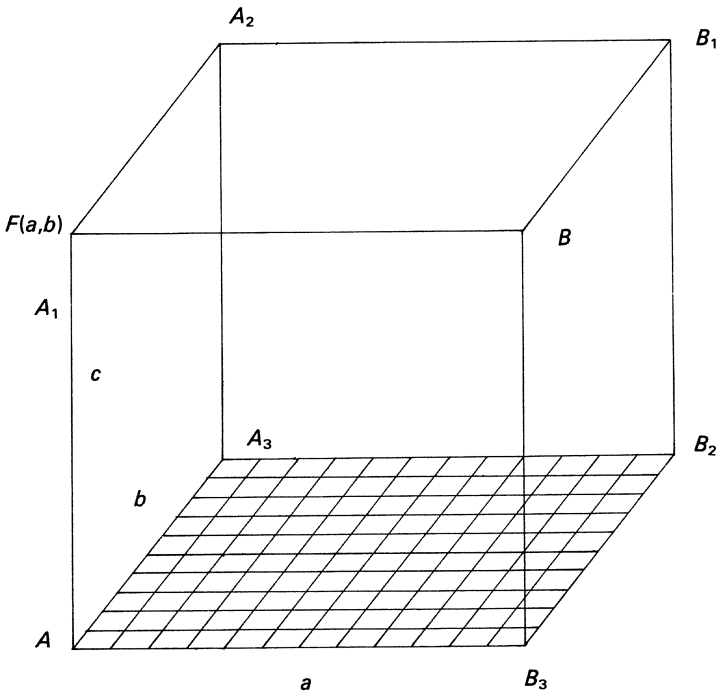


Fig. A2.2. Graphical representation of the module of function $F(a, b)$.

for each pair and on the corresponding square and draw a column of height equal to the value of the function. We obtain a stepped shape which represents the function and its volume represents the module of the function.

Suppose now that variables a and b are changing continuously and independently on the interval $[0, 1]$. The function $C = F(a, b)$ is continuous for all $0 \leq a \leq 1$ and $0 \leq b \leq 1$, since it is a superposition of maxima and minima of continuous functions on this interval. The function $F(a, b)$ can have four different values: $a, 1 - a, b, 1 - b$. Their geometrical interpretations are (see Figure A2.2):

- of a — plane AA_3BB_1 ;
- of $1 - a$ — plane $A_1A_2B_3B_2$;
- of b — plane $AA_2B_3B_1$;
- of $1 - b$ — plane $A_1A_3BB_2$.

Therefore, all values of function $F(a, b)$ are located on the four planes directed at an angle of 45° to the base.

We can show that when $n \rightarrow \infty$ the stepped shape tends to the shape of the function $F(a, b)$. To do this we draw upper and lower 'rounding surfaces' for a given stepped function $F(a, b)$. As an illustration we shall consider the function $F(a, b) = a$ and its rounding surfaces in projection on side of the cube (Figure A2.3).

It is easy to see that a correct consideration for one column of this function is true for one column of any other function. A graph of continuous function is AB ; a graph of a stepped function is $CC_1 \dots C_{10}D$. AC is the upper rounding; BD is the lower rounding. Let us choose an interval i . Suppose P_i is a volume of a column, shaped by arbitrary continuous function $F(a, b)$; p_i^s is a volume of a column shaped by a stepped function; p_i^+ is a volume of a column shaped by an upper rounding; p_i^- is a volume of a column shaped by a lower rounding.

It is easy to see that the following relations are true:

$$p_i^+ > p_i^s > p_i^-,$$

$$p_i^+ - p_i^- = \frac{1}{n^2} \cdot \frac{1}{n-1}.$$

Therefore, volumes determined by the stepped functions tend towards the volume determined by the continuous function.

In other words

$$p^s \rightarrow \int_0^1 \int_0^1 F(a, b) da db,$$

when $n \rightarrow \infty$.

5. SEPARATION OF SIGNIFICANTLY DIFFERENT FORMULAE

Let us call a monomial the expression

$$u_1 \cdot u_2 \cdot \dots \cdot u_i \cdot \dots \cdot u_k,$$

where u_i can have any value from the set $a, \bar{a}, b, \bar{b}$. We shall call 'a canonical expression' a record of propositional formula such as a finite sum of elements within the set

$$M_1, M_2, \dots, M_m, \dots,$$

where M_1 is the sum of one-letter monomials, M_2 is the sum of two-letter monomials, etc.

For example,

$$\underbrace{a + a + \bar{a} + b}_{M_1} + \underbrace{a \cdot b + \bar{a} \cdot b}_{M_2} + \underbrace{a \cdot b \cdot c}_{M_3}$$

THEOREM A2.1. *Any propositional formula can be written as a canonical expression.*

Proof. 1. If a formula contains exponential expressions, we can substitute for them a sum such as

$$f_{1^2}^f = \bar{f}_2 + f_1.$$

2. If symbols of addition and multiplication are under a bar in a linear expression, they can be removed by the following rules:

$$\overline{f_1 + f_2} = \overline{f_1} \cdot \overline{f_2},$$

$$\overline{f_1 \cdot f_2} = \overline{f_1} + \overline{f_2}.$$

3. By collecting first one-letter expressions, then two-letter ones, etc., we will get the expression sought. $\square$

THEOREM A2.2. *Any propositional formula can be expressed as a sum of elements within the set*

$$M_1, M_2, M_3, M_4.$$

Proof. Suppose that a canonical representation of a given propositional formula is

$$M_{k_1} + \cdots + M_{k_i} + \cdots + M_{k_r},$$

where $k_r > 4$ and $k_1 < k_2 < \dots < k_r$.

Consider each item belonging to M_{k_i} : it contains the product of k symbols; no more than four of them are different ($a, \bar{a}, b, \bar{b}$). By using the equation $f \cdot f = f$ we can reduce the number of factors to four. If we complete this procedure for every $k_i > 4$, we will obtain the expression sought. $\square$

Therefore, the number of algebraically different formulae is finite.

Let us call a canonical form minimal, if neither the number of items, nor the number of letters in each item can be reduced by identical transformation.

We can significantly reduce a number of considered formulae if we take into consideration that an individual is expressed as, a^W , where W is any propositional formula. This expression can be transformed into $a + \bar{W}$.

After reducing $\bar{W}$ to canonical representation, we can eliminate all monomials containing a , since

$$a + af = a(1 + f) = a.$$

Let us now note that the module of the formula will not change in any scale if we change b to $\bar{b}$ and $\bar{b}$ to b or simultaneously change a to b and b to a .

In view of the above, we can separate the following formulae and find their modules for scales 2 and 3, which are given in ninths.

	$n = 2$	$n = 3$
1. $a + \bar{a} + b + \bar{b}$	9	8.5
2. $a + \bar{a} + b$	9	8
3. $a + \bar{a} + b \cdot \bar{b}$	9	7.5
4. $a + b + \bar{a} \cdot \bar{b}$	9	7.5
5. $a + \bar{a}$	9	7.5
6. $a + \bar{a} \cdot b + \bar{a} \cdot \bar{b}$	9	7
7. $a + \bar{a} \cdot b + \bar{a} \cdot \bar{b} + b \cdot \bar{b}$	9	7
8. $a + b$	6.75	6.4
9. $a + \bar{a} \cdot b$	6.75	5.9
10. $a + \bar{a} \cdot b + b \cdot \bar{b}$	6.75	5.9
11. $a + b \cdot \bar{b}$	4.5	5
12. $a + \bar{a} \cdot b \cdot \bar{b}$	4.5	5
13. a	4.5	4.5

We have thirteen different formulae. For some formulae (in scales 2 and 3) the modules coincide. And if the relation $|f'| \geq |f''|$ is true in one scale, it is also true in the other one.

THEOREM A2.3. *If the module of f' is equal to the module of f'' in scale 3, this equality is true for any $n > 3$.*

Proof. We can separate from the table four groups of formulae having the same module.

$$(1) \quad |f_3| = |f_4| = |f_5|$$

$$(2) \quad |f_6| = |f_7|$$

$$(3) \quad |f_9| = |f_{10}|$$

$$(4) \quad |f_{11}| = |f_{12}|$$

(1) Let us show that $|f_3| = |f_5|$ in any scale.

The formulae

$$f_3 = a + \bar{a} + b \cdot \bar{b}, \quad f_5 = a + \bar{a}$$

have a common part. It is easy to see that any value of

$$a + \bar{a} = \max(a, 1 - a) \geq \frac{1}{2}.$$

On the other hand,

$$b \cdot \bar{b} = \min(b, 1 - b) \leq \frac{1}{2}.$$

Thus,

$$f_3 = a + \bar{a} + b \cdot \bar{b} = \max(a + \bar{a}, b \cdot \bar{b}) = a + \bar{a} = f_5.$$

Let us show now that $|f_4| = |f_5|$ for any scale.

We will operate, not with the functions, but rather with their negations. Transform the formulae $\bar{f}_4$ and $\bar{f}_5$ as follows

$$\begin{aligned}\bar{f}_4 &= \overline{a + b + \bar{a} \cdot \bar{b}} = \overline{(a + b)} \cdot (a + b) \\ \bar{f}_5 &= \overline{a + \bar{a}} = \bar{a} \cdot a.\end{aligned}$$

Consider the shapes of these functions on a single cube (Figure A2.2). Function $\bar{f}_4$ is symmetrical about the diagonal plane $AA_1B_1B_2$. In the area $a \geq b$ the shape of function $\bar{f}_4$ coincides with the shape of function $\bar{f}_5$. The function $\bar{f}_5$ is not symmetrical about the diagonal plane, but this plane divides its volume into two congruent parts. Thus, we can conclude that the halves of coinciding volumes are equal and consequently the volumes are also. Thus, the volumes corresponding to functions $\bar{f}_4$ and $\bar{f}_5$, and therefore, those corresponding to f_4 and f_5 are also equal for any scale, and point (1) is proven.

(2) Let us prove that $f_6 = f_7$.

$$f_6 = \max(a, \min(\bar{a}, \max(b, \bar{b}))), \quad \max(b, \bar{b}) \geq \frac{1}{2}.$$

If $\bar{a} \geq \frac{1}{2}$, then

$$\min(\bar{a}, \max(b, \bar{b})) \geq \frac{1}{2}, \quad \text{and} \quad f_6 \geq \frac{1}{2}.$$

If $\bar{a} < \frac{1}{2}$, then

$$\min(\bar{a}, \max(b, \bar{b})) = \bar{a}, \quad a > \frac{1}{2}, \quad \text{and} \quad f_6 > \frac{1}{2}.$$

Therefore, always $f_6 \geq \frac{1}{2}$.

Since

$$f_7 = f_6 + b \cdot \bar{b} = \max(f_6, \min(b, \bar{b})),$$

and

$$\min(b \cdot \bar{b}) \leq \frac{1}{2},$$

then $f_7 = f_6$.

(3) Let us prove that $f_9 = f_{10}$. Suppose $a \geq \frac{1}{2}$, then $f_9 = f_{10} = a$. Suppose $a < \frac{1}{2}$, then there are four possibilities:

- (a) $\bar{b} \leq a < \bar{a} \leq b, \quad f_9 = \bar{a} > \frac{1}{2},$
 $f_{10} = f_9 + b \cdot \bar{b} = f_9.$
- (b) $a \leq \bar{b} \leq b \leq \bar{a}, \quad f_9 = b,$
 $f_{10} = f_9 + b \cdot \bar{b} = \bar{b} + b \cdot \bar{b} = b \cdot (1 + \bar{b}) = b, \quad f_9 = f_{10}.$
- (c) $b \leq a < \bar{a} \leq \bar{b}, \quad f_9 = a,$
 $f_{10} = f_9 + b \cdot \bar{b} = a + b \cdot \bar{b} = a, \quad f_9 = f_{10}.$
- (d) $a \leq b \leq \bar{b} \leq \bar{a}, \quad f_9 = b,$
 $f_{10} = f_9 + b \cdot \bar{b} = b + b \cdot \bar{b} = b, \quad f_9 = f_{10}.$

Therefore, always $f_9 = f_{10}$.

(4) Let us prove that $f_{11} = f_{12}$. Suppose $a \geq \frac{1}{2}$, then

$$f_{11} = f_{12} = a.$$

Suppose $a < \frac{1}{2}$, then

$$f_{12} = \max(a, \min(\bar{a}, b \cdot \bar{b})).$$

Since $\bar{a} > \frac{1}{2}$ and $b \cdot \bar{b} \leq \frac{1}{2}$, then

$$f_{12} = \max(a, b \cdot \bar{b}) = a + b \cdot \bar{b} = f_{11}.$$

Therefore, always $f_{11} = f_{12}$.

The proof of Theorem A1.3 is complete $\square$

We found eight formulae whose modules may be different at $n \geq 3$.

- (1) $a + \bar{a} + b + \bar{b}$
- (2) $a + \bar{a} + b$
- (3) $a + \bar{a}$
- (4) $a + \bar{a} \cdot b + \bar{a} \cdot \bar{b}$
- (5) $a + b$
- (6) $a + \bar{a} \cdot b$
- (7) $a + b \cdot \bar{b}$
- (8) a

The next step is to prove that their modules *will* be different for any $n \geq 3$ and that their order is preserved as the modules descend in quantity.

6. LIMITING VALUES

Let us find limiting values for the eight formulae in the previous paragraph with $n \rightarrow \infty$. In order to do this we calculate the integrals

$$p = \int_0^1 \int_0^1 f(a, b) da db$$

for each of the eight formulae.

Table A2.1 shows the modules of the formulae for $n = 2$, $n = 3$, $n = \infty$.

TABLE A2.1

A list of the formulae of $a + W$ type for which
modules are different at $n \geq 3$.

Formulae	$n = 2$	$n = 3$	$n = \infty$
1. $a + \bar{a} + b + \bar{b}$	1	$\frac{8.5}{9}$	$\frac{20}{24}$
2. $a + \bar{a} + b$	1	$\frac{8}{9}$	$\frac{19}{24}$
3. $a + \bar{a}$	1	$\frac{7.5}{9}$	$\frac{18}{24}$
4. $a + \bar{a} \cdot b + \bar{a} \cdot \bar{b}$	1	$\frac{7}{9}$	$\frac{17}{24}$
5. $a + b$	$\frac{3}{4}$	$\frac{6.4}{9}$	$\frac{16}{24}$
6. $a + \bar{a} \cdot b$	$\frac{3}{4}$	$\frac{5.9}{9}$	$\frac{15}{24}$
7. $a + b \cdot \bar{b}$	$\frac{1}{2}$	$\frac{5}{9}$	$\frac{13}{24}$
8. a	$\frac{1}{2}$	$\frac{4.5}{9}$	$\frac{12}{24}$

TABLE A2.2
The values of modules of formulae of $a + \bar{W}$ types in scales $n \leq 50$.*

	2	3	4	5	6	7	8	9	10	11
8. a	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
7. $a + b \cdot \bar{b}$	0.5000	0.5556	0.5417	0.5500	0.5444	0.5476	0.5446	0.5463	0.5444	0.5455
6. $a + \bar{a} \cdot b$	0.7500	0.6667	0.6667	0.6500	0.6500	0.6429	0.6429	0.6389	0.6389	0.6364
5. $a + b$	0.7500	0.7222	0.7083	0.7000	0.6944	0.6905	0.6875	0.6852	0.6833	0.6818
4. $a + \bar{a} \cdot b + \bar{a} \cdot \bar{b}$	1.0000	0.7778	0.7917	0.7500	0.7556	0.7381	0.7411	0.7315	0.7333	0.7273
3. $a + \bar{a}$	1.0000	0.8333	0.8333	0.8000	0.8000	0.7857	0.7857	0.7778	0.7778	0.7727
2. $a + \bar{a} + b$	1.0000	0.8889	0.8750	0.8500	0.8444	0.8333	0.8304	0.8241	0.8222	0.8182
1. $a + \bar{a} + b + \bar{b}$	1.0000	0.9444	0.9167	0.9000	0.8889	0.8810	0.8750	0.8704	0.8667	0.8636

	12	13	14	15	16	17	18	19	20	21	22	23	24
8	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
7	0.5442	0.5449	0.5440	0.5444	0.5437	0.5441	0.5436	0.5439	0.5434	0.5437	0.5433	0.5435	0.5432
6	0.6364	0.6346	0.6346	0.6333	0.6333	0.6324	0.6324	0.6316	0.6316	0.6310	0.6310	0.6304	0.6304
5	0.6806	0.6795	0.6786	0.6778	0.6771	0.6765	0.6759	0.6754	0.6750	0.6746	0.6742	0.6739	0.6736
4	0.7285	0.7244	0.7253	0.7222	0.7229	0.7206	0.7211	0.7193	0.7197	0.7183	0.7186	0.7174	0.7177
3	0.7727	0.7692	0.7692	0.7667	0.7667	0.7647	0.7647	0.7632	0.7632	0.7619	0.7619	0.7609	0.7609
2	0.8169	0.8141	0.8132	0.8111	0.8104	0.8088	0.8083	0.8070	0.8066	0.8056	0.8052	0.8043	0.8040
1	0.8611	0.8590	0.8571	0.8556	0.8542	0.8529	0.8519	0.8509	0.8500	0.8492	0.8485	0.8478	0.8472

8	25	26	27	28	29	30	31	32	33	34	35	36	37
7	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
6	0.5433	0.5431	0.5432	0.5430	0.5431	0.5429	0.5430	0.5428	0.5429	0.5428	0.5429	0.5427	0.5428
5	0.6300	0.6300	0.6296	0.6296	0.6293	0.6293	0.6290	0.6290	0.6288	0.6288	0.6286	0.6286	0.6284
4	0.6733	0.6731	0.6728	0.6726	0.6724	0.6722	0.6720	0.6719	0.6717	0.6716	0.6714	0.6713	0.6712
3	0.7167	0.7169	0.7160	0.7163	0.7155	0.7157	0.7151	0.7152	0.7146	0.7148	0.7143	0.7144	0.7140
2	0.7600	0.7600	0.7593	0.7593	0.7586	0.7586	0.7581	0.7581	0.7576	0.7576	0.7571	0.7571	0.7568
1	0.8033	0.8031	0.8025	0.8022	0.8017	0.8015	0.8011	0.8009	0.8005	0.8004	0.8000	0.7999	0.7995
	0.8467	0.8462	0.8457	0.8452	0.8448	0.8444	0.8441	0.8434	0.8434	0.8431	0.8429	0.8426	0.8423
8	38	39	40	41	42	43	44	45	46	47	48	49	50
7	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
6	0.5427	0.5427	0.5426	0.5427	0.5426	0.5426	0.5425	0.5426	0.5425	0.5426	0.5425	0.5425	0.5424
5	0.6284	0.6282	0.6282	0.6280	0.6280	0.6279	0.6279	0.6278	0.6278	0.6277	0.6277	0.6275	0.6275
4	0.6711	0.6709	0.6708	0.6707	0.6706	0.6705	0.6705	0.6704	0.6703	0.6702	0.6701	0.6701	0.6700
3	0.7141	0.7137	0.7138	0.7134	0.7135	0.7132	0.7133	0.7130	0.7130	0.7128	0.7128	0.7126	0.7126
2	0.7568	0.7564	0.7564	0.7561	0.7561	0.7558	0.7558	0.7556	0.7556	0.7553	0.7553	0.7551	0.7551
1	0.7994	0.7991	0.7990	0.7988	0.7987	0.7984	0.7984	0.7981	0.7981	0.7979	0.7978	0.7976	0.7975
	0.8421	0.8419	0.8417	0.8415	0.8413	0.8411	0.8409	0.8407	0.8406	0.8404	0.8403	0.8401	0.8400

* The formulae correspond to those on p. 125.

We can see that the order of formulae is preserved as their modules descend from $n = 3$ to $n = \infty$. We will show that the same is true for any $n > 3$.

We use the fact that the sequence of modules of a given formula converges to the double integral of this formula when $n \rightarrow \infty$. We see that the minimal difference between two limiting values is equal to $\frac{1}{24}$. From (A2.1),

$$|p - p_s| < \frac{1}{n-1}$$

We can find a number n^* such that the modules of all formulae at $n > n^*$ differ from each other by less than $\frac{1}{48}$. In this case the order is preserved for all $n > n^*$. From evaluation (A2.1) it follows that n^* is a minimal integer for which the inequality

$$\frac{1}{n^* - 1} \leq \frac{1}{48}$$

is true. Therefore, $n^* = 49$. Thus, we have proved that for all $n > 49$ the order of modules is preserved.

For modules at $n \leq 49$ we have made a computer verification. Table A2.2 gives the modules of the eight formulae for all $n \leq 50$. It can be seen that the order for these values of n is also invariable.

7. THE MAIN FORMAL RESULTS

- (1) If two formulae have equal modules at $n = 3$, they will have equal modules at any $n > 3$.
- (2) If at $n = 3$ the module of one formula is greater than the module of another one, the relation ' $>$ ' will be true for any $n > 3$.
- (3) The relation ' $>$ ', true for the modules of two formulae at $n = 2$, is true for any $n > 2$.

Therefore, we have demonstrated that if one is interested only in comparing the order of the modules of the formulae, the case with $n = 3$ gives the complete relevant information.

Let us note that the obtained results are true only for formulae representing individuals; for any other two-letter formula this problem is still open.

8. A COMPARISON OF ETHICAL STATUSES OF INDIVIDUALS IN TWO-VALUED AND THREE-VALUED LOGICS

In the previous section of this appendix it was shown that some formulae with equal modules in two-element logic may have different modules in three-element logic. In this section we will see the consequences of this fact, i.e., the difference between ethical systems for $n = 2$ and $n = 3$.

We shall call a *reflexive* structure an object constructed as follows:

- (1) A finite graph-tree with no more than two branches at each node is given.
- (2) Each node is labeled either a or b .
- (3) Two branches from the same node are called neighboring branches. Two nodes that are ends of neighboring branches are called neighboring nodes. Neighboring nodes are labeled by different letters. If a node does not have a neighbor, it can be labeled by any letter.

Thus, a reflexive structure is a tree, each node of which has a designation a or b . It is easy to see that for each two-letter formula one can find a corresponding reflexive structure. For example, the formula

$$a\bar{a}^a + \bar{b} + \bar{b}$$

corresponds to the reflexive structure shown in Figure A2.4.

A reflexive structure will be designated by the letter S with auxiliary

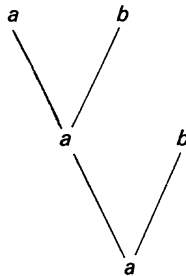


Fig. A2.4. An example of reflexive structure.

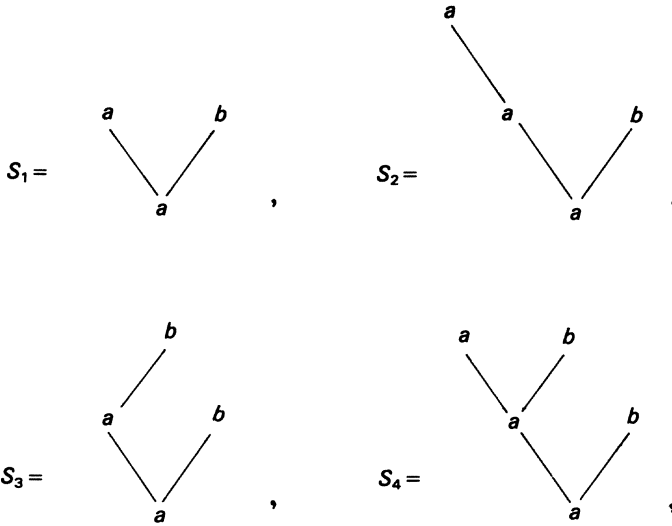


Fig. A2.5. Designation of reflexive structures.

subscripts. We limit ourselves to studying the four reflexive structures of Figure A2.5.

Now let us construct a special logical language (which is close to the one proposed by Alpersen (1975)) to interpret the formulae. We introduce abbreviations for the following statements:

- an individual has a correct evaluation of himself α ;
- an individual has an incorrect evaluation of himself $\bar{\alpha}$;
- an individual doubts his evaluation of himself β_1 ;
- an individual does not doubt his evaluation of himself $\bar{\beta}_1$;
- an individual doubts his evaluation of his partner β_2 ;
- an individual does not doubt his evaluation of his partner $\bar{\beta}_2$.

We shall use a logical description. For example, for the formula

$$a^{a\bar{a} + b + b}$$

we have the description:

$$S_4 \& \alpha \& \beta_1 \& \bar{\beta}_2.$$

The common description for the two formulae

$$a^{a\bar{a} \cdot b \cdot b} \quad \text{and} \quad a^{a\bar{a} \cdot \bar{b} \cdot b}$$

is

$$S_4 \& \alpha \& (\beta_1 \& \beta_2 \vee \bar{\beta}_1 \& \beta_2).$$

TABLE A.2.3

A connection between reflexive structure and ethical status for scales of $n = 2$ and $n = 3$ under conditions that individuals have only ' + ' in their inner worlds.

Formula	Reflexive structure	Ethical status	
		$n = 2$	$n = 3$
1. $a^{\bar{a} + b}$	S_1		
2. $a^{\bar{a}a + b}$	S_2	$\frac{3}{4}$	$\frac{5.9}{9}$
3. $a^{\bar{a}\bar{b} + b}$	S_3		
4. $a^{\bar{a}\bar{a} + b + b}$	S_4		
5. $a^{\bar{a}a + \bar{b} + b}$	S_4		
6. $a^{\bar{a}a + \bar{b} + b}$	S_4		
7. $a^{\bar{a}b + b}$	S_3		$\frac{5}{9}$
8. $a^{\bar{a}a + b + b}$	S_4		
9. $a^{\bar{a}a + b}$	S_2		
10. $a^{\bar{a} + b}$	S_1	$\frac{1}{2}$	
11. $a^{\bar{a}a + b}$	S_2		
12. $a^{\bar{a}a + b}$	S_2		$\frac{4.5}{9}$
13. $a^{\bar{a}b + b}$	S_3		
14. $a^{\bar{a}\bar{b} + b}$	S_3		
15. $a^{\bar{a}\bar{a} + b + b}$	S_4		
16. $a^{\bar{a}\bar{a} + b + b}$	S_4		
17. $a^{\bar{a}a + \bar{b} + b}$	S_4		
18. $a^{\bar{a}a + \bar{b} + b}$	S_4		

$$\alpha \& [(S_3 \& \beta_2) \vee (S_4 \& \beta_1 \& \beta_2)]$$

$$\bar{\alpha} \vee (S_2 \& \beta_1)$$

In this way we can give the general logical description of any finite set of formulae.

In the formulae above all signs are the same: either ‘+’ or ‘.’. We will not introduce special designation for signs.

Let us see what kind of disturbances of equality of ethical statuses take place by passing from $n = 2$ to $n = 3$. Compare individuals of the first ethical system who are ‘in conflict’. To do this we consult Table A2.3. The ‘positive ideal’ does not change: all individuals with the ethical status $\frac{3}{4}$ obtain status $\frac{5.9}{9}$. However, ‘negative individuals’ with the status $\frac{1}{2}$ do change: some of them obtain status $\frac{4.5}{9}$, others obtain status $\frac{5}{9}$. Thus, some of them become a little ‘better’. Which of them become better? We interpret the logical expression describing the individuals with status $\frac{5}{9}$.

That individual

- has a correct evaluation of himself;
- either does not doubt the correctness of his evaluation of his adversary (if his structure is S_3) or doubts neither the correctness of his evaluation of himself nor of his adversary (if his structure is S_4).

An individual with the status equal to $\frac{4.5}{9}$

- either has an incorrect evaluation of himself;
- or does not doubt the correctness of his evaluation of himself (if his structure is S_2).

Now we can see that the two types of individuals who have a correct evaluation of themselves and *do not have doubts* come to be a little better than those with incorrect evaluation of themselves. Thus, for $n = 3$ a fanatic excels an ‘individual being mistaken’ (this is not the case for $n = 2$).

Consider now the ‘individuals in conflict’ belonging to the second ethical system. To do this we consult Table A2.4. ‘Negative individuals’ with the status $\frac{3}{4}$ at $n = 2$ receive status $\frac{6.4}{9}$ at $n = 3$. No partition occurs. However, the ‘positive ideal’ with status equal to 1 at $n = 2$ changes. The highest status $\frac{8}{9}$ belongs to the individual who

- has a correct evaluation of himself;
- either does not ‘think’ about the correctness or incorrectness of his evaluations of himself and his adversary (structure S_1), or doubts the correctness of his evaluation of himself (structure S_2).

A lower status $\frac{7.5}{9}$ belongs to the individual who

- has a correct evaluation of himself;

TABLE A.2.4

A connection between reflexive structure and ethical status for scales of $n = 2$ and $n = 3$, under conditions that individuals have only ‘.’ in their inner worlds.

Formula	Reflexive structure	Ethical status	
		$n = 2$	$n = 3$
1. $a^a \cdot b$	S_1	1	$\frac{8}{9}$
2. $a^{aa} \cdot b$	S_2		$\frac{7.5}{9}$
3. $a^{ab} \cdot b$	S_3		
4. $a^{a\bar{a}} \cdot b \cdot b$	S_4		
5. $a^{aa} \cdot b$	S_2	$\frac{3}{4}$	$\frac{6.4}{9}$
6. $a^{a\bar{b}} \cdot b$	S_3		
7. $a^{aa} \cdot b \cdot b$	S_4		
8. $a^{aa} \cdot b \cdot b$	S_4		
9. $a^{aa} \cdot b \cdot b$	S_4		
10. $a^{\bar{a}} \cdot b$	S_1		
11. $a^{\bar{a}a} \cdot b$	S_2		
12. $a^{\bar{a}\bar{a}} \cdot b$	S_2		
13. $a^{\bar{a}b} \cdot b$	S_3		
14. $a^{\bar{a}\bar{b}} \cdot b$	S_3		
15. $a^{\bar{a}a} \cdot b \cdot b$	S_4		
16. $a^{\bar{a}\bar{a}} \cdot b \cdot b$	S_4		
17. $a^{\bar{a}a} \cdot \bar{b} \cdot b$	S_4		
18. $a^{\bar{a}\bar{a}} \cdot \bar{b} \cdot b$	S_4		

—either does not doubt the correctness of his evaluation of his adversary (structure S_3), or doubts the correctness of his evaluation of himself and does not doubt that of his adversary (structure S_4).

We see that reflexive structures play a significant role in the partition: for S_1 and S_2 the status is equal to $\frac{8}{9}$, and for S_3 and S_4 the status is equal to $\frac{7.5}{9}$.

Let us compare these pairs (Figure A2.5). In the structures S_3 and S_4 the letter b is located on the third tier, but in the structures S_1 and S_2 this letter is absent on the third tier. This means that an individual with the structure S_3 or S_4 *sees himself seeing* his adversary, and an individual with the structure S_1 or S_2 *does not see himself seeing* his adversary. Thus, consideration about one's picture of one's adversary, even if it is connected with a 'conviction of correctness' of this picture leads to a decrease in the ethical status.

Therefore, we see that in three-valued logic the differences between the individuals become more refined than in two-valued logic. The problem of what kind of logic is realized in real systems can be, in the future, a subject of a special empirical investigation.

9. A CORRELATION BETWEEN SUFFERING AND FEELING GUILTY FOR ARBITRARY SCALES

In Chapter XIII we demonstrated that in the two-element scale there is a formal correlation between feeling guilty and suffering, which for the simplest individual may be expressed by two inequalities:

$$|a^{a \cdot b} \cdot b| < |a^{a+b} + b|,$$

$$|a^{a \cdot b}| > |a^{a+b}|.$$

Let us now show that this formal relation holds for an arbitrary scale. The second inequality remains true for an arbitrary scale in view of the result proven above about the preservation of inequalities connecting modules of individuals during transition from a two-element scale to a multi-element scale. However, we cannot expand this result to the first inequality, because inner worlds may be expressed by formulae which cannot be represented as $a + R$, and the preservation of inequalities was proven only for formulae which can be represented as $a + R$.

Note that if we have a system of inequalities

$$\alpha_i \leq \beta_i, \quad i = 1, 2, \dots, k,$$

then it is obvious that

$$\sum_{i=1}^k \alpha_i \leq \sum_{i=1}^k \beta_i.$$

Let us suppose now that for some i^* , $\alpha_{i^*} < \beta_{i^*}$; then it is obvious that the nonstrict inequality would turn into the strict inequality

$$\sum_{i=1}^k \alpha_i < \sum_{i=1}^k \beta_i.$$

For any two values of an arbitrary scale the following relations are true:

$$a^{a \cdot b} \cdot b = \min(a^{a \cdot b}, b) \leq b.$$

$$a^{a+b} + b = \max(a^{a+b}, b) \geq b.$$

Therefore,

$$a^{a \cdot b} \cdot b \leq a^{a+b} + b$$

for any two values.

In order to examine modules, we substitute for a and b in the left- and right-hand sides of the inequality all possible pairs of values belonging to a given scale, then find a sum of values separately for the left- and right-hand sides, and divide them into a number of different pairs.

Let us now show that there is a pair of values which makes the inequality strict:

$$a^{a \cdot b} \cdot b < a^{a+b} + b.$$

Such a pair is, for example, $a = 1, b = 0$, which exists in any scale.

According to our auxiliary remark, we come to the inequality

$$|a^{a \cdot b} \cdot b| < |a^{a+b} + b|.$$

Similar considerations can be made for the pairs $(a, \bar{b})$, $(\bar{a}, b)$, and $(\bar{a}, \bar{b})$. Adding separately the left and right sides of the four inequalities and determining their respective means, we get

$$|a^{a \cdot b} \cdot b| < |a^{a+b} + b|.$$

Therefore, the correlation between feeling guilty and suffering for the case of the simplest three-tier individual is true for an arbitrary scale.

APPENDIX 3

SELF-GENERATION OF ENVIRONMENTS

In representing situations as automata (Chapter V), we assumed that the frequency of good directions does not depend on the activity of these automata. It is possible, however, to analyze a case where the frequency of directions at the automaton's inputs depends on the previous history of interaction between the automaton and the environment. We will call this frequency the *parameter* of the environment.

Consider the scheme in Figure A3.1.

This system works in steps. At the first step we state the environment with parameter p_1 . As a result the sequence of outputs of situations is generated; its parameter is p_2 . The second step is the changing of the environment: the parameter p_2 starts to work as a parameter of the environment; it generates a new environment with a parameter p_3 , which starts to work at the third step, etc. What is the tendency of the environment to change? We will call 'balanced' a state of the system in which the frequencies of input and output are equal; in this case there is no change of environment. We prove further that any sequence of environments in the system represented in Figure A3.1

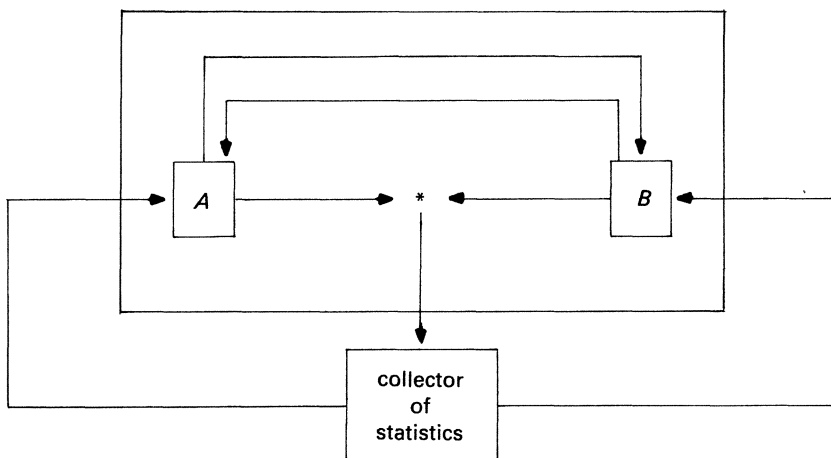


Fig. A3.1. A scheme for generating the parameter of environment by a situation.

TABLE A3.1

A list of different boolean functions of two independent variables.

<i>a</i>	<i>b</i>	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	1	1	0	0	0	1	1	1	0	0	0	0	1	1	1	1	0
0	1	0	1	0	0	1	0	0	1	1	0	1	0	1	1	1	0
1	0	0	0	1	0	0	1	0	0	1	1	1	1	0	1	1	0
0	0	0	0	0	1	0	0	1	1	0	1	1	1	1	0	1	0

tends to balance; we will give a classification of possible kinds of balance as well as a classification of various cases of *transformation* into a balanced state or of the tendency to asymptotically approach a balanced state.

In Boolean algebra of two variables sixteen different functions can exist. The combinations of values of these functions and their arguments are shown in Table A3.1.

The simplest formulae corresponding to these functions are as follows:

- | | |
|---|---|
| (1) $a \cdot b$ | (9) $\bar{a} \cdot b + a \cdot \bar{b}$ |
| (2) $\bar{a} \cdot b$ | (10) $\bar{b}$ |
| (3) $a \cdot \bar{b}$ | (11) $\bar{a} + \bar{b}$ |
| (4) $\bar{a} \cdot \bar{b}$ | (12) $a + \bar{b}$ |
| (5) b | (13) $\bar{a} + b$ |
| (6) a | (14) $a + b$ |
| (7) $a \cdot b + \bar{a} \cdot \bar{b}$ | (15) 1 |
| (8) $\bar{a}$ | (16) 0. |

Since individuals are represented by formulae of the type a, a^w, b, b^w or $a, a + \bar{w}, b, b + \bar{w}$, only six functions from those given above are suitable: 5, 6, 12, 13, 14, 15. Two functions are specific for individual A : $a, a + \bar{b}$; two functions for individual B : $b, b + \bar{a}$; and two functions may represent both individuals a and b : $1, a + b$. Those functions corresponding to situations of interaction of different pairs of individuals may be represented by two tables. The lines correspond to the various functions representing individual A , the columns correspond to those representing individual B . Table A3.2 represents the relationships ‘+’ between the individuals.

Table A3.3 represents the relationships ‘.’ between the individuals.

Now, given p_n , we can calculate p_{n+1} for any two interacting individuals. For example, let us consider the interaction

$$a^{a+b} + b^{\bar{b}+a},$$

TABLE A3.2

Boolean functions corresponding to situations
 $A + B$.

+	1	b	$b + a$	$b + \bar{a}$
1	1	1	1	1
a	1	$a + b$	$a + b$	1
$a + b$	1	$a + b$	$a + b$	1
$a + \bar{b}$	1	1	1	1

TABLE A3.3

Boolean functions corresponding to situations
 $A \cdot B$.

$\cdot$	1	b	$b + a$	$b + \bar{a}$
1	1	b	$a + b$	$\bar{a} + b$
a	a	$a \cdot b$	a	$a \cdot b$
$a + b$	$a + b$	b	$a + b$	b
$a + \bar{b}$	$a + \bar{b}$	$a \cdot b$	a	$a \cdot b + \bar{a} \cdot \bar{b}$

simplified to $a + b$.

Let a and b independently have the value 1, with the probability p_n . Then, $a + b$ has the value 1, with the probability

$$p_{n+1} = 2p_n - p_n^2.$$

Thus, we have connected the parameters of consecutive environments by a recurrence-formula. In a balanced state $p_{n+1} = p_n$, and by solving the equation $p = 2p - p^2$, we will find the parameters of two balanced states:

$$p' = 0 \quad p'' = 1.$$

Recurrence-formulae can be formed for any two functions from Tables A3.2 and A3.3. (These formulae are given in Tables A3.4 and A3.5.) We also need some knowledge about quadratic recurrence-formulae.

Consider the quadratic recurrence-formula

$$p_{n+1} = ap_n^2 + 2bp_n + c.$$

The substitution

$$u_n = ap_n + b, \quad c' = ac - b^2 + b$$

reduces the formula to the form

$$u_{n+1} = u_n^2 + c',$$

or

$$u_{n+1} - u_n = u_n^2 - u_n + c'.$$

The roots x_1 and x_2 of equation

$$u_n^2 - u_n + c' = 0$$

are:

$$x_1 = \frac{1}{2} + \sqrt{\frac{1}{4} - c'} = k, \quad x_2 = 1 - k.$$

Now we can write

$$u_{n+1} - u_n = (u_n - k) \cdot (u_n - 1 + k).$$

T. W. Chaundy and Eric Phillips (1936) studied this sequence. Their results relevant to our study are:

- (1) the limit u can only be k or $1 - k$;
- (2) if $|u_1| < k$ and $\frac{1}{2} \leq k \leq \frac{3}{2}$, $u_n \rightarrow 1 - k$: monotonically if $\frac{1}{2} \leq k \leq 1$.

All the situations from Tables A3.2 and A3.3 correspond to six different recurrence-formulae:

- (1) $p_{n+1} = 1$;
- (2) $p_{n+1} = -p_n^2 + 2p_n$; $p_1 \neq 0$; $p_n \rightarrow 1$.
- (3) $p_{n+1} = p_n^2 - p_n + 1$; $p_n \rightarrow 1$.
- (4) $p_{n+1} = p_n$; $p_n = p_1$.
- (5) $p_{n+1} = 2p_n^2 - 2p_n + 1$; $p_1 \neq 1$; $p_n \rightarrow \frac{1}{2}$.
- (6) $p_{n+1} = p_n^2$; $p_1 \neq 1$; $p_n \rightarrow 0$.

TABLE A3.4
Recurrent formulae corresponding to situations $A + B$.

+	1	b	$b + a$	$b + \bar{a}$
1	$p_{n+1} = 1$	$p_{n+1} = 1$	$p_{n+1} = 1$	$p_{n+1} = 1$
a	$p_{n+1} = 1$	$p_{n+1} = -p_n^2 + 2p_n$ $p_n \rightarrow 1$ $\tilde{p} = 0$	$p_{n+1} = -p_n^2 + 2p_n$ $p_n \rightarrow 1$ $\tilde{p} = 0$	$p_{n+1} = 1$
$a + b$	$p_{n+1} = 1$	$p_{n+1} = -p_n^2 + 2p_n$ $p_n \neq 1$ $\tilde{p} = 0$	$p_{n+1} = -p_n^2 + 2p_n$ $p \neq 1$ $\tilde{p} = 0$	$p_{n+1} = 1$
$a + \bar{b}$	$p_{n+1} = 1$	$p_{n+1} = 1$	$p_{n+1} = 1$	$p_{n+1} = 1$

TABLE A3.5
Recurrent formulae corresponding to situations $A \cdot B$.

.	1	b	$b + a$	$b + \bar{a}$
1	$p_{n+1} = 1$	$p_{n+1} = p_1$	$p_{n+1} = -p_n^2 + 2p_n$ $p_n \rightarrow 1$ $\tilde{p} = 0$	$p_{n+1} = p_n^2 - p_n + 1$ $p_n \rightarrow 1$
a	$p_{n+1} = p_1$	$p_{n+1} = p_n^2$ $p_n \rightarrow 0$ $\tilde{p} = 1$	$p_{n+1} = p_1$	$p_{n+1} = p_n^2$ $p_n \rightarrow 0$ $\tilde{p} = 1$
$a + b$	$p_{n+1} = -p_n^2 + 2p_n$ $p_n \rightarrow 1$ $\tilde{p} = 0$	$p_{n+1} = p_1$	$p_{n+1} = -p_n^2 + 2p_n$ $p_n \rightarrow 1$ $\tilde{p} = 0$	$p_{n+1} = p_1$
$a + \bar{b}$	$p_{n+1} = p_n^2 - p_n + 1$ $p_n \rightarrow 1$	$p_{n+1} = p_n^2$ $p_n \rightarrow 0$ $\tilde{p} = 1$	$p_{n+1} = p_1$	$p_{n+1} = 2p_n^2 - 2p_n + 1$ $p_n \rightarrow \frac{1}{2}$ $\tilde{p} = 1$

We see that formula (1) corresponds to the process of immediate transformation to the 'good environment' (which produces only 1's independently of the initial environment) at the first step. In this case the balanced environment is a 'good environment'. Formula 2 corresponds to the process of asymptotic approximation to a 'good environment' if $p_1 \neq 0$. Formula 3 corresponds to the process of asymptotic approximation to a 'good environment' independently of the initial environments. In these cases too the balanced environment is a good environment.

Formula 4 corresponds to the process of conservation of the initial environment: the system keeps supporting any environment which existed from the beginning. Any initial environment is a balanced environment in this case.

Formula 5 corresponds to the transformation to a randomized environment, with parameter $p = \frac{1}{2}$, if $p_1 \neq 1$. The balanced environment in this case is a randomized environment.

Finally, Formula 6 corresponds to the transformation to a 'bad environment' if $p_1 \neq 1$.

These six processes exhaust all possible cases. We can see that the system either transforms itself into a balanced state or asymptotically tends to it. However, a quadratic recurrence-formula may correspond to *two* different balanced environments. Because of this it is necessary to introduce the concept of stability of balance. Suppose the system is in a balanced state with parameter p . We transform the system from state p into the state with parameter $p + \varepsilon$. Let $p + \varepsilon$ not be a balanced state. If the system tends to transform back into the previous balanced state, this balanced state is a

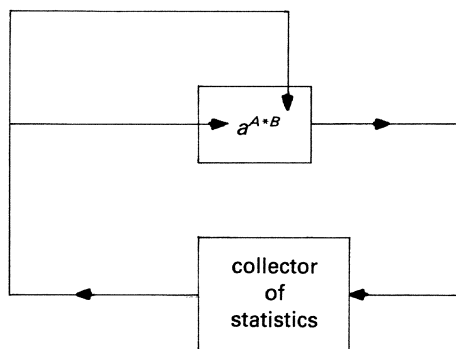


Fig. A3.3. A scheme for generating the parameter of environment by an individual.

stable balance. If the system begins to transform itself into another state, the state from which we took it was an *unstable balance*.

Therefore, any system can be either in a state of balance or can tend to a state of stable balance.

In Tables A3.4 and A3.5, the recurrence-formulae corresponding to all possible interactions of individuals as well as transformations of environments are given. The symbol $\tilde{p}$ stands for the parameter of unstable balance.

The case where an individual himself generates the parameters of the environment can be considered in a similar way. To do so we must connect the inputs with the output through a collector of statistics (Figure. A3.2).

In this case, the evolution of the environment's parameter is described in the first line of Table A3.5. This case may be regarded as a model of self-developing 'stream of consciousness', in which an individual sends demands to himself, reacts to them, and in this way sends new demands and so on.

APPENDIX 4

A METHOD OF CALCULATING MEAN ETHICAL STATUSES

1. REDUCING THE PROBLEM OF CALCULATING MEAN ETHICAL STATUSES TO THE PROBLEM OF CALCULATING THE MODULE OF A FORM

We will call an expression *W* a *form*, if only symbols **a** and **b** are used in its notation. The result of substituting letters $a, \bar{a}, b, \bar{b}$ for **a** and **b** we will call a *formula*. Thus, each form *W* corresponds to 2^M formulae, where *M* is the joint number of **a**'s and **b**'s. Forms of the type **a**, **a^W**, **b**, **b^W** are called *generalized individuals*. Forms of the type $A \cdot B$ and $A + B$ are called *generalized situations*.

We shall demonstrate that if instead of symbols **a** and **b** constituting a form, we substitute various configurations of 0's and 1's and calculate the module of the form (as we calculated the module of a formula), then the obtained number will be equal to the mean of the modules of the formulae that correspond to the given form.

Each form corresponds to a finite number of formulae $F_1, F_2, \dots, F_{2^M}$. The *mean* of their modules is

$$P = \frac{1}{2^M} \sum_{i=1}^{2^M} |F_i|.$$

We introduce the concept of *module of form*. We will call *substitution* of the form the result of replacing each symbol in the form *W* by 0 or 1. The number of all substitutions is finite and equal to 2^M . For each substitution we can calculate the value of the form. By *module of form* we will call the number $|W| = K/2^M$, where *K* indicates how many times the value of the form is equal to 1 (under the condition that *all* values of the form for *all* various substitutions are calculated).

Let us consider an arbitrary formula and fix values $a = \alpha$ and $b = \beta$; then replace all letters $a, \bar{a}, b, \bar{b}$ in the formula by corresponding values of $\alpha, \bar{\alpha}, \beta, \bar{\beta}$. The obtained expression will be called a substitution of the formula.

THEOREM A4.1. *For any form W , $P = |W|$.*

Proof. First we shall prove that each substitution of the form is a

substitution for four different formulae. In an arbitrary substitution of the form we take an arbitrary symbol α (which may be either 1 or 0), substituting any **a** in the form, and an arbitrary symbol β , substituting any **b** in the form. If we suppose that $a = \alpha$ and $b = \beta$ and substitute $a, b, \bar{a}, \bar{b}$ for $\alpha, \beta, \bar{\alpha}, \bar{\beta}$ in the form, we obtain the first formula. Now, we replace α by $\bar{a}$ and β by $\bar{b}$ and obtain the second formula. Similarly, two more formulae can be obtained by replacing α, β by the two remaining pairs $a, \bar{b}$ and $\bar{a}, b$.

It is clear that no other formulae different from the four mentioned above and corresponding to a given substitution of the form can exist. Therefore, each substitution of the form corresponds to four substitutions of the formulae. Note that different substitutions of the form correspond to non-intersecting quadruples of equivalent substitutions of the formulae, and each substitution of the formula belongs to a particular quadruple.

Consider all substitutions of the formula $F_i (i = 1, \dots, 2^M)$ which make the formula equal to 1. The number of all substitutions is equal to $\sum_{i=1}^{2^M} k_i$, where k_i is the number of substitutions under which formula F_i is equal to 1. The number of corresponding substitutions of the form (where it is equal to 1) is four times smaller:

$$k = \frac{1}{4} \sum_{i=1}^{2^M} k_i.$$

Now we shall find the module of the form:

$$|W| = \frac{1}{2^M} \cdot \frac{1}{4} \sum_{i=1}^{2^M} k_i,$$

but

$$\frac{1}{2^M} \cdot \frac{1}{4} \sum_{i=1}^{2^M} k_i = \frac{1}{2^M} \sum_{i=1}^{2^M} \frac{k_i}{4} = \frac{1}{2^M} \sum_{i=1}^{2^M} |F_i| = P.$$

Therefore, $P = |W|$. □

We call an γ -algebra a system with the following rules:

- (1) $0 \leq \gamma \leq 1$,
- (2) $\bar{\gamma} = 1 - \gamma$,
- (3) $\gamma_1 \cdot \gamma_2$ is the regular operation of multiplication of the real numbers,

$$(4) \quad \gamma_1 \oplus \gamma_2 = \overline{\gamma_1 \cdot \gamma_2} = \gamma_1 + \gamma_2 - \gamma_1 \cdot \gamma_2,$$

$$(5) \quad \gamma_1^{\gamma_2} = \bar{\gamma}_2 \oplus \gamma_1.$$

We call Boolean functions φ_1 and φ independent if they do not have common arguments. The following equations are true:

$$(i) \quad |\bar{\varphi}| = \overline{|\varphi|}$$

$$(ii) \quad |\varphi_1 \cdot \varphi_2| = |\varphi_1| \cdot |\varphi_2|$$

$$(iii) \quad |\varphi_1 + \varphi_2| = |\varphi_1| \oplus |\varphi_2|$$

$$(iv) \quad |\varphi_1^{\varphi_2}| = |\varphi_1|^{|\varphi_2|}.$$

Proof of (i). By definition $|\varphi| = K/2^p$, where p is the number of independent arguments of function φ , and K is the number of cases where function φ has value 1. The number of cases where φ has value 0 is equal to $2^p - K$. This number is equal to the number of cases where function $\bar{\varphi}$ has the value 1. Therefore,

$$|\bar{\varphi}| = \frac{2^p - K}{2^p} = 1 - \frac{K}{2^p} = 1 - |\varphi| = \overline{|\varphi|}. \quad \square$$

Proof of (ii). Let function φ_1 have p_1 different independent arguments, and function φ_2 have p_2 different independent arguments. Let the sets of arguments of functions φ_1 and φ_2 not intersect. Then the number of different pairs of values of functions φ_1 and φ_2 generating a set of values of function $\varphi_1 \cdot \varphi_2$ is equal to $2^{p_1 + p_2}$. Function $\varphi_1 \cdot \varphi_2$ is equal to 1, iff both multiplicands are equal to 1. The number of pairs of values where both multiplicands are equal to 1 is $k_1 \cdot k_2$, where k_1 is the number of those combinations of arguments' values (for function φ_1) which make $\varphi_1 = 1$, and k_2 is the number of those combinations of arguments' values (for function φ_2) which make $\varphi_2 = 1$. Therefore,

$$|\varphi_1 \cdot \varphi_2| = \frac{K_1 \cdot K_2}{2^{p_1 + p_2}} = \frac{K_1}{2^{p_1}} \cdot \frac{K_2}{2^{p_2}} = |\varphi_1| \cdot |\varphi_2|. \quad \square$$

Proof of (iii). $\varphi_1 + \varphi_2 = 0$ iff $\varphi_1 = 0$ and $\varphi_2 = 0$. φ_1 has the value 0 ($2^{p_1} - K_1$) times, and φ_2 has the value 0 ($2^{p_2} - K_2$) times. Therefore, $\varphi_1 + \varphi_2$ has

the value 0 $(2^{P_1} - K_1) \cdot (2^{P_2} - K_2)$ times. So, $\varphi_1 + \varphi_2$ has the value 1 $2^{P_1 + P_2} - (2^{P_1} - K_1) \cdot (2^{P_2} - K_2)$ times, and thus

$$\begin{aligned} |\varphi_1 + \varphi_2| &= \frac{2^{P_1 + P_2} - (2^{P_1} - K_1) \cdot (2^{P_2} - K_2)}{2^{P_1 + P_2}} \\ &= 1 - \left(1 - \frac{K_1}{2^{P_1}}\right) \cdot \left(1 - \frac{K_2}{2^{P_2}}\right) = \overline{|\varphi_1|} \cdot \overline{|\varphi_2|} \\ &= |\varphi_1| + |\varphi_2| - |\varphi_1| \cdot |\varphi_2|. \end{aligned} \quad \square$$

Proof of (iv).

$$\begin{aligned} \left| \varphi_1^{\varphi_2} \right| &= |\bar{\varphi}_2 + \varphi_1| = |\bar{\varphi}_2| \oplus |\varphi_1| \\ &= \overline{|\varphi_2|} \oplus |\varphi_1| = |\varphi_1|^{\varphi_2}. \end{aligned} \quad \square$$

Let us consider generalized individuals A_1 of five different sorts:

$$\begin{aligned} A_1 &\equiv \mathbf{a}, & A_1 &\equiv \mathbf{a}^{A_2}, & A_1 &\equiv \mathbf{a}^{B_2}, \\ A_1 &\equiv \mathbf{a}^{A_2 \cdot B_2}, & A_1 &\equiv \mathbf{a}^{A_2 + B_2}. \end{aligned}$$

It is easy to see that the following equations are true:

$$\begin{aligned} |A_1| &= |\mathbf{a}|, & |A_1| &= |\mathbf{a}|^{A_2}, & |A_1| &= |\mathbf{a}|^{B_2}, \\ |A_1| &= |\mathbf{a}|^{A_2 \cdot B_2}, & |A_1| &= |\mathbf{a}|^{A_2 \oplus B_2}. \end{aligned}$$

Similar equalities may be written for the modules of the generalized individual B_1 . Let us call the procedure of transition from the left side of the equality to its right side a *step*.

Consider an arbitrary generalized individual A . To each letter in the formula of a generalized individual we put into correspondence a generalized individual for whom this given letter is a root. We say that a generalized individual belongs to level i if its root is located on i 's level of an individual A .

Suppose all generalized individuals from i 's level are included in the 'lines of module' $||$. After making a step we come to an expression where all

generalized individuals from $(i + 1)$'s level (if they exist) are included into the lines of module, and all signs of relationships on $(i + 1)$'s level (if they exist) are replaced by the signs of γ -algebra. Let an individual A have n levels. Then by including an individual A into the lines of module we move on the $(n - 1)$ th step to the expression where each letter is replaced by its module and all signs of operations of Boolean algebra are replaced by the signs of the corresponding operations of γ -algebra. As a result, we have a procedure for calculating the module of the form corresponding to a given generalized individual, and knowing this procedure we can calculate modules of generalized situations

$$|A \cdot B| = |A| \cdot |B| \quad \text{and} \quad |A + B| = |A| \oplus |B|.$$

Thus we have a method for calculating the modules of arbitrary forms.

Now we can demonstrate how we calculated the ethical statuses in Chapter XIII. The mean module of an image of oneself, $\mathbf{a}^{\mathbf{a} + \mathbf{b}}$, is

$$\begin{aligned} |\mathbf{a}^{\mathbf{a} + \mathbf{b}}| &= \frac{1}{2}^{1/2 \oplus 1/2} = \frac{1}{2}^{1/2 + 1/2 - 1/2 \cdot 1/2} = \\ &= \frac{1}{2}^{3/4} = (1 - \frac{3}{4}) \oplus \frac{1}{2} = \frac{1}{4} + \frac{1}{2} - \frac{1}{4} \cdot \frac{1}{2} = \frac{5}{8}. \end{aligned}$$

The mean ethical status of the individual represented by the form $\mathbf{a}^{\mathbf{a} \cdot \mathbf{b} \cdot \mathbf{b}}$ is equal to

$$\begin{aligned} |\mathbf{a}^{\mathbf{a} \cdot \mathbf{b} \cdot \mathbf{b}}| &= \frac{1}{2}^{1/2 \cdot \frac{1}{2} \cdot \frac{1}{2}} = \frac{1}{2}^{\frac{1}{4}} \\ &= \frac{1}{2}^{(3/4 \oplus 1/2) \cdot 1/2} = \frac{1}{2}^{(3/4 + 1/2 - 3/4 \cdot 1/2) \cdot 1/2} = \frac{1}{2}^{7/16} \\ &= (1 - \frac{7}{16}) \oplus \frac{1}{2} = \frac{9}{16} + \frac{1}{2} - \frac{9}{16} \cdot \frac{1}{2} = \frac{25}{32} \approx 0.781. \end{aligned}$$

The mean module of the situation is calculated as follows.

$$\mathbf{a}^{\mathbf{a} \cdot \mathbf{b} \cdot \mathbf{b}} \cdot \mathbf{b}^{\mathbf{b} \cdot \mathbf{a} \cdot \mathbf{a}}$$

is a form of interaction of two heroes. The module of this expression is

$$|\mathbf{a}^{\mathbf{a} \cdot \mathbf{b} \cdot \mathbf{b}}| \cdot |\mathbf{b}^{\mathbf{b} \cdot \mathbf{a} \cdot \mathbf{a}}| \approx 0.610.$$

The modules in the straight brackets are equal to each other, since they differ only in the names of symbols. The mean module of an individual was calculated in the previous example. Therefore, the module of the situation is $\left(\frac{25}{32}\right)^2 \approx 0.610$.

The remaining mean modules are calculated similarly.

2. LIMITING EXPRESSIONS

Now we will study the behavior of modules as the number of levels of form increases without bound. Let us analyze the three following recurrent expressions generating three different groups of forms:

$$W_{n+1} \equiv a^{W_n}, \quad W_1 = \mathbf{a}, \quad (\text{A4.1})$$

$$W_{n+1} \equiv a^{W_n + W_n}, \quad W_1 = \mathbf{a}, \quad (\text{A4.2})$$

$$W_{n+1} \equiv a^{W_n \cdot W_n}, \quad W_1 = \mathbf{a}. \quad (\text{A4.3})$$

To simplify the representation we do not use the symbol **b** in the expressions (A4.2) and (A4.3). Instead of **b** we use the symbol **a**, which is equivalent to **b** in everything, since each instance of a letter in these expressions is understood as an independent variable. By using the results of the previous part of the Appendix we can find the modules of expressions (A4.1), (A4.2), (A4.3):

$$|W_{n+1}| = |a|^{|W_n|},$$

$$|W_{n+1}| = |a|^{|W_n| \oplus |W_n|},$$

$$|W_{n+1}| = |a|^{|W_n| \cdot |W_n|}.$$

Or by changing notation:

$$U_{n+1} = \frac{1}{2}U_n,$$

$$U_{n+1} = \frac{1}{2}U_n \oplus U_n,$$

$$U_{n+1} = \frac{1}{2}U_n \cdot U_n,$$

and transforming them to the usual appearance:

$$U_{n+1} = -\frac{1}{2}U_n + 1, \quad (\text{A4.1}')$$

$$U_{n+1} = \frac{1}{2}U_n^2 - U_n + 1, \quad (\text{A4.2}')$$

$$U_{n+1} = -\frac{1}{2}U_n^2 + 1. \quad (\text{A4.3}')$$

Now we can find the limits of the sequences represented by these recurrence-formulae if $n \rightarrow \infty$.

It is easy to see (Appendix 3), that such limits exist and are equal to:

$$\lim_{n \rightarrow \infty} U_n = \frac{2}{3}. \quad (\text{A4.1}'')$$

$$\lim_{n \rightarrow \infty} U_n = 2 - \sqrt{2}, \quad (\text{A4.2}'')$$

$$\lim_{n \rightarrow \infty} U_n = \sqrt{3} - 1. \quad (\text{A4.3}'')$$

3. THE CHARACTER OF CHANGING MODULES OF FORMULAE WHEN CHANGING SIGNS

Let us analyze the limiting values of the sequences (A4.2) and (A4.3) calculated above. The sequence (A4.2) consists of forms with the sign ‘+’ only. The limit of the modules of such forms is $2 - \sqrt{2} = 0.586$.

TABLE A4.1

Limit values of individual's modules, of their images of themselves, inner worlds, and situations.

Sign	+	.
Individual	$2 - \sqrt{2} = 0.586$	$\sqrt{3} - 1 = 0.732$
Image of himself		
Inner world	$(2 - \sqrt{2}) \oplus (2 - \sqrt{2}) = 0.829$	$(\sqrt{3} - 1)^2 = 0.535$
Situation		

Sequence (A4.3) consists of only those forms with sign ‘.’. The limit of modules of these forms is $\sqrt{3} - 1 = 0.732$.

According to the theorem on reduction (proved in the first section of this appendix) we can regard these limits as the mean ethical statuses of individuals corresponding to these forms. We can see that the limit of the means of the ethical statuses of individuals with the sign ‘.’ only is larger than that of individuals with the sign ‘+’ only. Thus, by changing his relation with his partner from the sign ‘+’ to the sign ‘.’ an individual increases, on the average, his ethical status not only for the simplest cases, but with an unlimited increase in ‘levels of awareness’ also.

In Table A4.1 we give the limits of mean modules of individuals, images of himself, inner worlds, and situations.

In conclusion, let us note that we have proved *only* that starting from a certain level the relation ‘greater than’ between the modules will not change as the number of levels increases. We did not evaluate the number of that initial level. The proof solves the problem ‘in principle’, but examination is necessary in each concrete case. Only the first three levels are psychologically interesting. Direct verification of these cases has demonstrated that the discovered regularity is correct for them.

APPENDIX 5

TYPES OF ADEQUACY OF REFLEXION

In this Appendix we shall give a classification of the types of correspondence between images and originals. We shall formulate and prove some important statements related to our classification.

We designate the root of individual A as 'Root A ' and the reflexive structure of individual A as 'Ref A '. (The concept of reflexive structure was introduced in Appendix 2.) Then we shall analyze the two expressions:

$$a^{A_2} \equiv A_1, \quad a^{B_2} * B_1.$$

The first one will be used for the analysis of one's self-reflexion, the second one for the analysis of one's partner's reflexion.

We shall distinguish the following types of correspondence between an image and an original.

1. CORRECTNESS

A has a correct image of himself if $\text{Root } A_1 = \text{Root } A_2$.

A has a correct image of his partner if $\text{Root } B_1 = \text{Root } B_2$.

The sign '=' means the coincidence of symbols on the right and on the left. We have used such an interpretation of correctness throughout. An individual may have correct images both of himself and of his partner. Two interacting individuals may have correct images of each other.

2. REFLEXIVE ADEQUACY

A has a reflexive adequate image of himself if $\text{Ref } A_1 = \text{Ref } A_2$.

A has a reflexive adequate image of his partner if $\text{Ref } B_1 = \text{Ref } B_2$.

The sign '=' means a coincidence of reflexive structures on the right and on the left.

We call a *route* a way without return from a root to an arbitrary node of a tree of a reflexive structure. We call the *height* of a tree the length of the maximal route. Let us consider the equation $A_1 \equiv a^{A_2}$.

It is easy to see that if the height of tree A_2 is equal to k , then the height of

tree A_1 is equal to $k + 1$. It follows from this that an individual cannot have a reflexive adequate image of himself.

Let an individual A have a reflexive adequate image of an individual B . Then an individual B cannot have a reflexive adequate image of A . Suppose the opposite is true. Let both individuals have reflexive adequate images of each other. Since A has a reflexive adequate image of B , the height of tree A is greater than that of tree B . On the other hand, since B has a reflexive adequate image of A , the height of tree B is greater than that of tree A . From this contradiction it follows that two individuals cannot have reflexive adequate images of each other at the same time.

3. ETHICAL ADEQUACY

Individual A has an ethically adequate image of himself if $|A_1| = |A_2|$.

A has an ethically adequate image of his partner if $|B_1| = |B_2|$.

It is easy to see that individuals with an ethical status of $\frac{1}{2}$ or $\frac{3}{4}$ may have ethically adequate images of themselves.

Examples:

$$A \equiv a^{\bar{a}a}; \quad |A| = \frac{1}{2}; \quad |\bar{a}^a| = \frac{1}{2}.$$

$$A \equiv a^{ab}; \quad |A| = \frac{3}{4}; \quad |a^b| = \frac{3}{4}.$$

In Chapter XII, we demonstrated that a perfect individual (an individual with ethical status equal to 1) cannot have an image of himself as a perfect individual. This means that a perfect individual cannot have an ethically adequate image of himself (but his image of himself is always correct!).

Two interacting individuals may have ethically adequate images of each other. For example,

$$a^{b\bar{a}} + b^{a\bar{b}}, \quad |a^{b\bar{a}}| = |a^{\bar{b}}| = \frac{3}{4}, \quad |b^{a\bar{b}}| = |b^{\bar{a}}| = \frac{3}{4}.$$

4. FUNCTIONAL ADEQUACY

A has a functionally adequate image of himself if $A_1 = A_2$.

A has a functionally adequate image of his partner if $B_1 = B_2$.

Sign '=' means logical equality.

THEOREM A5.1. *If individual A has a functionally adequate image of B , then B may have a functionally adequate image of A .*

Proof. Statements of this theorem may be written down as a system of two Boolean equations:

$$\varphi_A = a^{\varphi_B} \quad \varphi_B = b^{\varphi_A}$$

where φ_A and φ_B are Boolean functions corresponding to individuals A and B .

It is easy to see that this system has two simple solutions:

$$(1) \quad \varphi_A = a \quad \varphi_B = b^a$$

$$(2) \quad \varphi_A = a^b \quad \varphi_B = b.$$

□

THEOREM A5.2. *An individual cannot have a functionally adequate image of himself.*

Proof. Let φ be a Boolean function corresponding to individual A . The statement of the theorem is reduced to the statement that the Boolean equation $\varphi = a^\varphi$, where φ is unknown, does not have a solution.

When $a = 0$, $\varphi = \bar{\varphi}$. This contradiction proves the theorem.

□

The statement of Theorem A5.2 is equivalent to the similar statement for reflexive structure. However, we also have the following unexpected result:

THEOREM A5.3. *Individual A may have an image of B , which possesses a functionally adequate image of A .*

Proof. Let φ be a Boolean function corresponding to individual A . The theorem is reduced to the statement that the Boolean equation $\varphi = a^{b^\varphi}$ has at least one solution.

Let us show that $\varphi = a$ is a solution. Put this value into the right part of one equation and get $\varphi = a^{b^a}$. After taking the logarithm twice, we have

$$a^{b^a} = LLa^{b^a} = Lb^aLa = aLb + a = a(Lb + 1) = a.$$

□

Thus, Theorems A5.2 and A5.3 indicate that individual A cannot have a functionally correct image of himself, but he may have an image of another individual who possesses a functionally correct image of A . It is interesting that this 'other one' may be A himself. To prove this statement let us consider the Boolean equation $\varphi = a^{a^\varphi}$; its simplest solution is $\varphi = a$. Thus, individual A cannot have a functionally correct image of himself, but he may see some individual B or himself possessing a functionally correct image of A .

5. ABSOLUTE ADEQUACY

A has an absolutely adequate image of his partner if $B_1 \equiv B_2$, where ' $\equiv$ ' is a sign of semantic equality (see Chapter IV). An individual cannot have an absolutely adequate image of himself, since to accomplish this the following identity has to be true: $A \equiv a^A$. But this identity is false, since its right-hand part cannot be turned into its left-hand part by using commutative transformations. (Commutative transformations do not change the number of tiers. But in the identity above the left-hand part has K tiers, and the right-hand part has $K + 1$ tiers.)

Let us now examine the logical connections between various types of adequacy. If an individual has an absolutely adequate image of his partner, this image is correct, reflexively adequate, ethically adequate, and functionally adequate. If an individual has a functionally adequate image of himself or of his partner, this image is also ethically adequate. However, correctness, ethical adequacy, and reflexive adequacy do not imply other types of adequacy.

APPENDIX 6

SCHEMES OF EMPIRICAL PROCEDURES

To construct a formula of interaction we need representations of personae. To get this we ask an expert to answer a set of binary questions. Here is a scheme for constructing a person's structure (Fig. A6.1).

All questions, except the first and second, depend on others. This is shown with arrows. If a question is answered 'NO', its dependent questions are not asked.

As a sample we construct a structure for Hamlet (See p. 46 of this book). The author himself was the expert.

Answer	Construction of formula
1. YES	a^a
2. YES	$a^{a?b}$ relationship unknown
3. YES	a^{a+b}
4. YES	a^{a+b} no change
5. YES	a^{a+b} no change
6. YES	a^{aa+b}
7. YES	$a^{aa?b+b}$
8. NO	$a^{aa?b+b}$ no change
9. NO	$a^{aa?b+b}$ no change
10. YES	a^{aa+b+b}
12. YES	a^{aa+b+b}
13. YES	$a^{aa+\tilde{b}+b}$

Finally $H \equiv a^{aa+\tilde{b}+b}$.

All other formulae for literary personages are constructed similarly.

The experiments described in Chapter X were conducted with similar questionnaires. However, the above scheme is too complex. The author made some statements *a priori* before the experiments. In Figure A6.2 the questions in oval frames were asked, and the questions in rectangular frames were supposed to have an *a priori* 'YES'.

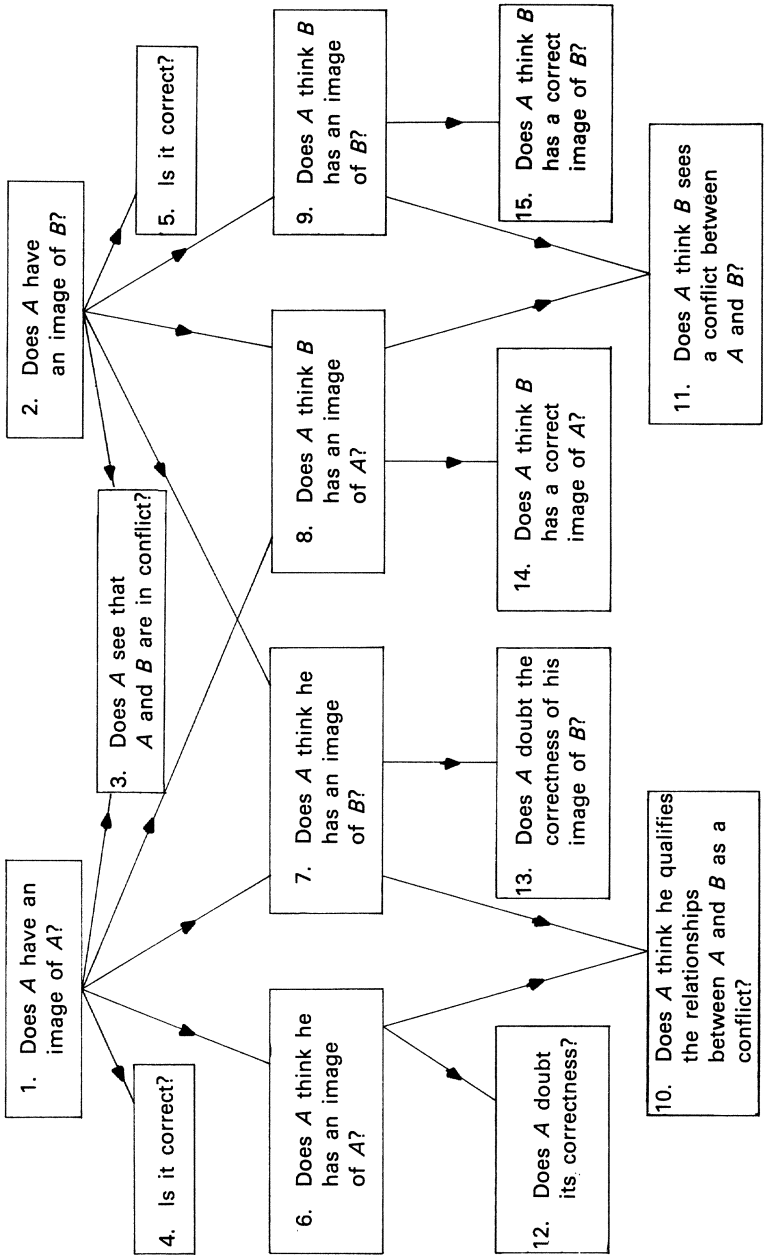


Fig. A6.1. An algorithm for constructing a formula in experiments 'Tom and John' and 'Pavlik Morozov'.

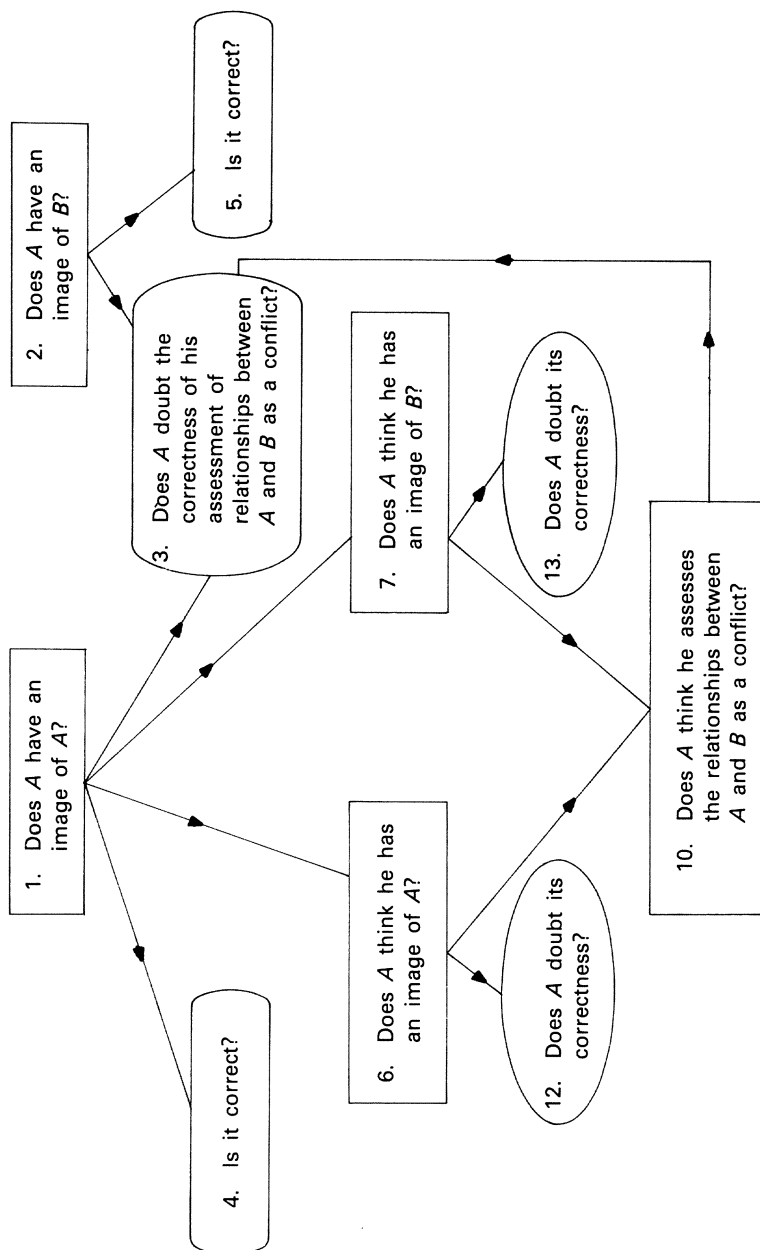


Fig. A6.2. An algorithm for constructing a formula in the 'Kennedy-Nixon' experiment.

For example,

Answer	Formula
1. YES <i>a priori</i>	a^a
2. YES <i>a priori</i>	$a^a?b$
4. NO	$a^{\bar{a}}?b$
5. YES	$a^{\bar{a}}?b$ no change
6. YES <i>a priori</i>	$a^{\bar{a}}a?b$
7. YES <i>a priori</i>	$a^{\bar{a}}a?b?b$
12. NO	$a^{\bar{a}}\bar{a}?b?b$
13. YES	$a^{\bar{a}}\bar{a}?b?b$
10. YES <i>a priori</i>	$a^{\bar{a}}\bar{a}+b?b$
3. YES	$a^{\bar{a}}\bar{a}+b.b$

Finally $H \equiv a^{\bar{a}\bar{a}+b.b}$.

It is supposed for constructing Figure A6.2 that questions 8 and 9 of Figure A6.1 have answer NO *a priori*.

What is important is that answer 3 is given by contrast with answer 10. If answer 3 is YES (doubt), this means A sees a union; if answer 3 is NO (no doubts), this means A sees a conflict.

APPENDIX 7

TABLES

TABLE A7.1

All possible different three-tier structures representing an individual *A*, who thinks he is in conflict with *B* and knows that both he and *B* see their relationships as one of conflict.

First ethical system		Second ethical system	
Formula	Ethical status	Formula	Ethical status
1. $a^{aa+b+bb+a}$	$\frac{1}{2}$	$a^{aa \cdot b, bb \cdot a}$	$\frac{1}{2}$
2. $a^{a\bar{a}+b+bb+a}$	$\frac{1}{2}$	$a^{a\bar{a} \cdot b, bb \cdot a}$	$\frac{3}{4}$
3. $a^{aa+\bar{b}+bb+a}$	$\frac{1}{2}$	$a^{aa \cdot \bar{b}, bb \cdot a}$	$\frac{1}{2}$
4. $a^{a\bar{a}+\bar{b}+bb+a}$	$\frac{1}{2}$	$a^{a\bar{a} \cdot \bar{b}, bb \cdot a}$	$\frac{3}{4}$
5. $a^{aa+b+bb+\bar{a}}$	$\frac{1}{2}$	$a^{aa \cdot b, bb \cdot \bar{a}}$	$\frac{1}{2}$
6. $a^{aa+b+bb+a\bar{a}}$	$\frac{1}{2}$	$a^{aa \cdot b, bb \cdot \bar{a}}$	$\frac{1}{2}$
7. $a^{aa+b+bb+\bar{a}\bar{a}}$	$\frac{1}{2}$	$a^{aa \cdot b, bb \cdot \bar{a}}$	$\frac{3}{4}$
8. $a^{a\bar{a}+b+bb+a}$	$\frac{3}{4}$	$a^{a\bar{a} \cdot b, bb \cdot a}$	$\frac{3}{4}$
9. $a^{a\bar{a}+b+bb+a\bar{a}}$	$\frac{3}{4}$	$a^{a\bar{a} \cdot b, bb \cdot \bar{a}}$	$\frac{3}{4}$
10. $a^{a\bar{a}+b+bb+\bar{a}\bar{a}}$	$\frac{3}{4}$	$a^{a\bar{a} \cdot b, bb \cdot \bar{a}}$	1
11. $a^{aa+\bar{b}+bb+a}$	$\frac{3}{4}$	$a^{aa \cdot \bar{b}, bb \cdot a}$	$\frac{1}{2}$
12. $a^{aa+\bar{b}+bb+\bar{a}}$	$\frac{3}{4}$	$a^{aa \cdot \bar{b}, bb \cdot \bar{a}}$	$\frac{1}{2}$
13. $a^{aa+\bar{b}+bb+\bar{a}\bar{a}}$	$\frac{3}{4}$	$a^{aa \cdot \bar{b}, bb \cdot \bar{a}}$	$\frac{3}{4}$
14. $a^{a\bar{a}+\bar{b}+bb+a}$	$\frac{3}{4}$	$a^{a\bar{a} \cdot \bar{b}, bb \cdot a}$	$\frac{3}{4}$
15. $a^{a\bar{a}+\bar{b}+bb+a\bar{a}}$	$\frac{3}{4}$	$a^{a\bar{a} \cdot \bar{b}, bb \cdot \bar{a}}$	$\frac{3}{4}$
16. $a^{a\bar{a}+\bar{b}+bb+\bar{a}\bar{a}}$	$\frac{3}{4}$	$a^{a\bar{a} \cdot \bar{b}, bb \cdot \bar{a}}$	$\frac{3}{4}$
17. $a^{\bar{a}\bar{a}+b+bb+a}$	$\frac{1}{2}$	$a^{\bar{a}\bar{a} \cdot b, bb \cdot a}$	$\frac{1}{2}$
18. $a^{\bar{a}\bar{a}+b+bb+a\bar{a}}$	$\frac{1}{2}$	$a^{\bar{a}\bar{a} \cdot b, bb \cdot a}$	$\frac{1}{2}$
19. $a^{\bar{a}\bar{a}+\bar{b}+bb+a}$	$\frac{1}{2}$	$a^{\bar{a}\bar{a} \cdot \bar{b}, bb \cdot a}$	$\frac{1}{2}$

Table A7.1 continued

First ethical system		Second ethical system	
Formula	Ethical status	Formula	Ethical status
20. $a\bar{a} + \bar{b} + b\bar{b} + a$	$\frac{1}{2}$	$a\bar{a} \cdot \bar{b} \cdot b\bar{b} \cdot a$	$\frac{1}{2}$
21. $a\bar{a} + b + b\bar{b} + a$	$\frac{1}{2}$	$a\bar{a} \cdot b \cdot b\bar{b} \cdot a$	$\frac{1}{2}$
22. $a\bar{a} + b + b\bar{b} + \bar{a}$	$\frac{1}{2}$	$a\bar{a} \cdot b \cdot b\bar{b} \cdot \bar{a}$	$\frac{1}{2}$
23. $a\bar{a} + b + b\bar{b} + \bar{a}$	$\frac{1}{2}$	$a\bar{a} \cdot b \cdot b\bar{b} \cdot \bar{a}$	$\frac{3}{4}$
24. $a\bar{a} + b + b\bar{b} + a$	$\frac{1}{2}$	$a\bar{a} \cdot b \cdot b\bar{b} \cdot a$	$\frac{1}{2}$
25. $a\bar{a} + b + b\bar{b} + \bar{a}$	$\frac{1}{2}$	$a\bar{a} \cdot b \cdot b\bar{b} \cdot \bar{a}$	$\frac{1}{2}$
26. $a\bar{a} + b + b\bar{b} + \bar{a}$	$\frac{1}{2}$	$a\bar{a} \cdot b \cdot b\bar{b} \cdot \bar{a}$	$\frac{3}{4}$
27. $a\bar{a} + \bar{b} + b\bar{b} + a$	$\frac{1}{2}$	$a\bar{a} \cdot \bar{b} \cdot b\bar{b} \cdot a$	$\frac{1}{2}$
28. $a\bar{a} + \bar{b} + b\bar{b} + \bar{a}$	$\frac{1}{2}$	$a\bar{a} \cdot \bar{b} \cdot b\bar{b} \cdot \bar{a}$	$\frac{1}{2}$
29. $a\bar{a} + \bar{b} + b\bar{b} + \bar{a}$	$\frac{1}{2}$	$a\bar{a} \cdot \bar{b} \cdot b\bar{b} \cdot \bar{a}$	$\frac{3}{4}$
30. $a\bar{a} + \bar{b} + b\bar{b} + a$	$\frac{1}{2}$	$a\bar{a} \cdot \bar{b} \cdot b\bar{b} \cdot a$	$\frac{1}{2}$
31. $a\bar{a} + \bar{b} + b\bar{b} + a$	$\frac{1}{2}$	$a\bar{a} \cdot \bar{b} \cdot b\bar{b} \cdot \bar{a}$	$\frac{1}{2}$
32. $a\bar{a} + \bar{b} + b\bar{b} + \bar{a}$	$\frac{1}{2}$	$a\bar{a} \cdot \bar{b} \cdot b\bar{b} \cdot \bar{a}$	$\frac{3}{4}$

TABLE A7.2

All possible different three-tier structures representing an individual A , who thinks that he is in conflict with B and that B does not have any images. Structures 1–8 correspond to cases where A does not doubt the correctness of his assessment of his relationships with B ; structures 9–16 correspond to cases where A has such doubts.

First ethical system		Second ethical system	
Formula	Ethical status	Formula	Ethical status
1. a^{aa+b+b}	$\frac{1}{2}$	$a^{aa \cdot b \cdot b}$	$\frac{3}{4}$
2. $a^{a\bar{a}+b+b}$	$\frac{3}{4}$	$a^{a\bar{a} \cdot b \cdot b}$	1
3. $a^{aa+\bar{b}+b}$	$\frac{3}{4}$	$a^{aa \cdot \bar{b} \cdot b}$	$\frac{3}{4}$
4. $a^{a\bar{a}+\bar{b}+b}$	$\frac{3}{4}$	$a^{a\bar{a} \cdot \bar{b} \cdot b}$	$\frac{3}{4}$
5. $a^{\bar{a}\bar{a}+b+b}$	$\frac{1}{2}$	$a^{\bar{a}\bar{a} \cdot b \cdot b}$	$\frac{3}{4}$
6. $a^{\bar{a}\bar{a}+\bar{b}+b}$	$\frac{1}{2}$	$a^{\bar{a}\bar{a} \cdot \bar{b} \cdot b}$	$\frac{3}{4}$
7. $a^{\bar{a}a+\bar{b}+b}$	$\frac{1}{2}$	$a^{\bar{a}a \cdot \bar{b} \cdot b}$	$\frac{3}{4}$
8. $a^{\bar{a}\bar{a}+\bar{b}+b}$	$\frac{1}{2}$	$a^{\bar{a}\bar{a} \cdot \bar{b} \cdot b}$	$\frac{3}{4}$
9. $a^{aa+b \cdot b}$	1	$a^{aa \cdot b+b}$	$\frac{1}{2}$
10. $a^{a\bar{a}+b \cdot b}$	1	$a^{a\bar{a} \cdot b+b}$	$\frac{1}{2}$
11. $a^{aa+\bar{b} \cdot b}$	$\frac{3}{4}$	$a^{aa \cdot \bar{b}+b}$	$\frac{1}{2}$
12. $a^{a\bar{a}+\bar{b} \cdot b}$	1	$a^{a\bar{a} \cdot \bar{b}+b}$	$\frac{1}{2}$
13. $a^{\bar{a}a+b \cdot b}$	$\frac{3}{4}$	$a^{\bar{a}a \cdot b+b}$	$\frac{1}{2}$
14. $a^{\bar{a}\bar{a}+b \cdot b}$	$\frac{3}{4}$	$a^{\bar{a}\bar{a} \cdot b+b}$	$\frac{1}{2}$
15. $a^{\bar{a}a+\bar{b} \cdot b}$	$\frac{3}{4}$	$a^{\bar{a}a \cdot \bar{b}+b}$	$\frac{1}{2}$
16. $a^{\bar{a}\bar{a}+\bar{b} \cdot b}$	$\frac{3}{4}$	$a^{\bar{a}\bar{a} \cdot \bar{b}+b}$	$\frac{1}{2}$

APPENDIX 8

PROBLEMS OF SUBSTANTIATING THE INITIAL AXIOMS IN AN ARBITRARY ENVIRONMENT

In this Appendix we shall generalize the substantiation of our model (see Conclusion) to the case of arbitrary frequencies of 'good' environmental demands.

In order to interpret and calculate expressions of the type $\mathbf{a}^{**b}$, we have to establish two commutative operations, $*_1$ and $*_2$ and a non-commutative algebra which allows calculation of exponents. In other words, we have to establish two Boolean functions $\varphi_1(a, b)$ and $\varphi_2(a, b)$ with commuting arguments and one Boolean function $f(a, b)$ with non-commuting arguments. Functions φ_1 and φ_2 model the normative evaluation of ethical confrontation and ethical compromise, and function f models a process of reflexion.

Let us define an ethical system by the three functions $(f, \varphi_1, \varphi_2)$ if it is known which one of φ_1 and φ_2 models confrontation and which compromise. Each triple of functions corresponds to the two ethical systems. There exist 16 different Boolean functions f and 28 different pairs of functions (φ_1, φ_2) , since the number of pairs is equal to the number of combinations of 2 from 8. Thus, the number of the different triples is equal to $16 \times 28 = 448$, and the number of ethical systems is $448 \times 2 = 996$.

We define a parameter of the environment p as the frequency of its 'good' demands presented to an individual.

We call the triple $(f, \varphi_1, \varphi_2)$ 'appropriate' in an environment with a parameter p , if the four following expressions are true at the same time:

- (1) $|a^a| > |a^{\bar{a}}|,$
- (2) $|a^{a\bar{a}}| > |a^{\bar{a}\bar{a}}|,$
- (3) $|a^{a\bar{a}}| > |a^{\bar{a}a}|,$
- (4) $|a^{a*b}| > |a^{\bar{a}*b}|.$

The requirements which we impose on the three functions differ from those we presented in the Conclusion for the randomized case. First, we

TABLE A8.1
Values of substitutions and modules of formulae α^e and α^s depending on frequency p and on the number of algebra.

Formula	Frequency	Substitution	Number of algebra from Table C1															
			1	2	3	4	⑤	6	⑦	8	⑨	⑩	11	12	⑬	⑭	⑮	16
α^a	p	$1^1 =$	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0
	$1-p$	$0^0 =$	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0
$\alpha^{\bar{a}}$		Module	1	p	$1-p$	0	1	p	$1-p$	0	1	p	$1-p$	0	1	p	$1-p$	0
	p	$1^0 =$	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
	$1-p$	$0^1 =$	1	1	1	1	0	0	0	1	1	1	1	0	0	0	0	0
		Module	1	1	1	1	p	p	p	p	$1-p$	$1-p$	$1-p$	$1-p$	0	0	0	0

added inequality (3), which may be read as follows: ‘Of the two individuals who doubt the correctness of their image of themselves, the one who has a correct image of himself has a higher ethical status than the one who has an incorrect image of himself.’ Second, we changed inequality $|a^{a*b}| > |\bar{a}^{a*b}|$ to inequality (4):

$$|a^{a*b}| > |\bar{a}^{a*b}|,$$

which may be read: “An individual who has correct images of himself and of his partner has a higher ethical status than another one who has incorrect images of himself and of his partner.”

We suppose that individual A is considered in a context of interaction with individual B with the root b . If B obtains ‘good’ demands from the environment with frequency p , then A who has an incorrect image of B would believe that ‘good’ demands from the environment come with frequency $1 - p$. It is worth noting that for a randomized environment, analyzed in the Conclusion, incorrectness of an image of a partner was not important, since in that case $p = 1 - p$.

THEOREM A8.1. *The triple $(f, \varphi_1, \varphi_2)$ is appropriate in an environment with parameter $p(0 \leq p < 1)$, iff f is the logical implication and φ_1, φ_2 are operations of Boolean algebra ‘+’ and ‘·’.*

Proof. The proof is cumbersome, but we present it completely because of the importance of the Theorem.

(1) We find all functions f which satisfy inequality (1). These are shown in Table A8.1 The number of each function corresponds to the number of an algebra in Table C1. We see that inequality (1) can be true for functions 5, 7, 9, 10, 13, 14, and 15.

(2) We find now all f which satisfy both expressions (1) and (2). These are shown in Table A8.2 for the functions found in Table A8.1. We see that inequalities (1) and (2) can be simultaneously satisfied by functions 5 and 13.

(3) We examine whether functions 5 and 13 satisfy inequality (3). Consider Table A8.3. We see that only function 5 satisfies inequality (3); therefore this is the only function satisfying all three inequalities (1), (2), (3). But function 5 is the logical implication; so the first statement of the Theorem is proven.

(4) We find all functions φ satisfying inequality (4). To do this, we examine Table A8.4. The numbers of the different functions correspond to the numbers of operations given in Table C3. Function f is the logical

TABLE A8.2

Values of substitutions and modules of formulae $a^{a\bar{a}}$ and $a^{\bar{a}a}$ depending on frequency p and on the number of algebra.

Formula	Frequency	Substitution	Number of algebra from Table C1						
			⑤	7	9	10	⑬	14	15
$a^{a\bar{a}}$	p	$1^{1^0} =$	1	0	0	0	0	0	0
	$1-p$	$0^{0^1} =$	1	1	1	1	1	0	1
		Module	1	$1-p$	$1-p$	$1-p$	$1-p$	0	$1-p$
$a^{\bar{a}a}$	p	$1^{0^0} =$	1	0	1	0	1	0	0
	$1-p$	$0^{1^1} =$	0	1	1	1	0	0	1
		Module	p	$1-p$	1	$1-p$	p	0	$1-p$

TABLE A8.3

Values of substitutions and modules of formulae $a^{a\bar{a}}$ and $a^{\bar{a}a}$ depending on frequency p and on the number of algebra.

Formula	Frequency	Substitution	Number of algebra from Table C1	
			⑤	13
$a^{a\bar{a}}$	p	$1^{1^0} =$	1	0
	$1-p$	$0^{0^1} =$	1	1
		Module	1	$1-p$
$a^{\bar{a}a}$	p	$1^{0^1} =$	1	0
	$1-p$	$0^{1^0} =$	0	1
		Module	p	$1-p$

implication. We see that only functions (operations) 2 and 7 satisfy inequality (4).

For function 2:

$$p + (1-p)^2 > p, \quad 0 \leq p < 1.$$

For function 7:

$$1 > p + (1-p)p,$$

which is equal to $(1-p)^2 > 0, 0 \leq p < 1$. This completes the proof. $\square$

TABLE A8.4
Values of substitutions and modules of formulae a^{a^*b} and a^{a^*b} depending on frequency p and the number of algebra.

Formula	Frequency	Substitution	Number of operation from Table C3	1	②	3	4	5	6	⑦	8
a^{a^*b}	$(1-p) \cdot p$	$0^{0*1} =$	0	0	1	0	0	1	0	1	1
	$(1-p)^2$	$0^{0*0} =$	0	1	0	0	0	0	1	1	1
	p^2	$1^{1*1} =$	1	1	1	1	1	1	1	1	1
	$p(1-p)$	$1^{1*0} =$	1	1	1	1	1	1	1	1	1
		Module	p	$p + (1-p)^2$	$p + (1-p) \cdot p$	$p + (1-p) \cdot p$	$p + (1-p) \cdot p$	$p + (1-p)^2$	$p + (1-p)^2$	1	1
a^{a^*b}	$(1-p) \cdot p$	$0^{1*0} =$	0	0	1	0	0	1	0	1	1
	$(1-p)^2$	$0^{1*1} =$	0	0	0	1	1	1	1	0	1
	p^2	$1^{0*0} = 1$	1	1	1	1	1	1	1	1	1
	$p \cdot (1-p)$	$1^{0*1} =$	1	1	1	1	1	1	1	1	1
		Module	p	p	$p + (1-p) \cdot p$	$p + (1-p)^2$	$p + (1-p) \cdot p$	1	$p + (1-p)^2$	$p + (1-p) \cdot p$	1

From the Theorem above it follows that only two different ethical systems exist, and they correspond to the same appropriate triple $(\supset, +, \cdot)$; this fact does not depend on the parameter of the environment if $0 \leq p < 1$. For the environment with $p = 1$ there is no appropriate set of $(f, \varphi_1, \varphi_2)$ at all, which is clear from a simple analysis of the Tables.

APPENDIX 9

ANOTHER METHOD OF REPRESENTING INDIVIDUALS

In this Appendix we point out the possibility of representing an individual using a method different from the one employed throughout this book. Consider an individual

$$A_1 = \mathbf{a}^{A_2 * B_2}.$$

Let us raise both parts of this logical equality to the power $A_2 * B_2$:

$$A_1^{A_2 * B_2} = (\mathbf{a}^{A_2 * B_2})^{A_2 * B_2} = \mathbf{a}^{A_2 * B_2}.$$

From this we can see that

$$A_1 = A_1^{A_2 * B_2}.$$

Let

$$A_2 = \mathbf{a}^{A_{3_1} * B_{3_1}} \quad \text{and} \quad B_2 = \mathbf{b}^{A_{3_2} * B_{3_2}}.$$

After a similar transformation of A_2 and B_2 we obtain

$$A_1 = A_1^{A_2^{A_{3_1} * B_{3_1}} * B_2^{A_{3_2} * B_{3_2}}}$$

If individuals A_{3_k} and B_{3_k} ($k = 1, 2$) are at least two-tiered, we can perform such a transformation with each of them and similarly with each newly-obtained individual until we exhaust all possible individuals. As a result we obtain a representation of the individual such that the hierarchical structure of his images is clearly seen.

$$A_1 = \mathbf{a}^{\bar{a}^{\bar{a} + \bar{b}} + b^{\bar{a}}}.$$

This individual contains the following elements:

$$\begin{array}{ll} A_2 = \bar{a}^{\bar{a} + \bar{b}}, & B_2 = b^{\bar{a}}, \\ A_{3_1} = \bar{a}, & B_{3_1} = \bar{b}, \\ A_{3_2} = \bar{a}, & B_{3_2} \text{ is absent.} \end{array}$$

We can represent the individual as follows:

$$A_1 = A_1^{A_2^{A_{3_1} + B_{3_1} + B_2^{A_{3_2}}}}.$$

Further expansion of this individual is impossible, since A_{3_1} , B_{3_1} and A_{3_2} consists of only a root. This is the limit depth for the expansion of individual A_1 . However, if we are interested only in individuals at the second tier, we may limit ourselves by the following expansion:

$$A_1 = A_1^{A_2 * B_2}.$$

THE PRINCIPLE OF COMPLEMENTARITY AND THE PHENOMENON OF INTERFERENCE IN THE ALGEBRAIC MODEL OF ETHICAL COGNITION

The principle of complementarity formulated by Niels Bohr appeared as the methodological result of the revolution in physics which took place in the first three decades of our century. This principle arose as a means of overcoming the paradoxical dualism found in the microworld. On the one hand, the electron behaves as a particle, yet on the other it has some of the properties of waves, i.e., interference and diffraction. On the one hand the electron is a point, on the other, it is distributed in space. It was impossible to combine these contradicting peculiarities of microparticles within the framework of the natural-scientific paradigm of the time. This is why Niels Bohr revised the concept of reality itself, the reality which underlies the traditional physical picture of the world. The traditional schemes of the interrelationship between the physicist and reality has assumed a clear dichotomy: on the one hand, there was the physicist with his research instruments, on the other hand, the reality being researched by the physicist with these instruments. Niels Bohr suggested that the paradoxical dualism is caused by neglecting the instruments' role in the cognitive scheme. Here is an approximate scheme of his thinking: If one wants to obtain information about peculiarities of the type α in the microworld, one has to create a special experimental situation A , in the framework of which the peculiarities of the type α may appear. If one wants to obtain information about peculiarities of the type β , one has to create another experimental situation, B , in the framework of which the peculiarities of the type β appear. However, there is no guarantee that one can construct an experimental situation C such that both types of peculiarities α and β might be investigated at the same time. If the situation C cannot exist, then the information about peculiarities α is connected with situation A , and the information about peculiarities β is connected with situation B . Thus, in one model we cannot obtain information about peculiarities of the α -type and the β -type *at the same time*.

Let us suppose that peculiarities of the α -type and the β -type are incompatible within the framework of any traditional model of reality. This incompatibility does not frighten us *logically*, since these contradicting peculiarities belong to different experimental situations, which cannot be

constructed at the same time. We may say that our reality has both types of peculiarities α and β and emphasize that they are *complementary*, i.e., they can be found only in two mutually-exclusive experimental situations.

By using the principle of complementarity, Niels Bohr succeeded to a significant degree in overcoming the conceptual difficulties which appeared after the discovery of quantum-wave dualism. However, Bohr understood that the methodological principle he introduced might not be limited to physical research.

The epistemological lesson of atomic physics has naturally, just as have earlier advances in physical science, given rise to renewed consideration of the use of our means of communication for objective description in other fields of knowledge. Not least the emphasis placed on the observational problem raises the question of the position of living organisms in the description of nature and of our own situation as thinking and active beings. Even though it was, to some extent, possible within the frame of classical physics to compare organisms with machines, it was clear that such comparisons did not take sufficient account of many of the characteristics of life. The inadequacy of the mechanical concept of nature for the description of man's situation is particularly evident in the difficulties entailed in the primitive distinction between soul and body (Niels Bohr, 1958, p. 91).

We will demonstrate that within the framework of the algebraic model of ethical cognition it is possible to find the realization of the principle of complementarity in the processes of self-observation. Our analysis will be strictly mathematical. (Note that in all works we could find which attempt to extend the principle of complementarity beyond the framework of physics, this extension is only a metaphor.)

Niels Bohr felt the phenomenon of complementarity in the observation in human psychology:

However unexpected this development may appear in the domain of physics, I am sure that many of you will have recognized the close analogy between the situation as regards the analysis of atomic phenomena, which I have described, and characteristic features of the problem of observation in human psychology. Indeed, we may say that the trend of modern psychology can be characterized as a reaction against the attempt at analyzing psychical experience into elements which can be associated in the same way as are the results of measurements in classical physics. In introspection it is clearly impossible to distinguish sharply between the phenomena themselves and their conscious perception, and although we may often speak of lending our attention to some particular aspect of a psychical experience, it will appear on closer examination that we really have to do, in such cases, with mutually exclusive situations. We all know the old saying that, if we try to analyze our own emotions, we hardly possess them any longer, and in that sense we recognize between psychical experiences, for the description of which words such as 'thoughts' and 'feelings' are adequately used, a complementary relationship similar to that between the experiences regarding the behavior of atoms obtained under different experimental arrangements and described by means of

different analogies taken from our usual ideas. By such a comparison it is, of course, in no way intended to suggest any closer relation between atomic physics and psychology, but merely to stress an epistemological argument common to both fields, and thus to encourage us to see how far the solution of the relatively simple physical problems may be helpful in clarifying the more intricate psychological questions with which human life confronts us, and which anthropologists and ethnologists so often meet in their investigations (Niels Bohr, 1958, p. 27).

We will demonstrate that within the framework of the algebraic model of ethical cognition the relationship between 'thoughts' and 'feelings', is complementary. Let us note here that while constructing the algebraic model of ethical cognition, the present author did not introduce into it any correlations of this type, and the appearance of complementarity was quite unexpected. It happens to be a consequence of the axioms, which are based directly or indirectly on empirical phenomena.

1. THE PHENOMENON OF COMPLEMENTARITY IN THE ALGEBRAIC MODEL OF ETHICAL COGNITION

In Chapter VI the analogues of human feelings were introduced. Here we will make a more detailed classification. We will distinguish between feelings connected with an individual's reflexion of evil intentions caused by the environmental demands (input feelings) and feelings connected with one's reflexion of the images of himself, of a partner, and of a situation (output feelings). For convenience, we introduce the following designation: let C be an individual; iC means that we consider the case of those values of a and b such that the root of the given individual C is equal to 0. Let W be either A , or B , or $A*B$; the expression Wi is tantamount to the equality $W = 0$. Consider the individual

$$A_1 \equiv a^{A_2 \cdot B_2},$$

where

$$A_2 \equiv a^{A_{31} \cdot B_{31}} \quad \text{and} \quad B_2 \equiv b^{A_{32} \cdot B_{32}}$$

The interpretation of the equality to 0 of his components is as follows.

1. *Input feelings and registrations*

- iA_2 – the individual feels an impulse of his own evil intention;
- iB_2 – the individual feels an impulse of his partner's evil intention;

- iA_{3_1} —the individual registers an impulse of his evil intention from his point of view;
- iA_{3_2} —the individual registers an impulse of his own evil intention from his partner's point of view
- iB_{3_1} —the individual registers an impulse of his partner's evil intention from his own point of view.
- iB_{3_2} —the individual registers an impulse of his partner's evil intention from his partner's point of view.

2. Output feelings and registrations

- A_2i —the individual feels an impulse of his own guilt;
- $A_{3_1}i$ —the individual registers an impulse of his own guilt from his own point of view;
- $A_{3_2}i$ —the individual registers an impulse of his own guilt from his partner's point of view;
- B_2i —the individual feels an impulse of his partner's guilt;
- $B_{3_1}i$ —the individual registers an impulse of his partner's guilt from his own point of view.
- $B_{3_2}i$ —the individual registers an impulse of his partner's guilt from his partner's point of view;
- $(A_2 * B_2)i$ —the individual feels an impulse of suffering;
- $(A_{3_1} * B_{3_1})i$ —the individual registers an impulse of suffering from his own point of view;
- $(A_{3_2} * B_{3_2})i$ —the individual registers an impulse of suffering from his partner's point of view.

We call each feeling of an impulse and each registration of an impulse *elementary observation*.

We call a set of elementary observations an *observation* if there exists an individual A_1 such that if for a certain pair of values $a = \alpha$ and $b = \beta$, every elementary observation from a given set is realized.

We call two observations *compatible* if there exists an individual in whom these two observations can be realized at the 'same time', i.e., for a certain pair of values $a = \alpha$ and $b = \beta$.

We call two observations *complementary* if each of them can be realized in some individual, but there is no individual in whom they can be realized at the same time.

We can state the following facts:

Fact 1. A complete set of all input feelings and their registrations is an observation. In other words, we can find an individual who feels the two possible input impulses and has four possible registrations of these input impulses. *For example*

$$a^{a^a + b + b^a + b}.$$

When $a = 0$ and $b = 0$, all feelings of input impulses and their registrations are equal to 0.

Elements $A_{3_1}, B_{3_1}, A_{3_2}, B_{3_2}, A_{3_1} * B_{3_1}$, and $A_{3_2} * B_{3_2}$ we call registrations of images belonging to A_1 . If the roots of the individuals $A_{3_1}, B_{3_1}, A_{3_2}$, and B_{3_2} become equal to 0 simultaneously with the roots of the individual A_2 and B_2 , we shall say that the individual has correct registrations of his images of himself and of his partner, both from his own point of view and from his partner's point of view. If for some pair (A_2, A_{3_1} or A_2, A_{3_2} or B_2, B_{3_1} or B_2, B_{3_2}) there is no simultaneous transformation of roots into zero, then we shall say that a given registration of an image is not correct. It is easy to see that registrations A_{3_k} and B_{3_k} ($k = 1, 2$) are correct only under the condition that their roots do not have a bar above the letter; otherwise these registrations are incorrect.

Fact 2. The set of all output feelings and their registrations is not an observation. In other words, we cannot find such an individual who simultaneously feels three possible output impulses and has six possible output registrations. The impossibility of simultaneous realization of these elementary observations follows, for example, from the fact that $A_2 i$ and $(A_{3_1} * B_{3_1})i$ cannot be realized at the same time. Indeed, let us consider individual A_2 :

$$A_2 \equiv a^{A_{3_1} * B_{3_1}}.$$

If $A_{3_1} * B_{3_1}$ is equal to zero, then $A_2 = a^0 = 1$.

However, we can say that observations $A_2 i$ and $(A_{3_1} * B_{3_2})i$ are *complementary*, since there is an individual

$$a^{a^a + b + b}$$

who can observe $A_2 i$ (and cannot observe $(A_{3_1} * B_{3_1})i$), when $a = 0$, and $b = 0$, and can observe $(A_{3_1} * B_{3_1})i$ (and cannot observe $A_2 i$) when $a = 1$ and $b = 0$. Thus, we conclude that the feeling of an impulse of one's guilt and the registration of one's suffering (from one's own point of view) are

complementary observations. (Compare with Niels Bohr's words: "We all know the old saying that, if we try to analyze our own emotions, we hardly possess them any longer.")

To conclude this section, we note that it is quite likely that the hypothesis of complementary relations lies at the base of psychotherapy. A therapist tends to realize structures in the patient's inner world, in which registrations of impulses exist. In this case, the possibility of feeling of impulses (which may be the reason for patient's depressive state) is blocked.

2. INTERFERENCE OF THE MICROPARTICLE AND ITS ANALOGUE – THE LAW OF ADDITION OF FREQUENCIES OF IMPULSES OF FEELINGS

Realization of the principle of complementarity in the algebraic model of ethical cognition motivated the author to look for deeper analogies

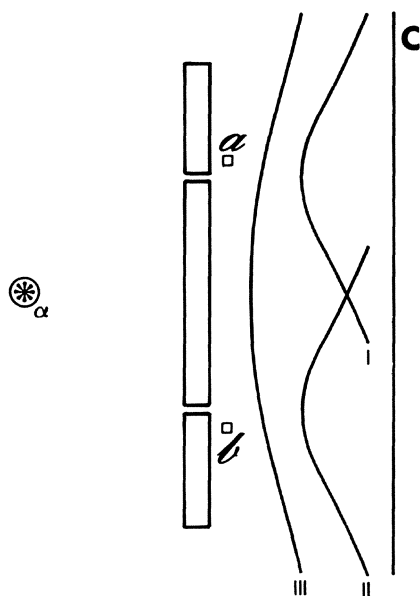


Fig. A10.1. A picture of addition of frequencies from two holes. *a* and *b*, the registering devices, switched on.

between the phenomena observed in the micro-world and phenomena reflected in the human inner world. We will show that the phenomenon of interference of the microparticle has its analogue in our model. In physics textbooks the phenomenon of interference is described as Figures A10.1 and A10.2.

A beam of electrons from the source α goes through two holes A and B to screen C . Device a registers each electron coming through hole A and device b registers each electron coming through hole B (Figure A10.1). After a large number of electrons has come onto the screen, we obtain the distribution of them as a curve III, which is the sum of elementary curves I and II (obtained as a result of dispersion of electrons coming through each hole A and B separately). Thus, the frequency of appearance of an electron at any one point is $r = p + q$, where p is the frequency of an electron appearing after passing through hole A , and q is the frequency of an electron appearing after

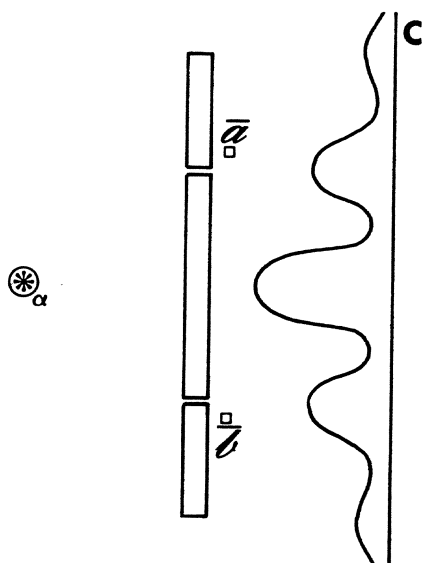


Fig. A10.2. A picture of interference from two holes. $\bar{a}$ and $\bar{b}$, registering devices, switched off.

passing through hole *B*. Now we switch off the registering devices. We obtain another picture of distribution. It turns out that the law of addition of frequencies (described above) cannot be applied in this case (Figure A10.2). To calculate the frequency of an electron appearing at some point, we have to introduce some fictitious wave process and find the sum of complex amplitudes of this process at the given point. The frequency is equal to the square of the module of the sum of these amplitudes. Let φ_1 be an amplitude corresponding to hole *A* and φ_2 be an amplitude corresponding to hole *B*. Then for each point of the screen, the following is true:

$$r = |\varphi_1 + \varphi_2|^2 = |\varphi_1|^2 + |\varphi_2|^2 + 2|\varphi_1| \cdot |\varphi_2| \cdot \cos(\theta_1 - \theta_2).$$

where $\theta_1 - \theta_2$ is the difference of the phases of the oscillatory process at a given point.

Let us return to our algebraic model of ethical cognition and consider an individual with no more than three tiers: $a^{A \cdot B}$. Let *a* become 0 with a frequency *x*, and let *b* become 0 with a frequency *y*. Let us represent the frequency of the impulses of suffering of this individual $\|A \cdot B\|$ as a function of the frequency of the impulses of his own guilt $\|A\|$, and of frequency of the impulses of his partner's guilt $\|B\|$, which is called the frequency of impulses of condemnation. There is a curious correlation connecting these variables, similar to the phenomenon of interference of particles and its disappearance in the case of registration of particles passing through. In a case where an individual does not have any single correct registration, either of his own image or of his partner's image, the frequencies are connected by the following equality:

$$\|A \cdot B\| = \|A\| + \|B\| - \|A + B\|,$$

where $\|A + B\| > 0$.

This equality we will regard as the analogue of interference in our model of cognition.

In the case where an individual has at least one correct registration either of his image of himself or of that of his partner, the frequencies are connected by the formula:

$$\|A \cdot B\| = \|A\| + \|B\|.$$

The precise formulation of this rule is given by the following theorem:

THEOREM. Let individual $a^{A \cdot B}$ be given. A is any formula from the set $a, a^a, a^b, a^{a \cdot b}$, and B is any formula from the set $b, b^b, b^a, b^{a \cdot b}$. Then:

(i) equation

$$\|A \cdot B\| = \|A\| + \|B\| - \|A + B\|,$$

where $\|A + B\| > 0$, is true for all cases, where A is any formula from the set $a, a^a, a^b, a^{a \cdot b}$, and B is any formula from the set $b, b^b, b^a, b^{a \cdot b}$;

(ii) in all other cases the equation

$$\|A \cdot B\| = \|A\| + \|B\|$$

is true.

Proof. Note that the frequency of A and B simultaneously becoming zero, is equal to the frequency of formula $A + B$ becoming zero. Let us consider A and B , mentioned in case (i). If $a = 0$ and $b = 0$, then $A + B = 0$ since all symbols which may appear at the second tier have a bar above. Thus, the set of cases where $A + B = 0$ is not empty, and $\|A + B\| > 0$.

Consider A and B , mentioned in case (ii). A and B cannot become equal to zero at the same time, since if $a = 0$ and $b = 0$ (which is the necessary condition for A and B simultaneously being zero) at least one letter at the second tier does not have a bar above and the value of the corresponding individual (when $a = 0$ and $b = 0$) becomes $0^0 = 1$. Thus the set of cases when $A + B = 0$ is empty, and $\|A + B\| = 0$.

Let us consider any individuals A and B allowed by these Theorem conditions. To make formula $A \cdot B$ equal to zero it is enough for one of the factors to become equal to zero. Thus, according to the general rule of addition of probabilities we can write down:

$$\|A \cdot B\| = \|A\| + \|B\| - \|A + B\|.$$

It was shown above that $\|A + B\| > 0$ is true for the cases where the second tier can have only symbols with a bar above, and $\|A + B\| = 0$ is true for those cases where at least one symbol at the second tier does not have a bar above. □

Therefore, the presence of interference in our sense is connected with the

absence of correct registration of images: if there is correct registration, there is no interference; if there is not correct registration, there is interference.

3. AN IMITATION OF THE EXPERIMENT WITH THE TWO HOLES WITH THE HELP OF THE ALGEBRAIC MODEL OF ETHICAL COGNITION

Now we relate our construction more closely to a physical experiment. Consider an individual $a^{A \cdot B}$, where $A = a^a \cdot b$ and $B = b^a \cdot b$, and draw the following parallels:

The Table of Correspondence between Physical and Cognitive Interpretations

Parts of formulae and their values	Physical interpretations	Cognitive interpretation
A	hole A	one's image of oneself
B	hole B	one's image of one's partner
$A \cdot B$	absorbing screen	one's image of the situation
a	counter which registers particles through hole A and shows the frequency of their coming	registration of one's image of oneself
$a = a$	counter is switched on	correct registration of one's image of oneself
$a = \bar{a}$	counter is switched off	incorrect registration of one's image of oneself
b	counter which registers particles through hole B and shows the frequency of their coming	registration of one's image of one's partner
$b = b$	counter is switched on	correct registration of one's image of one's partner
$b = \bar{b}$	counter is switched off	incorrect registration of one's image of one's partner
Ai	a particle reaching a screen when only hole A is open	an impulse of guilt
Bi	a particle reaching a screen when only hole B is open	an impulse of condemnation
$(A \cdot B)i$	a particle reaching a screen when both holes are open	an impulse of suffering
ai	counter a registers a particle	registration of an impulse of guilt
bi	counter b registers a particle	registration of an impulse of condemnation

(Table continued)

Parts of formulae and their values	Physical interpretations	Cognitive interpretation
$\ A\ $	probability of reaching a screen by a particle, when only hole A is open	frequency of impulses of guilt
$\ B\ $	probability of reaching a screen by a particle when only hole B is open	frequency of impulses of condemnation
$\ A \cdot B\ $	probability of reaching a screen by a particle when both holes are open	frequency of impulses of suffering
$\ a\ $	frequency of registration of particles shown at counter a	frequency of registration of impulses of guilt
$\ b\ $	frequency of registration of particles shown at counter b	frequency of registration of impulses of condemnation
x	probability that a particle from the source comes to hole A	frequency of the environmental demands on individual A to perform an evil action
y	probability that a particle from the source comes to hole B	frequency of the environmental demands on individual B to perform an evil action

Note that in our model not all of the particles coming to the hole (if only this hole is open) reach the screen; some particles may be absorbed somewhere. In addition, we have to introduce one more limitation: $x + y \leq 1$. (In the model of ethical cognition this limitation was not necessary.) We suppose also that, if a hole is closed or that not a single particle passing through the hole reaches the screen, the reading of a corresponding counter has no sense. Ai and $\|A\|$ have a physical interpretation if only one hole A is open, and Bi and $\|B\|$ have a physical interpretation if only one hole B is open.

Let us now conduct a few 'physical' experiments. First, we switch off all counters. Our experimental device corresponds to the individual

$$a^{\bar{a} \cdot \bar{b}} \cdot b^{\bar{a} \cdot \bar{b}}$$

If we open only hole A , the frequency of particles' appearance on the screen is:

$$\|A\| = \|a^{\bar{a} \cdot \bar{b}}\| = x \cdot y.$$

The calculations are made under the assumption that a becomes 0 with frequency x , and b becomes 0 with frequency y .

If we open only hole B , the frequency of particles' appearance on the screen is

$$\|B\| = \|b^{\bar{a} \cdot \bar{b}}\| = x \cdot y.$$

If we open both holes, then

$$\|A \cdot B\| = \|a^{\bar{a} \cdot \bar{b}} \cdot b^{\bar{a} \cdot \bar{b}}\| = x \cdot y.$$

We see that both holes being open works in the same way as any of them alone.

Now we derive the analytical expression connecting $\|A\|$ and $\|B\|$ with $\|A \cdot B\|$. It is easy to see that the following equation holds

$$\|A \cdot B\| = \|A\| + \|B\| - \sqrt{\|A\|} \cdot \sqrt{\|B\|}.$$

This formula is closely related to the standard formula of interference, which is obtained from completely different considerations. (This formula is a particular case of the formula for interference, under the condition that the amplitudes of summed oscillations are equal, and the difference of phases in a given point of the screen is $(2k+1)\pi \pm 1/3\pi$, where $k=0, 1, 2, \dots$). Let us note that we did not use the idea of a wave, nor any of its analytical analogues. Our formula in its origin is not connected in any way with the usual mathematical transformation for obtaining formulae of interference. Let us consider the individual with only counter b switched on

$$a^{\bar{a} \cdot b} \cdot b^{\bar{a} \cdot \bar{b}}.$$

Open both holes:

$$\|A \cdot B\| = \|a^{\bar{a} \cdot b} \cdot b^{\bar{a} \cdot \bar{b}}\| = x.$$

Then open one hole A :

$$\|A\| = \|a^{\bar{a} \cdot b}\| = x \cdot (1 - y).$$

Open only one hole B :

$$\|B\| = \|b^{\bar{a} \cdot \bar{b}}\| = x \cdot y.$$

Thus the equation

$$\|A \cdot B\| = \|A\| + \|B\|$$

holds. *The phenomenon of interference has disappeared.*

The reading $\|b\|$ of counter b is the probability of a particle coming to hole B .

Thus, our model predicts the phenomenon of interference when there is no registration of particles and the disappearance of interference when particles are registered. For the case of the absence of registration of particles the model indicates the following formula:

$$\|A \cdot B\| = \|A\| + \|B\| - \sqrt{\|A\|} \cdot \sqrt{\|B\|}, \quad (\text{A10.1})$$

and for the case with registration of particles:

$$\|A \cdot B\| = \|A\| + \|B\|. \quad (\text{A10.2})$$

If we remove the counters from the experimental device, we obtain the formula $a^{a \cdot b}$. In this case $A \equiv a$, and $B \equiv b$. If we open only hole A , we have the frequency of the appearance of the particle on the screen:

$$\|A\| = \|a\| = x,$$

if we open only hole B , we have the frequency

$$\|B\| = \|b\| = y.$$

If we open the two holes A and B , we have the frequency

$$\|A \cdot B\| = \|a \cdot b\| = x + y - xy. \quad (\text{A10.3})$$

But this is the formula for addition of probabilities of joined events.

Thus, we see that the only *reason* for a difference between the formula of interference (A10.1) and the formula of addition of probabilities (A10.3) is the presence of *counters switched off* in the experimental device.

Now we will try to find the analogues of the concepts 'amplitude of probability' and 'module of amplitude of probability', which are used in quantum mechanics. Let us consider the interference formula obtained above:

$$\|A \cdot B\| = \|A\| + \|B\| - \sqrt{\|A\|} \cdot \sqrt{\|B\|}.$$

This formula may be rewritten as

$$(\sqrt{\|A \cdot B\|})^2 = (\sqrt{\|A\|})^2 + (\sqrt{\|B\|})^2 - \sqrt{\|A\|} \cdot \sqrt{\|B\|}.$$

By comparing this formula with the standard physical expression, we can see that the Boolean function is an analogue of an amplitude of probability, that a square root of the module of this function is an analogue of the module of an amplitude of probability, and that the module of the Boolean function is probability itself.

If we look now in another way at our formulae describing individuals, situations, and their relations, we see that they are *the analogues of amplitude correlations in quantum mechanics*.

One may suppose that this parallel would be useful not only for the psychologist, but also for the physicist, since in physics neither amplitudes of probabilities, nor their modules (as it seems to the present author) have clear interpretations.

Let us consider the above formula and try to understand the meaning of modules of amplitudes $\sqrt{\|A\|}$ and $\sqrt{\|B\|}$. Suppose $x = y$, then

$$\sqrt{\|A\|} = x \quad \sqrt{\|B\|} = x.$$

As noted, x is the probability of a particle passing one open hole in the ideal case where there is *no possibility* in the experimental device to register particles. The individual $a^{a \cdot b}$ corresponds to this situation.

In other words, our 'modules of amplitudes of probabilities' are probabilities of passing the particles through one open hole in an ideal world where no counters exist. We may suppose that the obligatory existence of counters is not a removable limitation in any physical experimental situation; however, there is no reason to suppose the same about the human inner world.

Let us now try to construct a scheme for consecutive addition of holes.

Symbol $+$ corresponds to this. The first case is two consecutive holes with no counters a^{a+b} . It is easy to see that

$$\|a + b\| = \|a\| \cdot \|b\|.$$

The situation with two consecutive holes and counters switched off will be expressed as:

$$a^{a\bar{a} + \bar{b} + b\bar{a} + \bar{b}}.$$

where $A \equiv a^{\bar{a} + \bar{b}}$ and $B \equiv b^{\bar{a} + \bar{b}}$. One can prove that

$$\|A + B\| = \|A\| \cdot \|B\|.$$

Transforming this formula to an expression with the modules of amplitudes of probabilities:

$$\|A + B\| = (\sqrt{\|A\|})^2 \cdot (\sqrt{\|B\|})^2,$$

which is very close to the corresponding correlation in quantum mechanics.

Let us now clarify the meaning of the modules of amplitudes of probabilities for consecutive holes. If $x = y$,

$$\|A\| = x \quad \text{and} \quad \|B\| = x.$$

Therefore,

$$\sqrt{\|A\|} = \sqrt{x}, \quad \sqrt{\|B\|} = \sqrt{x},$$

and the meaning of the module of amplitude of probability for consecutive holes does not coincide with its meaning for parallel holes. Now, there are the *square roots* of the probabilities of passing the particle through each hole, under the condition that there are no counters at all in the experimental device, even switched off.

The phenomenon of complementary, which we found through formal calculation within the algebraic model of cognition is similar to the phenomenon of complementarity found in physics by Niels Bohr, who

predicted its possible existence in human cognition. Moreover, it was found that such specific physical correlations as 'registration of particle present—interference absent, registration absent—interference present' has its analogue in the algebraic model of cognition; specific quantum concepts such as amplitude of probability and square of its module interpreted as probability also have their analogues in the algebraic model of cognition.

The examined theoretical relations suggest to us that both in the micro-world and in human ethics the selfsame universal laws are manifested.

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